# 🛡️ Module 11: Reverse Engineering, Fuzzing and Binary Exploitation

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# Exercise 10: Reverse Engineering a Binary with Ghidra
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## Scenario

**Ghidra** is a state-of-the-art Software Reverse Engineering (**SRE**) framework developed by the National Security Agency (**NSA**). It provides an extensive suite of software analysis tools for decompiling, disassembling, scripting, and auditing compiled Portable Executables (**PE**), ELF binaries, and firmware across multiple processor architectures (x86, x64, ARM, MIPS, PowerPC).

Key reverse engineering workflows in Ghidra:
1. **Code Disassembly & Decompilation**: Transforming raw machine opcodes into structured C pseudo-code using Ghidra's decompiler engine ( `Ctrl+E` / Decompiler Window).
2. **Control Flow Graph Analysis**: Visualizing function basic block structures, execution branches, and conditional jumps using the **Function Graph** and **Function Call Tree / Graph**.
3. **Static Symbol & String Auditing**: Extracting hardcoded program strings (**Defined Strings** window), identifying standard library API imports ( `strcmp` , `scanf` , `malloc` ), and resolving local stack frame offsets ( `EBP - 0x18` , `EBP - 0x1c` ).
4. **Binary Patching & Equates**: Mapping memory addresses, setting symbolic equates (**Equates Table**), and customizing code listing displays.

In this exercise, we switch to the **Windows 11** VM (**Admin** / **Pa$$w0rd**), launch Ghidra ( `Ghidrarun.bat` ), import and analyze Portable Executable binaries **ABO2.exe** and **Crackme0x00.exe**, inspect function call graphs, disassemble stack frame alignments ( `PUSH EBP` , `MOV EBP, ESP` , `SUB ESP, 0x38` ), identify string comparison logic ( `strcmp` ), and successfully extract the hardcoded password (**250382**).

## Objective

The objective of this lab is to help students learn how to:
- Launch Ghidra SRE suite on Windows 11 target VM (**C:\Users\Admin\Downloads\Tools\Ghidra\Ghidrarun.bat**)
- Create non-shared Ghidra projects ( `test` , `test1` )
- Import PE binaries (**ABO2.exe** and **Crackme0x00.exe**) into CodeBrowser
- Configure Ghidra Core plugins, processor modules (x86, ARM), and keybindings
- Decompile binary functions into C pseudo-code using **Ctrl+E** / Decompiler view
- Audit execution flow through basic block Function Graphs (**Window -> Function Graph**)
- Analyze stack initialization sequences ( `PUSH EBP` -> `MOV EBP, ESP` -> `SUB ESP, 0x38` )
- Utilize Defined Strings window (**Window -> Defined Strings**) to uncover embedded binary literals
- Apply custom Equates (**Set Equate...**) and view symbol references
- Decompile string validation logic comparing input against embedded password (**250382**)
- Verify recovered password against **crackme0x00.exe** via Windows Command Prompt

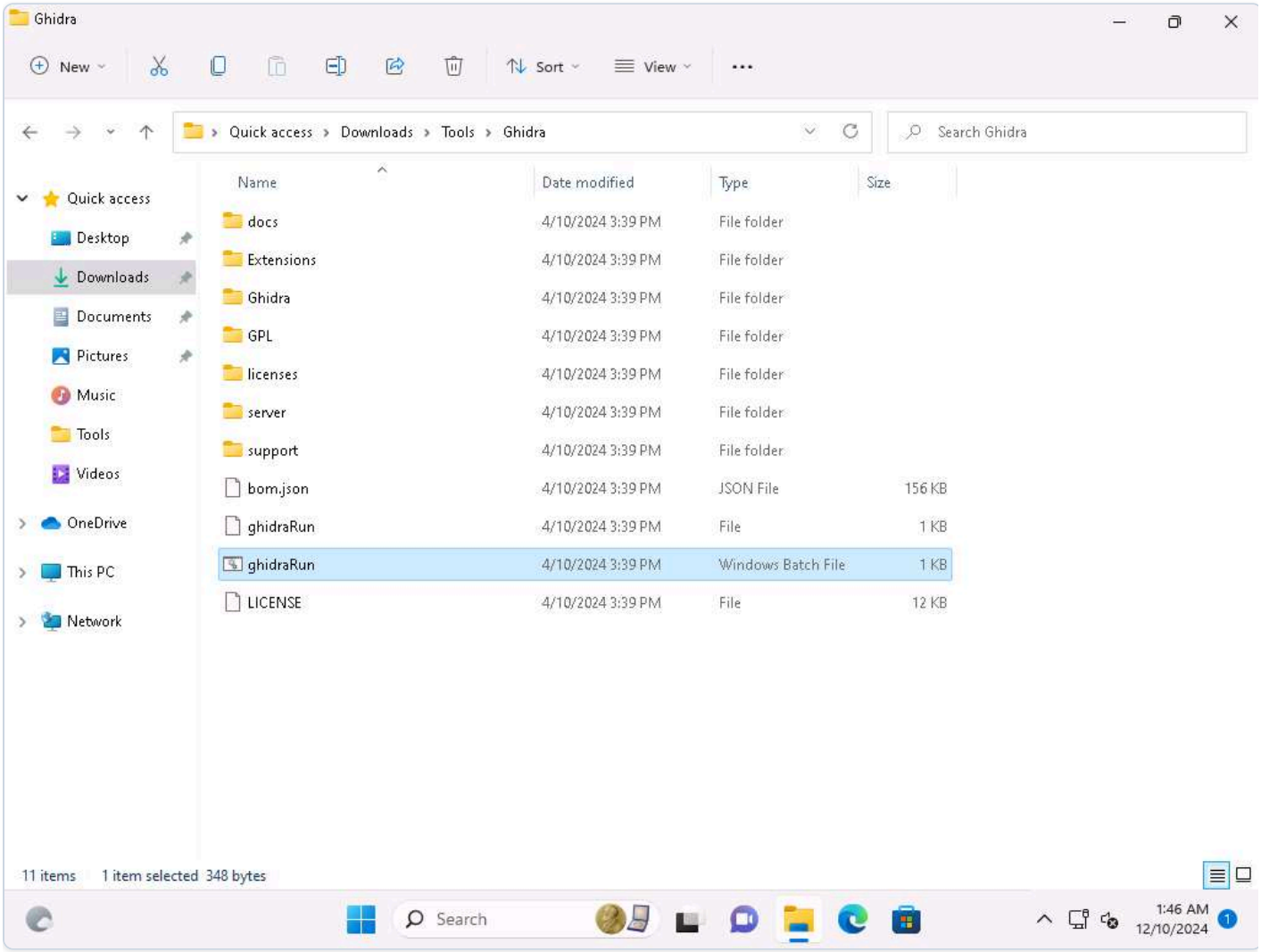
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## Task 1: Initialize Ghidra, Import ABO2.exe, and Decompile Functions

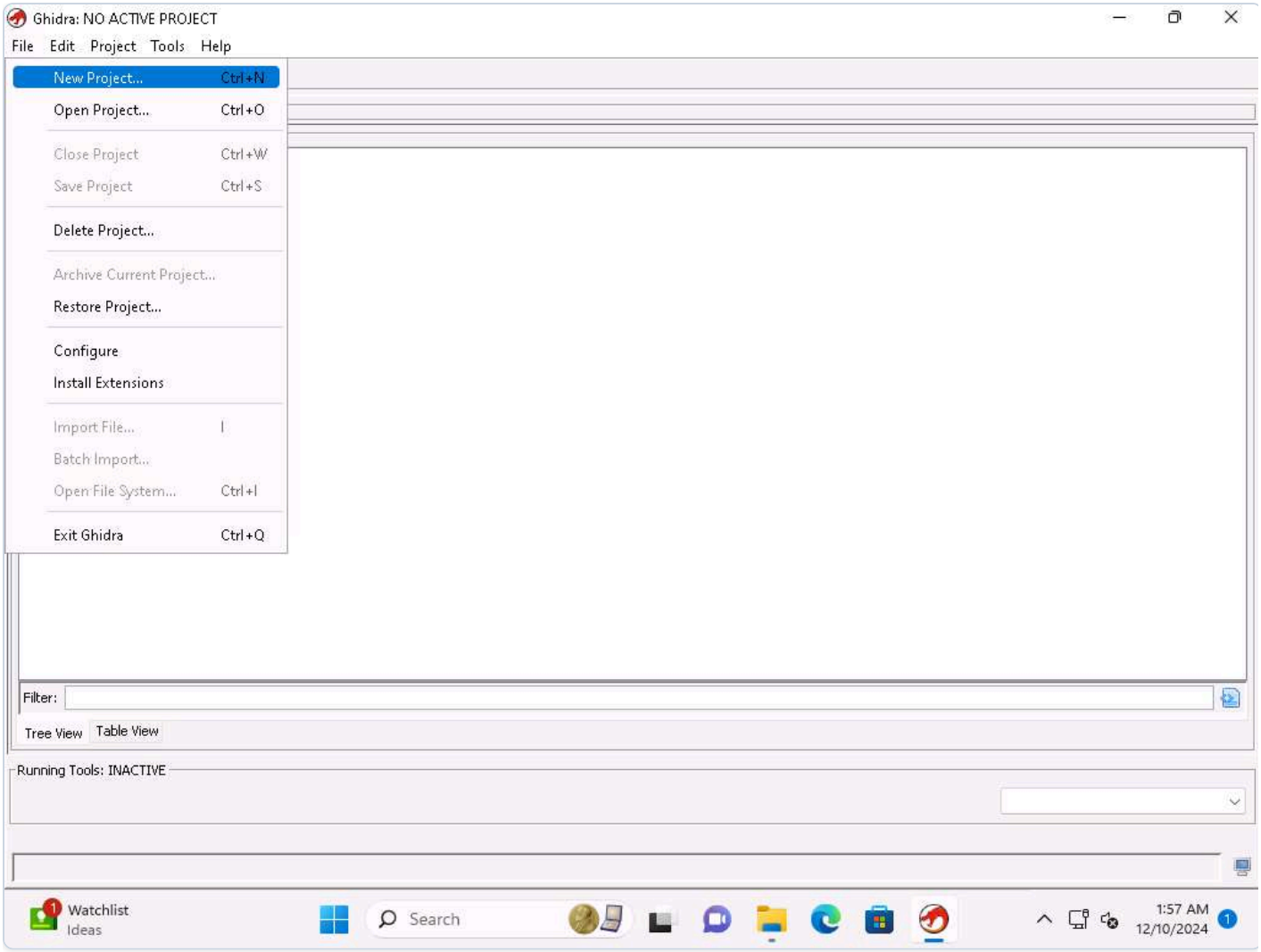
In this task, we launch Ghidra on Windows 11, create project `test`, import `AB02.exe`, run auto-analysis, explore function listings (`_scrt_common_main_seh`), inspect function graphs, and decompile C pseudo-code using `Ctrl+E`.

## Lab Tasks

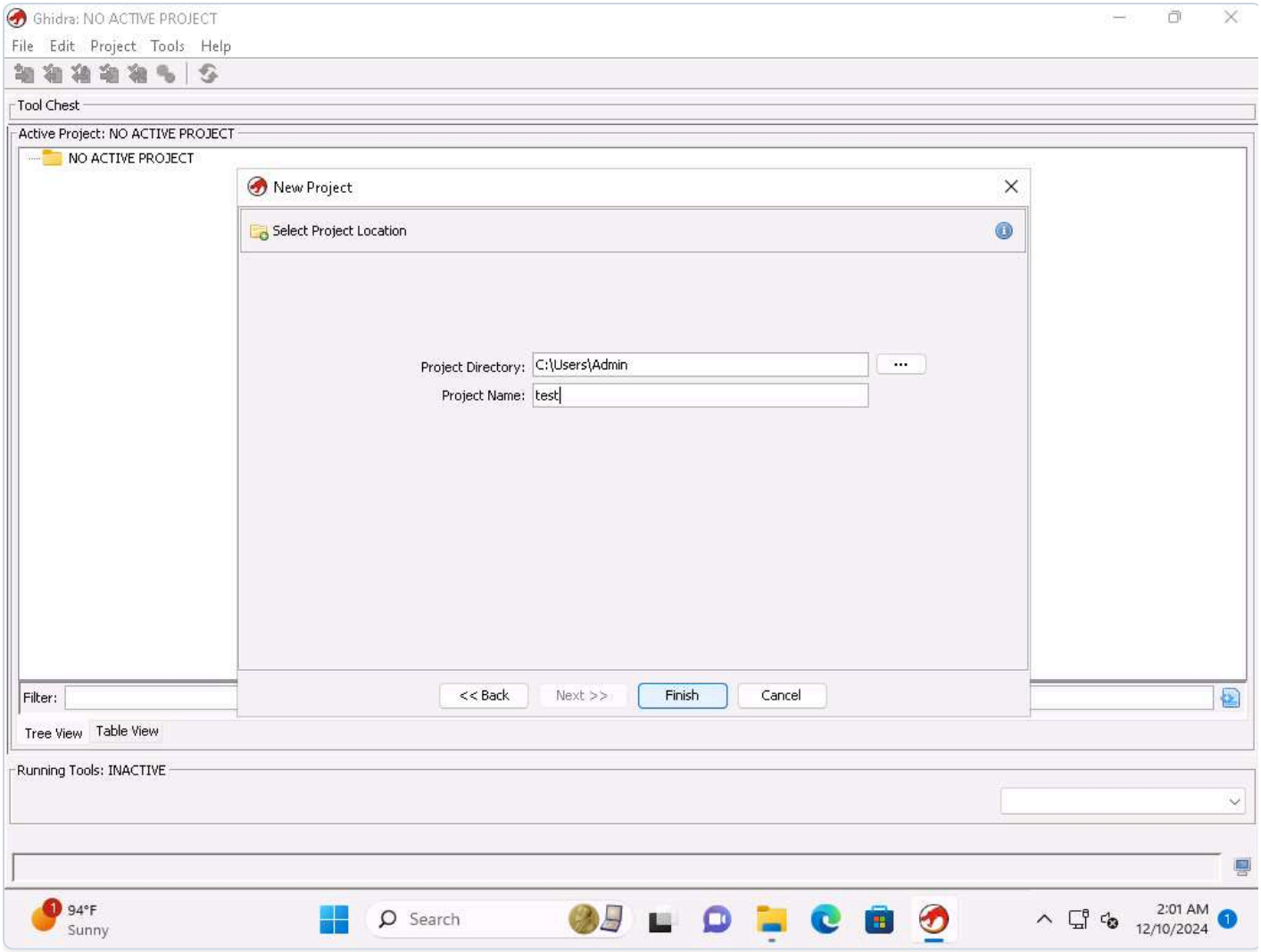
1. Switch to **Windows 11** VM, navigate to `C:\Users\Admin\Downloads\Tools\Ghidra`, and double-click **Ghidrarun.bat**.



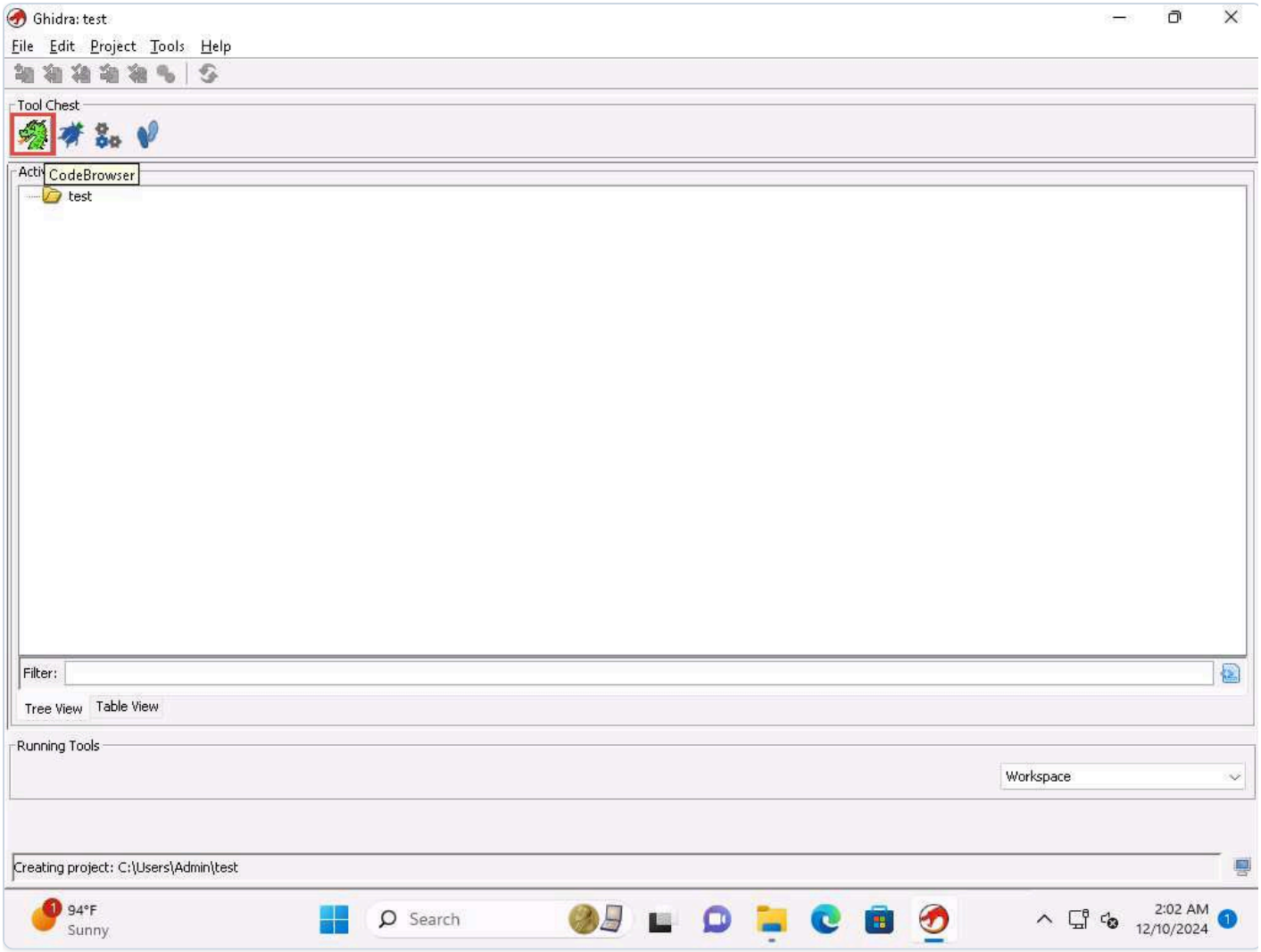

1. In Ghidra window, click **File -> New Project**.



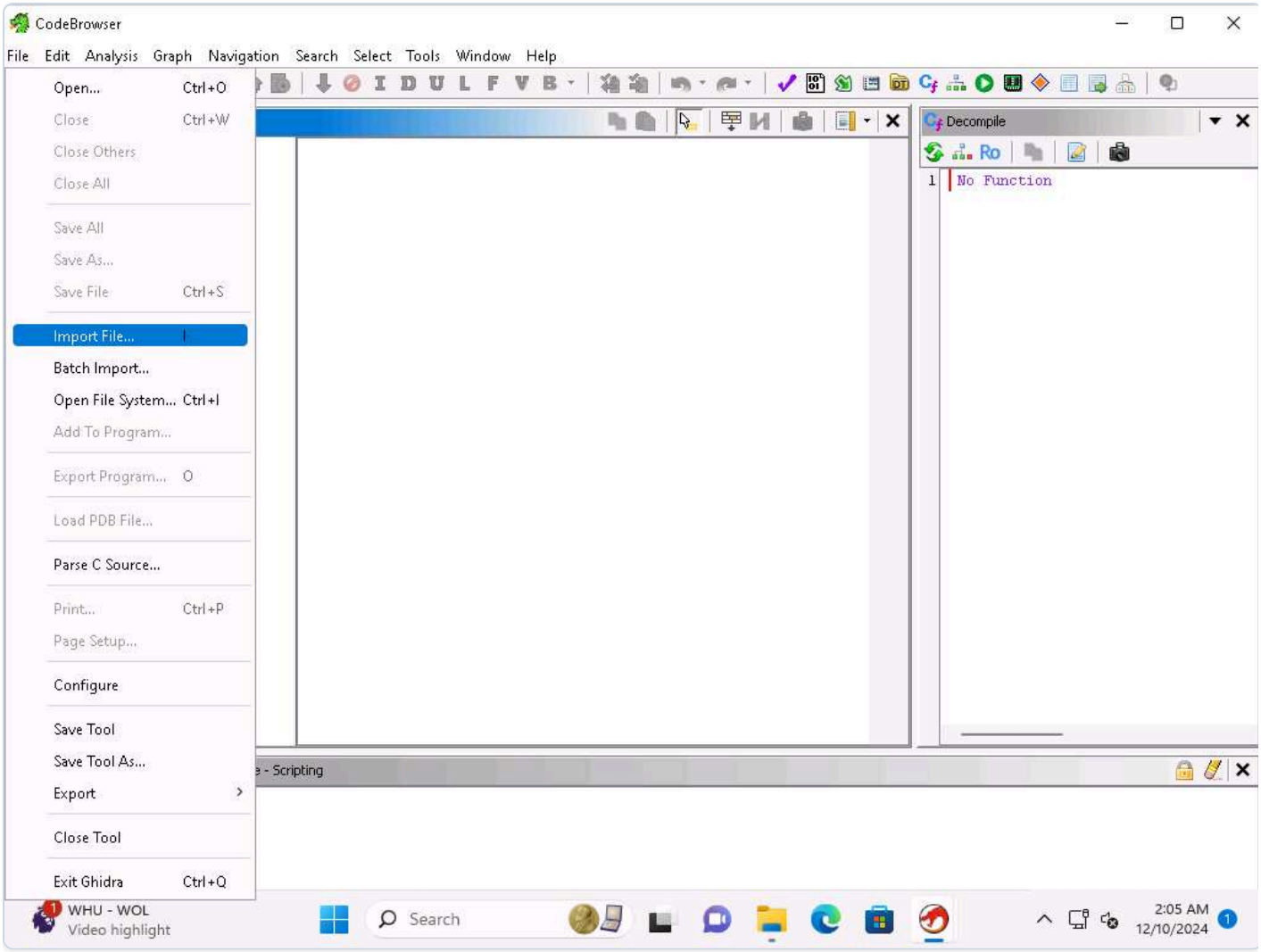

1. Select **Non-Shared Project**, click **Next**, enter project name `test`, and click **Finish**.

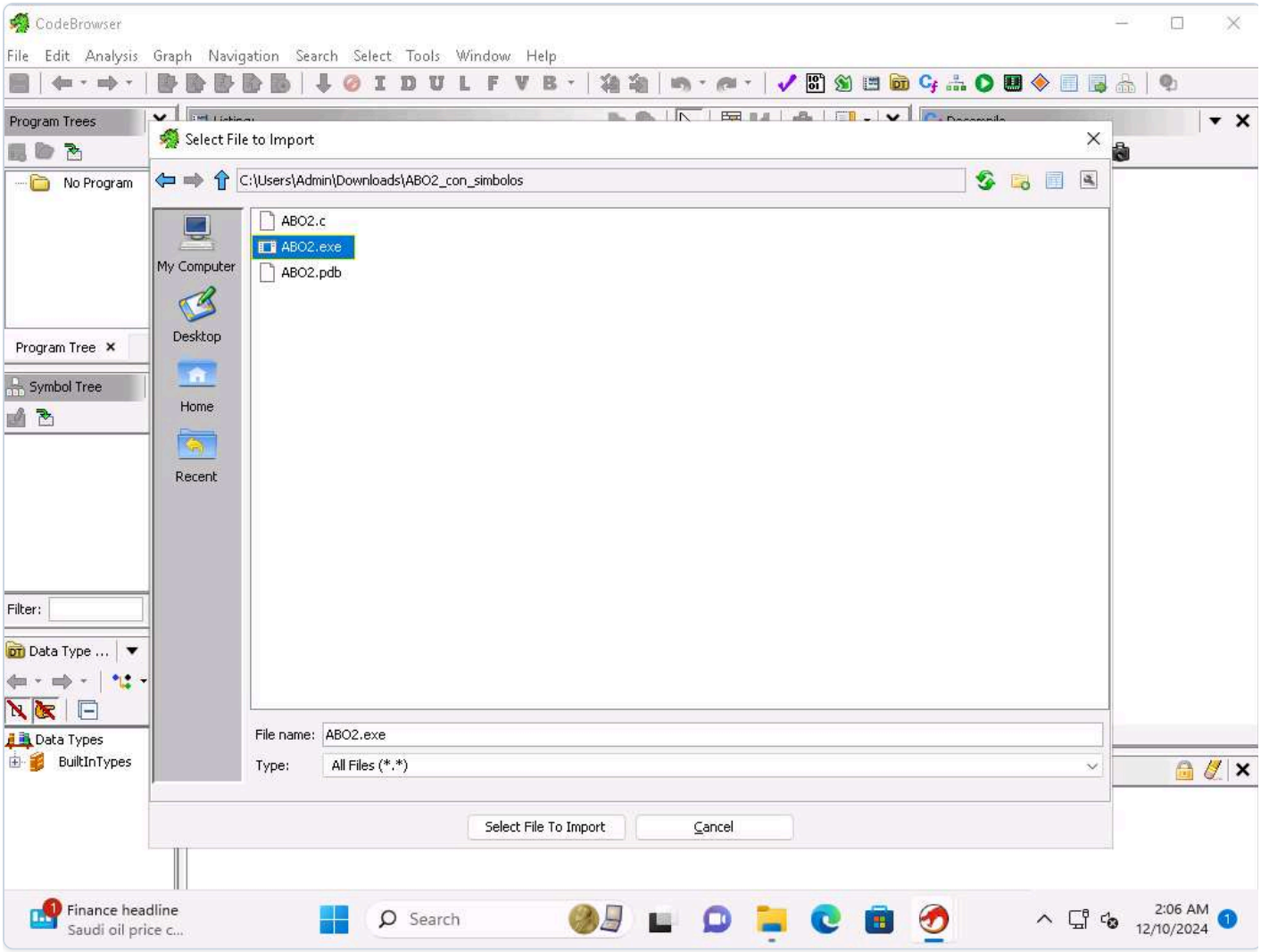



1. Click the **Code Browser** tool icon (dragon icon).

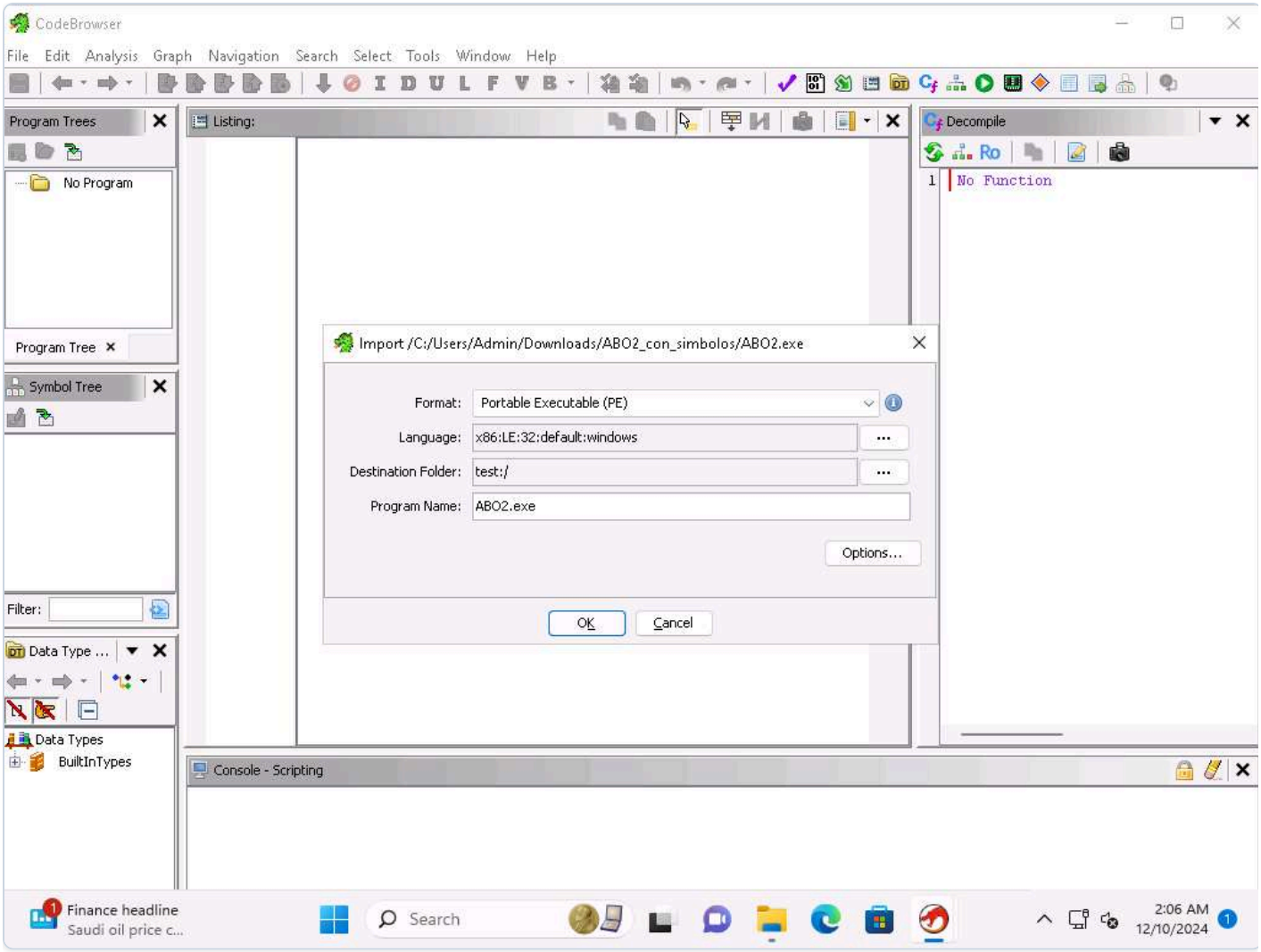



1. Click **File -> Import File**, navigate to `C:\Users\Admin\Downloads\ABO2_con_simbolos`, select **ABO2.exe**, and click **Select File to Import**.

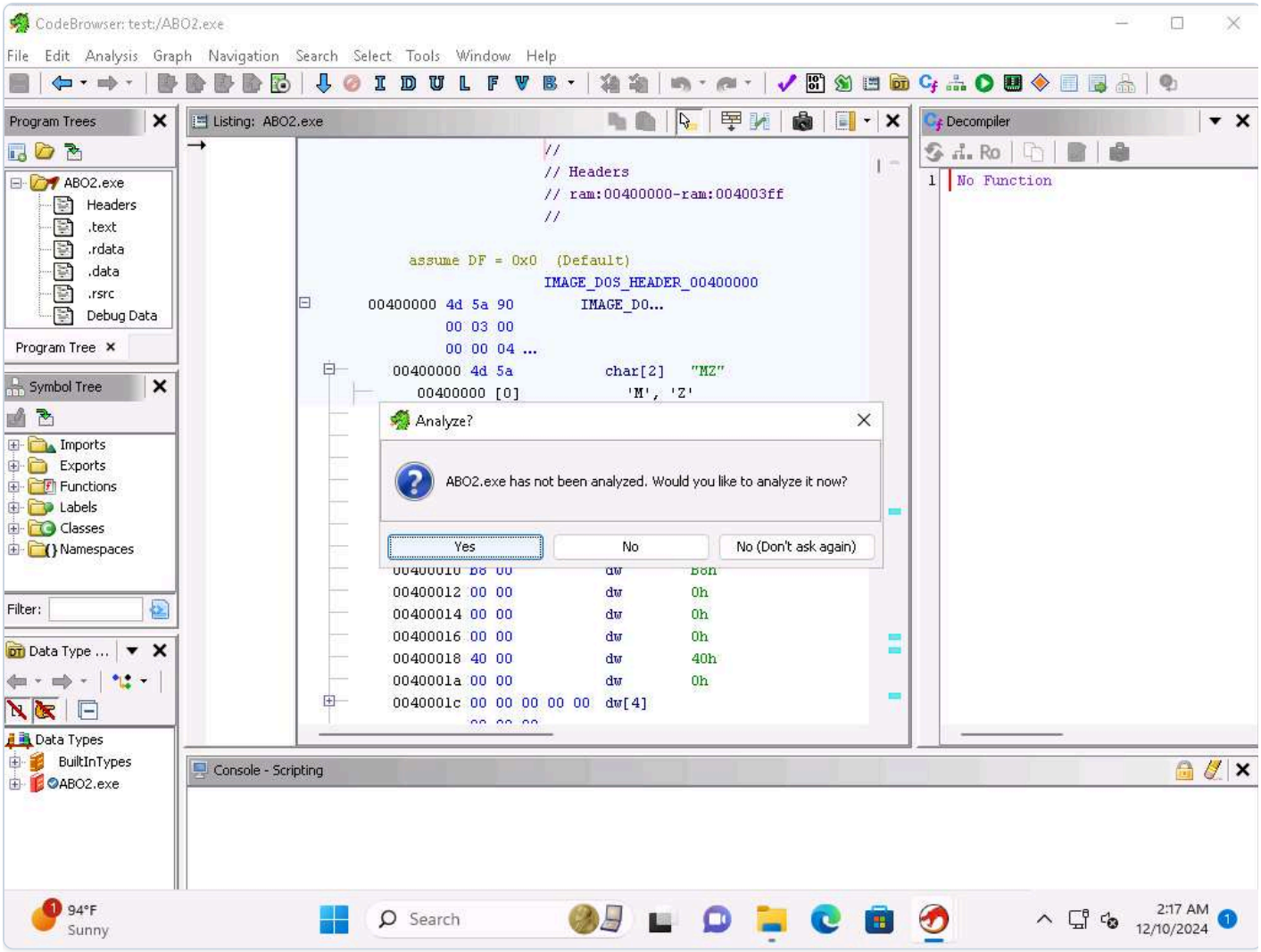

CodeBrowser
File Edit Analysis Graph Navigation Search Select Tools Window Help
Open... Ctrl+O
Close Ctrl+W
Close Others
Close All
Save All
Save As...
Save File Ctrl+S
Import File...
Batch Import...
Open File System... Ctrl+I
Add To Program...
Export Program... O
Load PDB File...
Parse C Source...
Print... Ctrl+P
Page Setup...
Configure
Save Tool
Save Tool As...
Export
Close Tool
Exit Ghidra Ctrl+Q
Decompile
1 No Function
- Scripting
WHU - WOL
Video highlight
Search
2:05 AM
12/10/2024



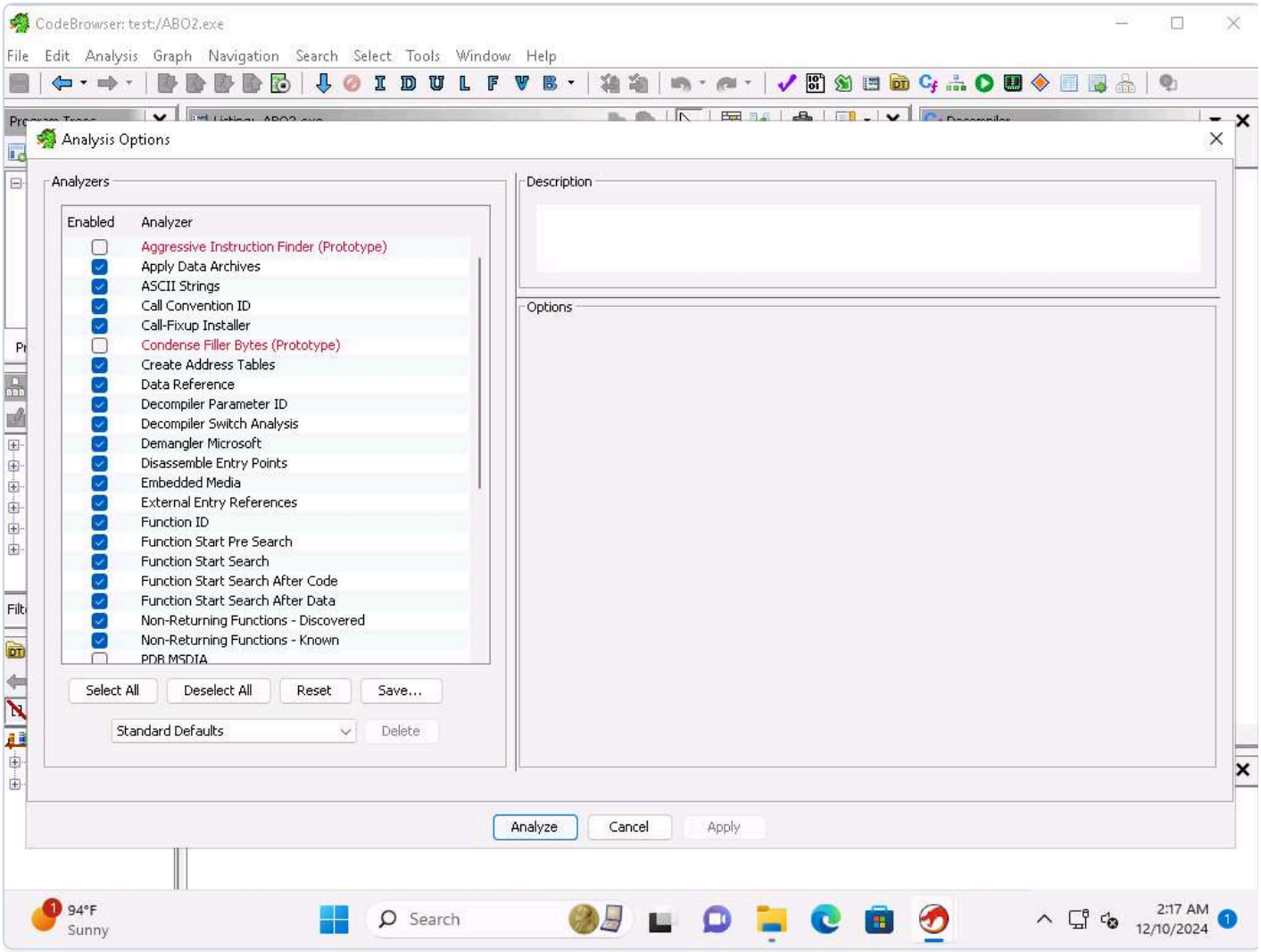

1. Confirm PE format details in Import pop-up and click **OK**. Accept Windows Defender firewall prompt if presented.



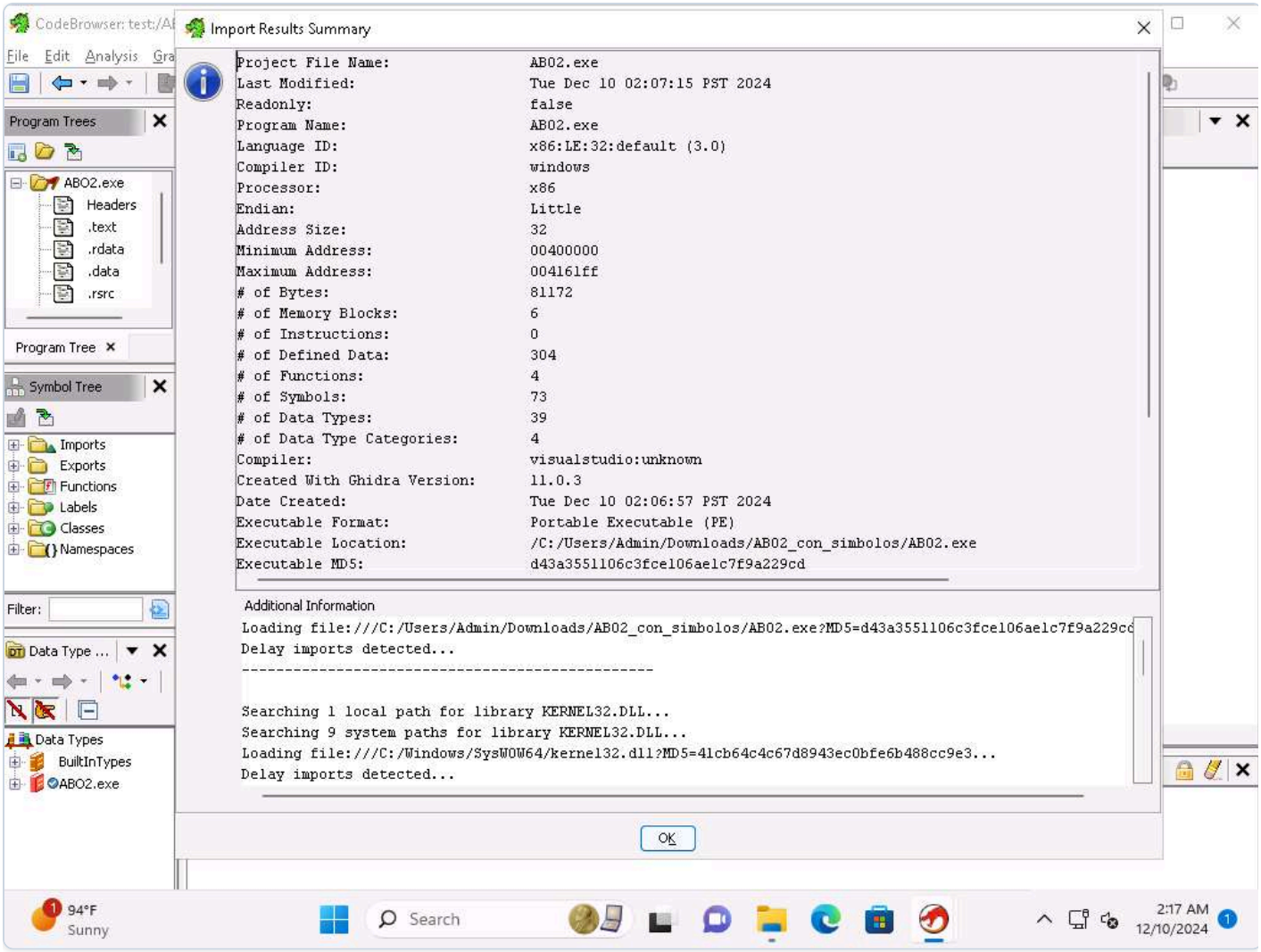

1. In **Analyze?** prompt, click **Yes**.



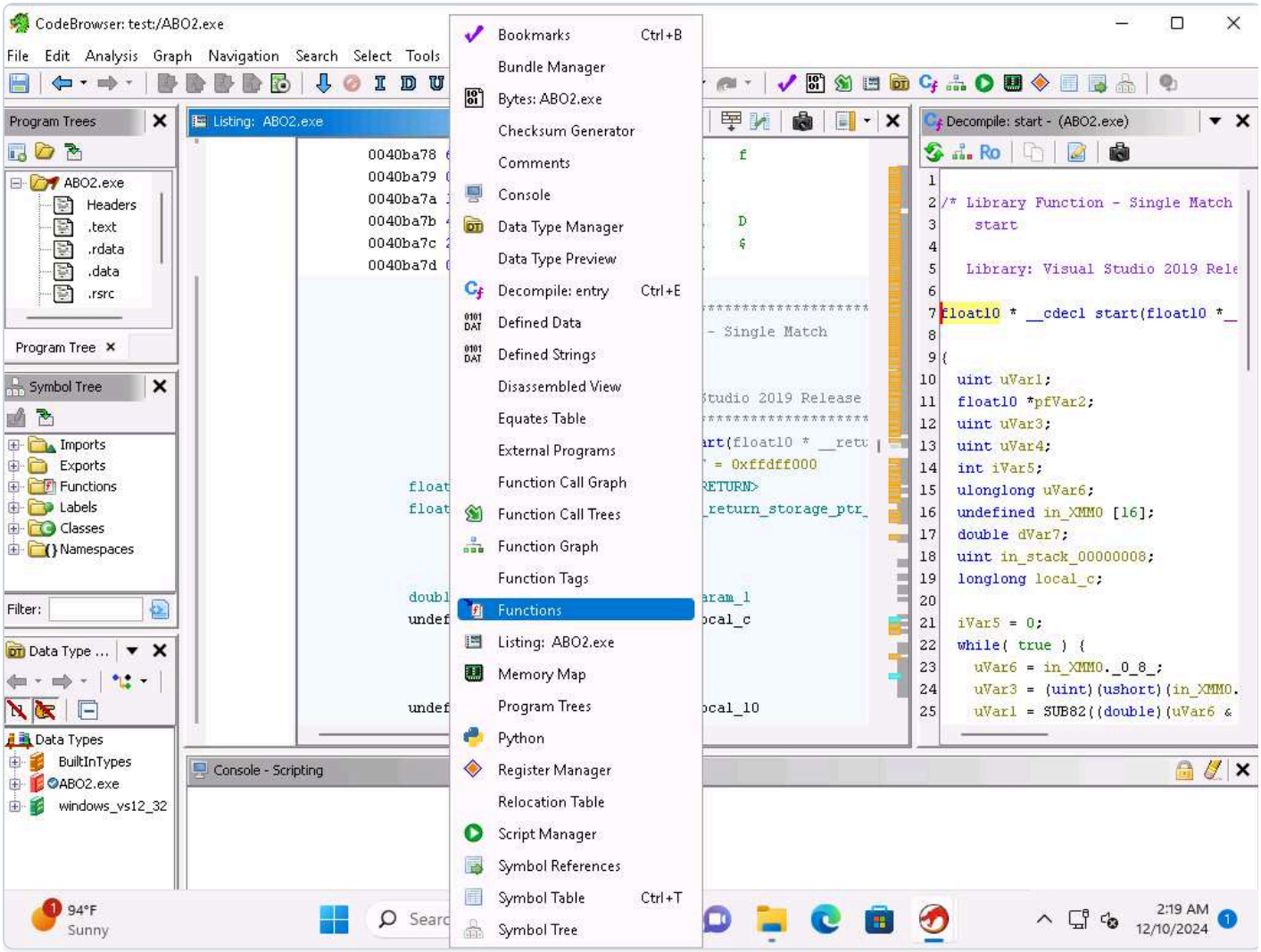

1. Keep default analyzer options and click **Analyze**.



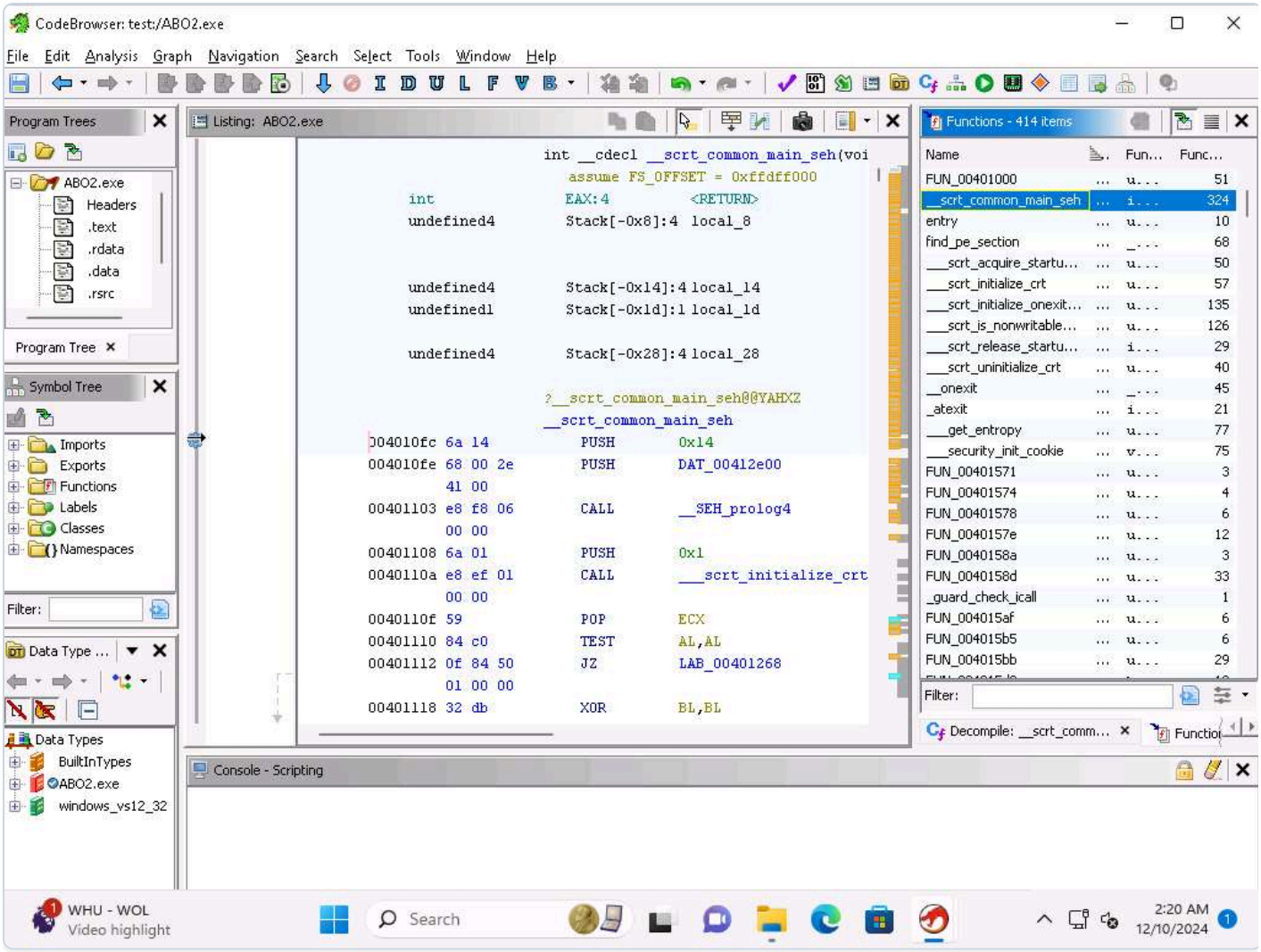

1. Click **OK** in **Import Results Summary**.



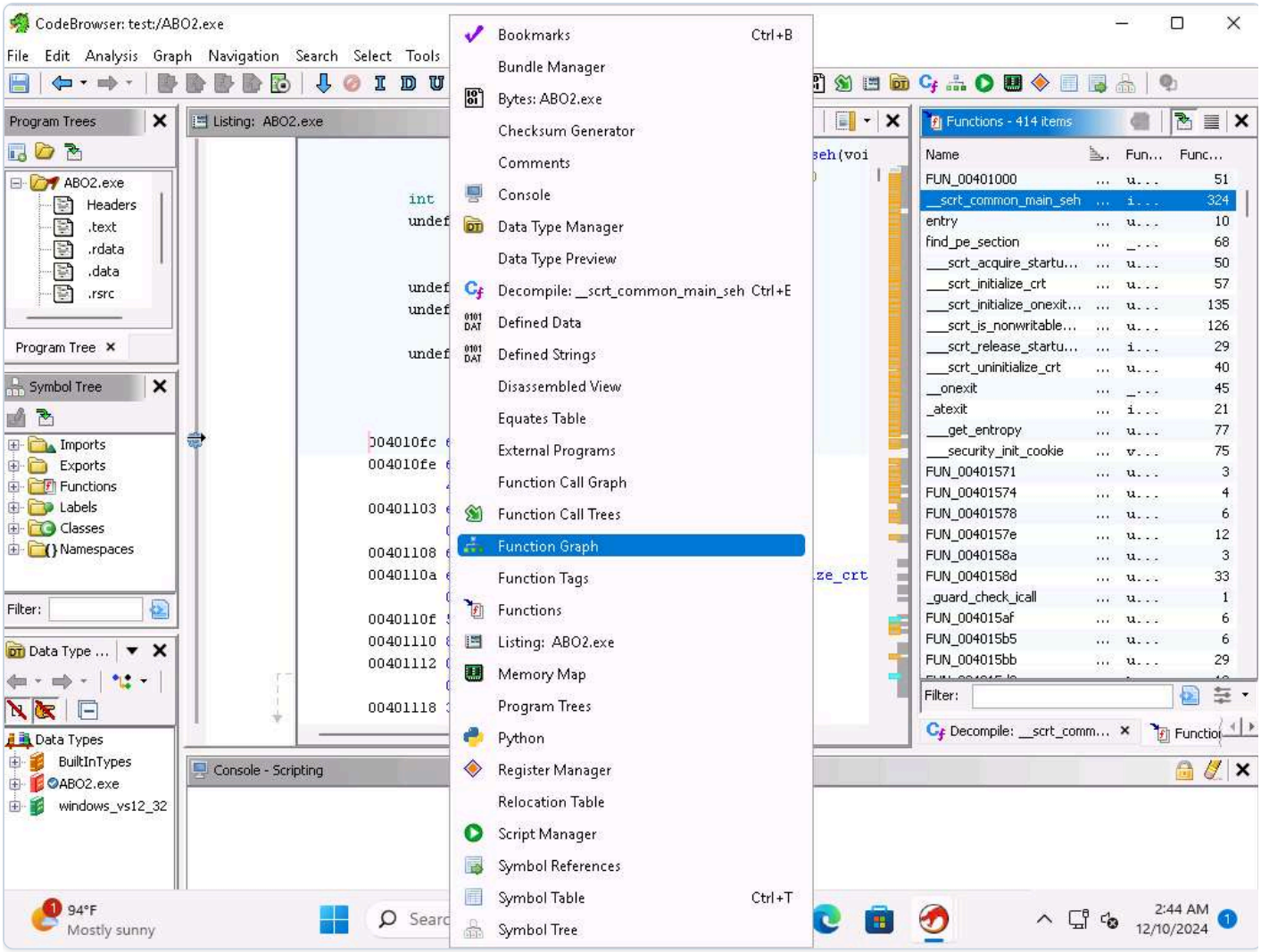

1. Click **Window -> Functions** from menu bar.



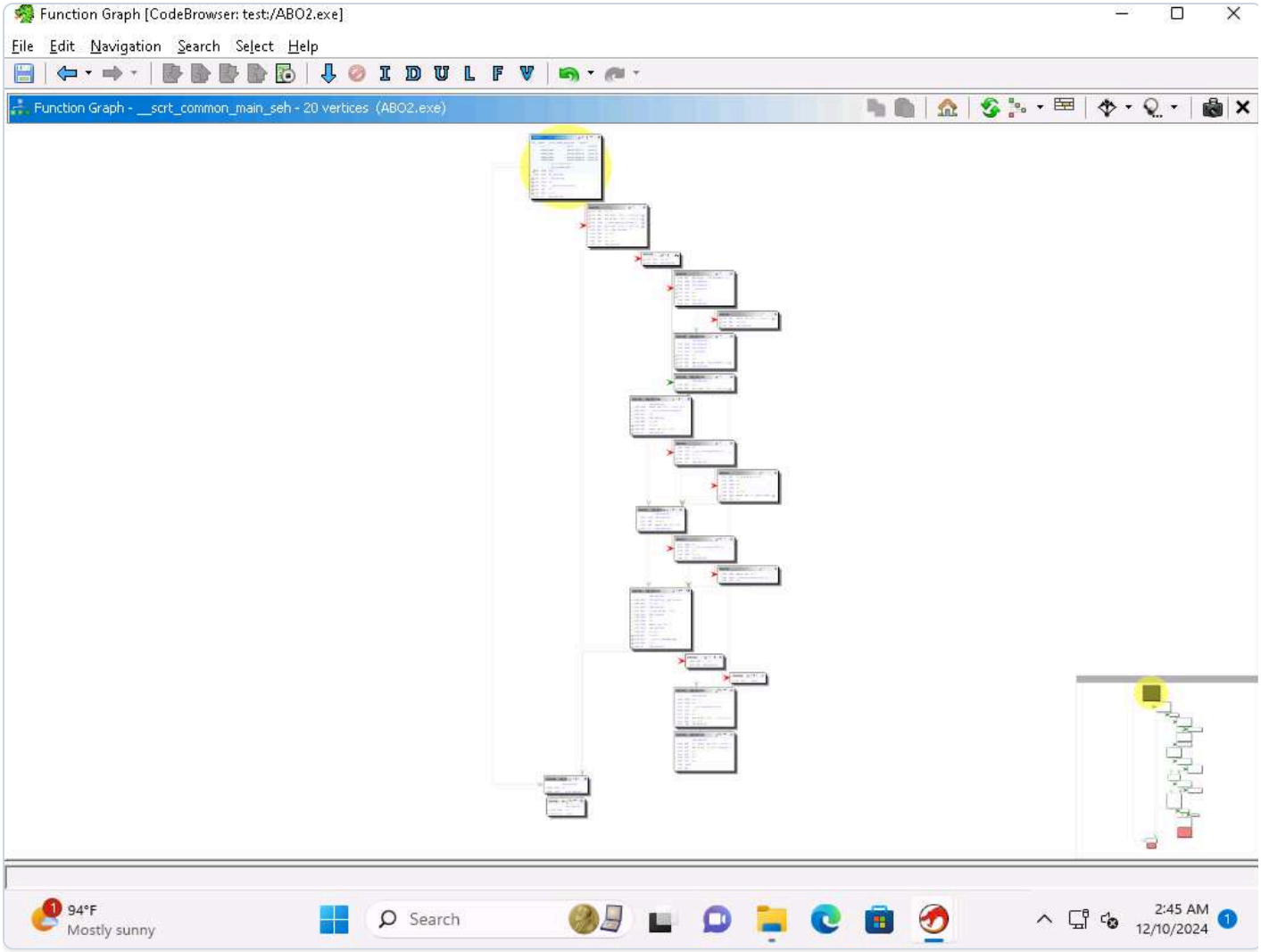

1. Double-click `_scrt_common_main_seh` in Functions pane.



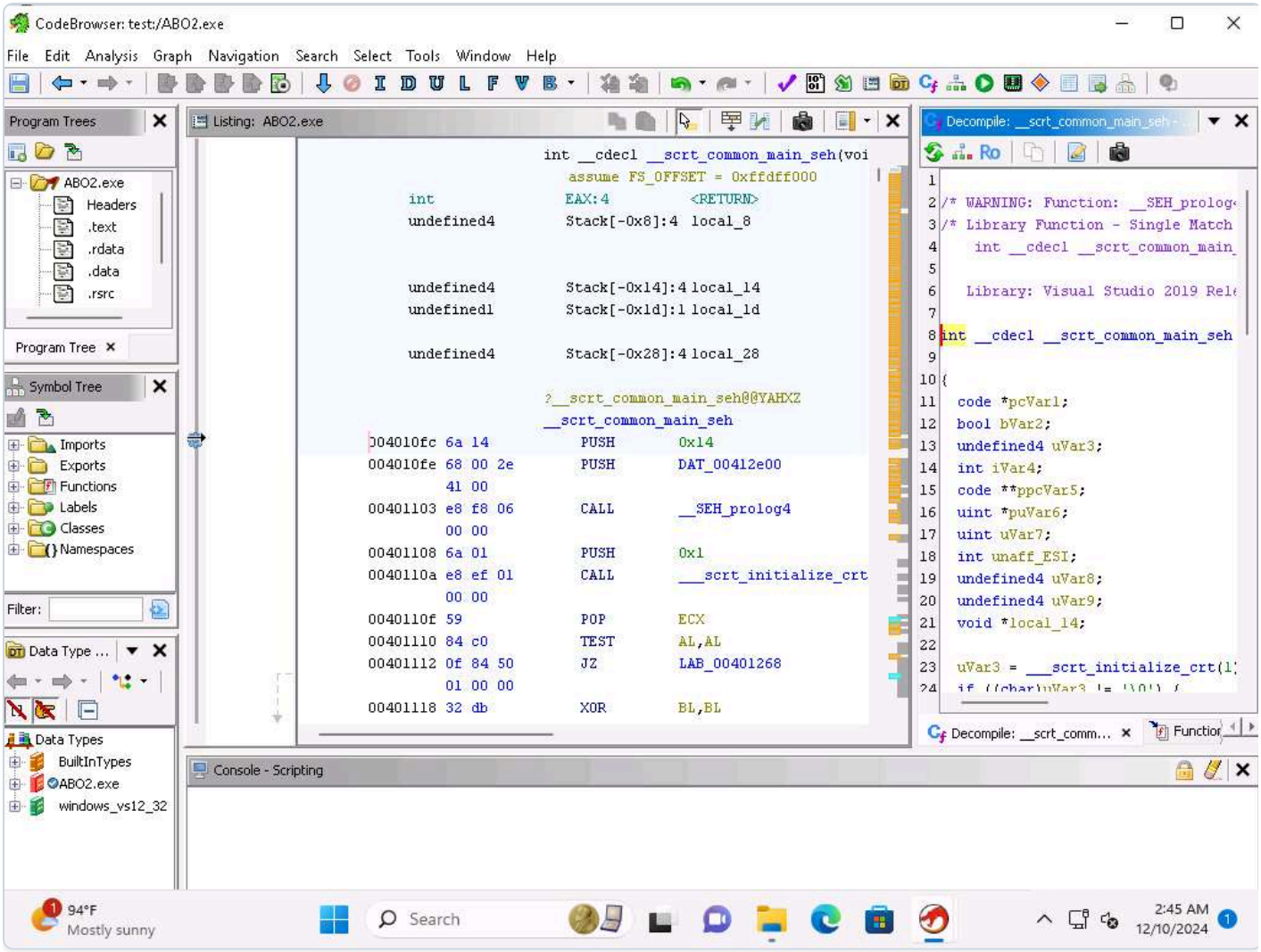

1. Click **Window -> Function Graph**.




1. Observe graphical representation of basic blocks and execution branches.



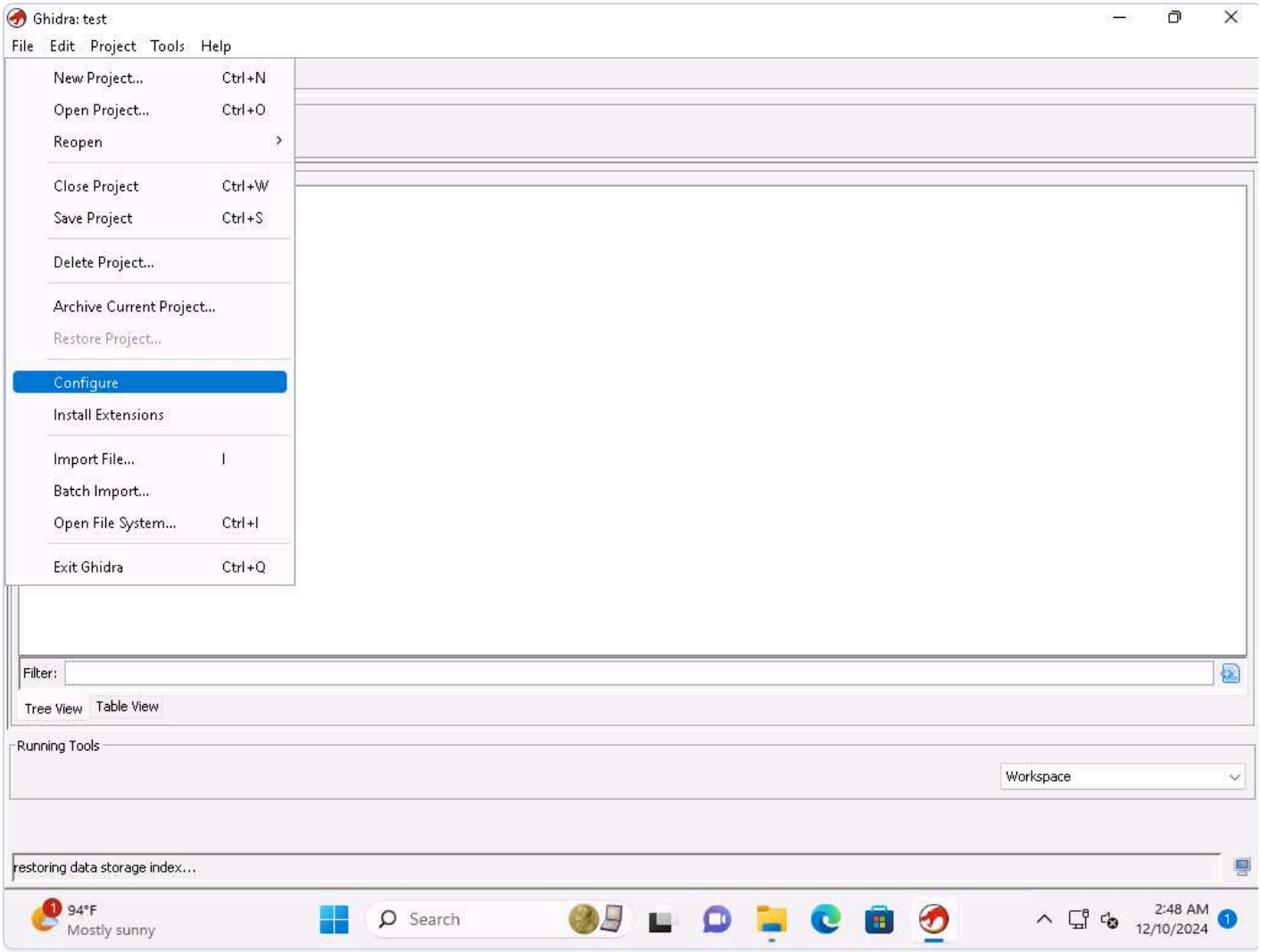

1. Press **Ctrl+E** to decompile assembly into C pseudo-code.



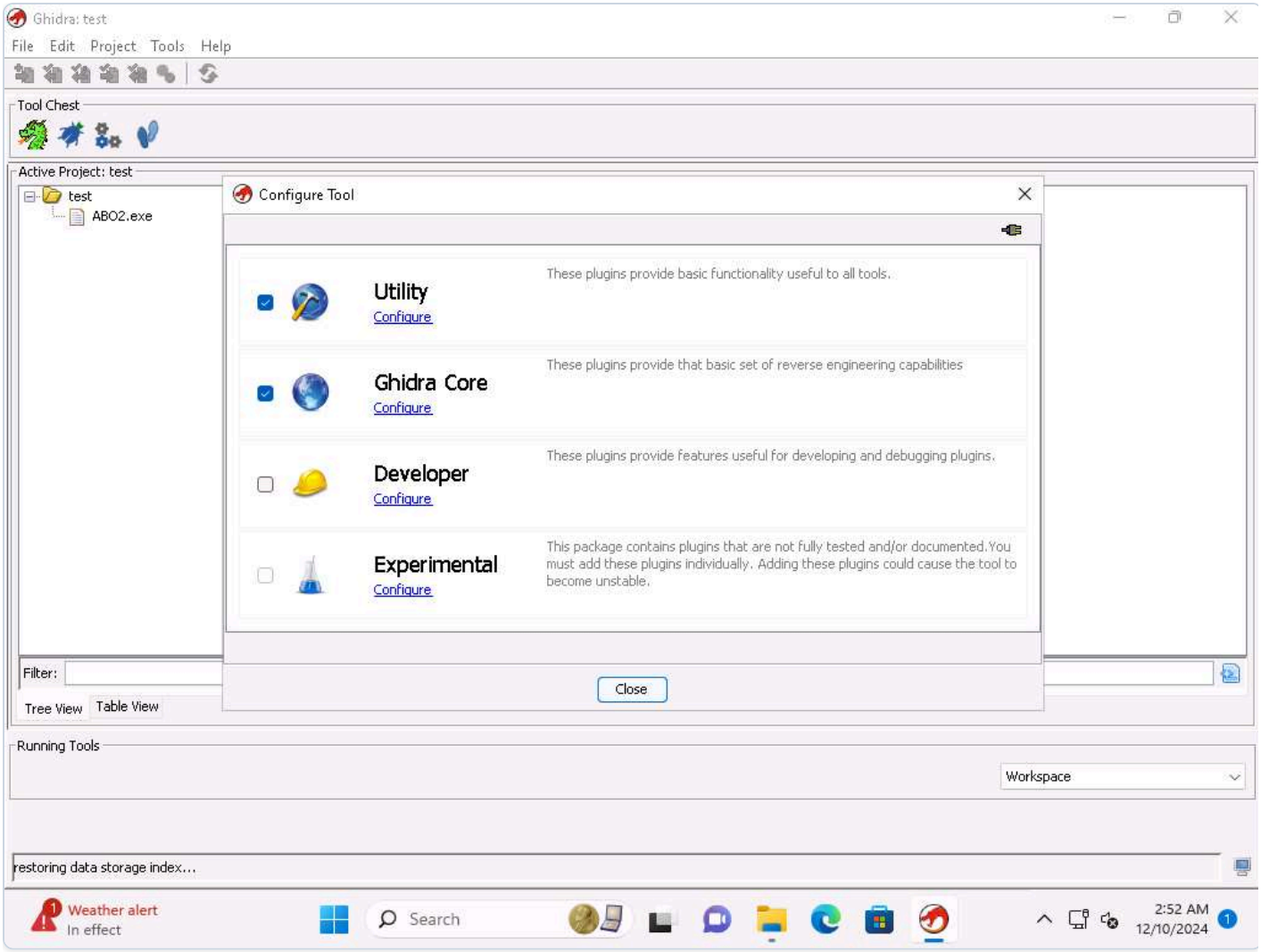

## Task 2: Configure Ghidra Plugins, Processor Modules, and Keybindings

In this task, we review Ghidra Core plugins, view installed processor architectures (ARM, x86, MIPS), check logs, and inspect keybinding configurations.

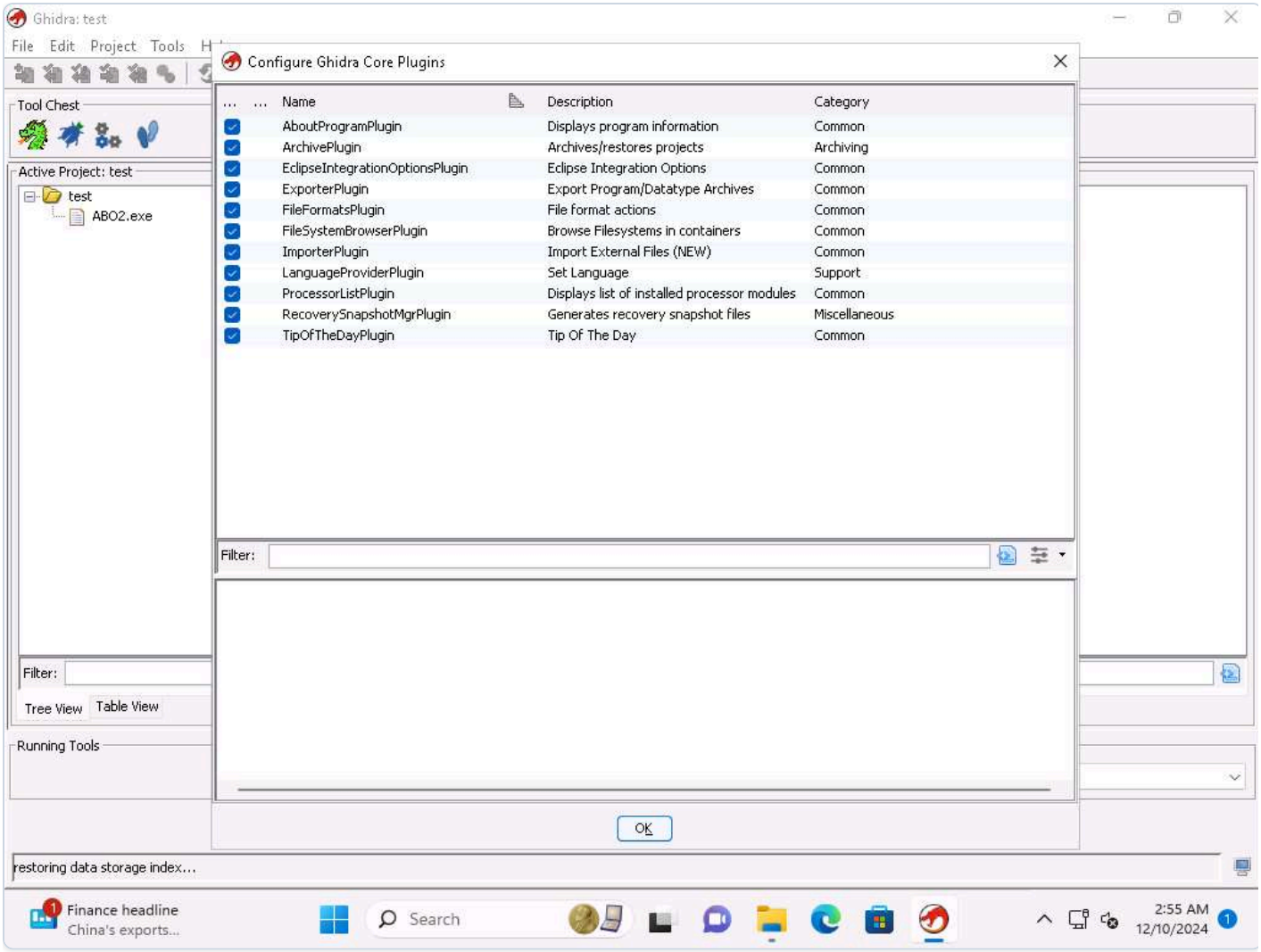
1. Click **File -> Configure** from main menu.



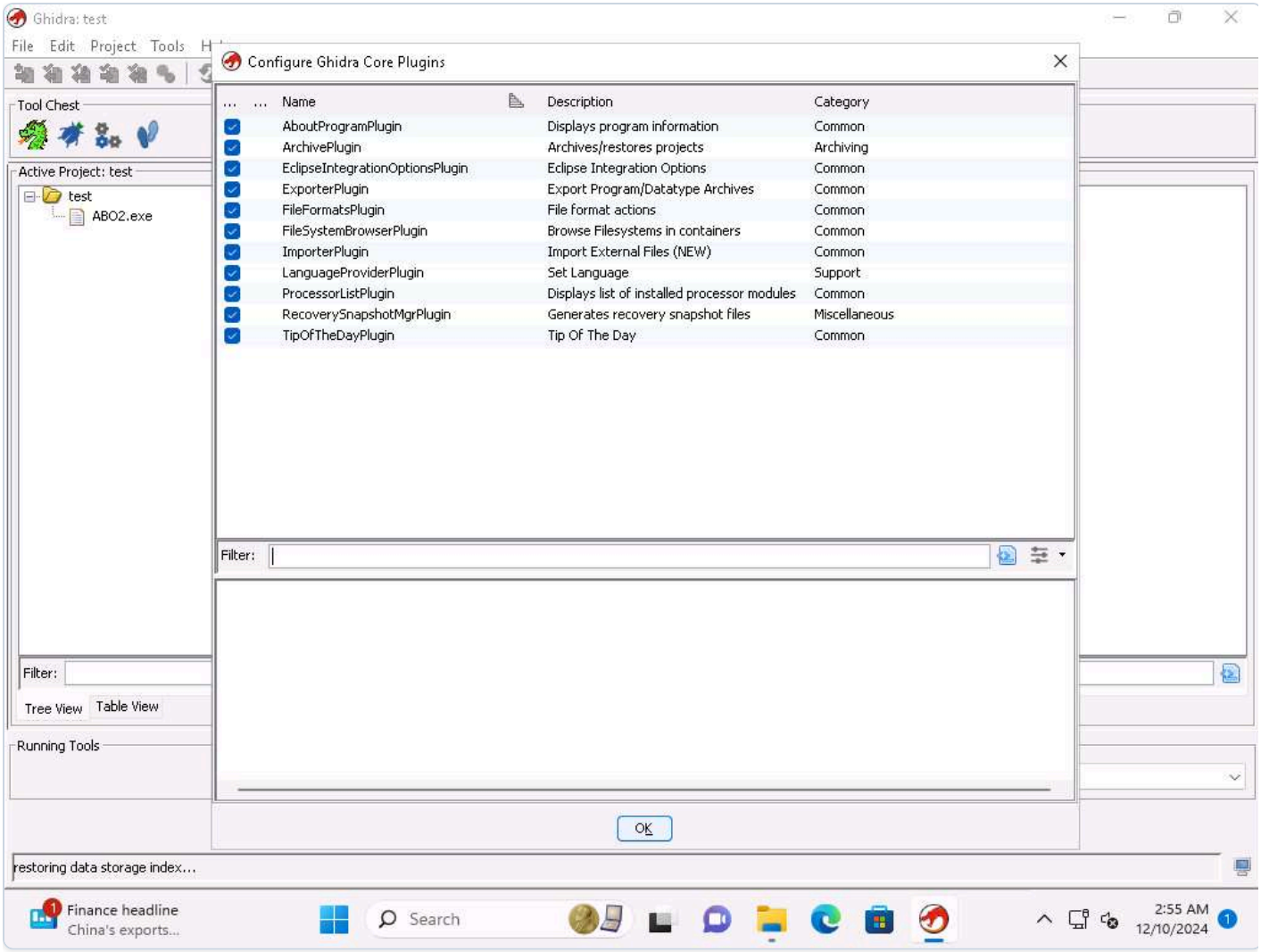

1. Ensure **Ghidra Core** checkbox is enabled.



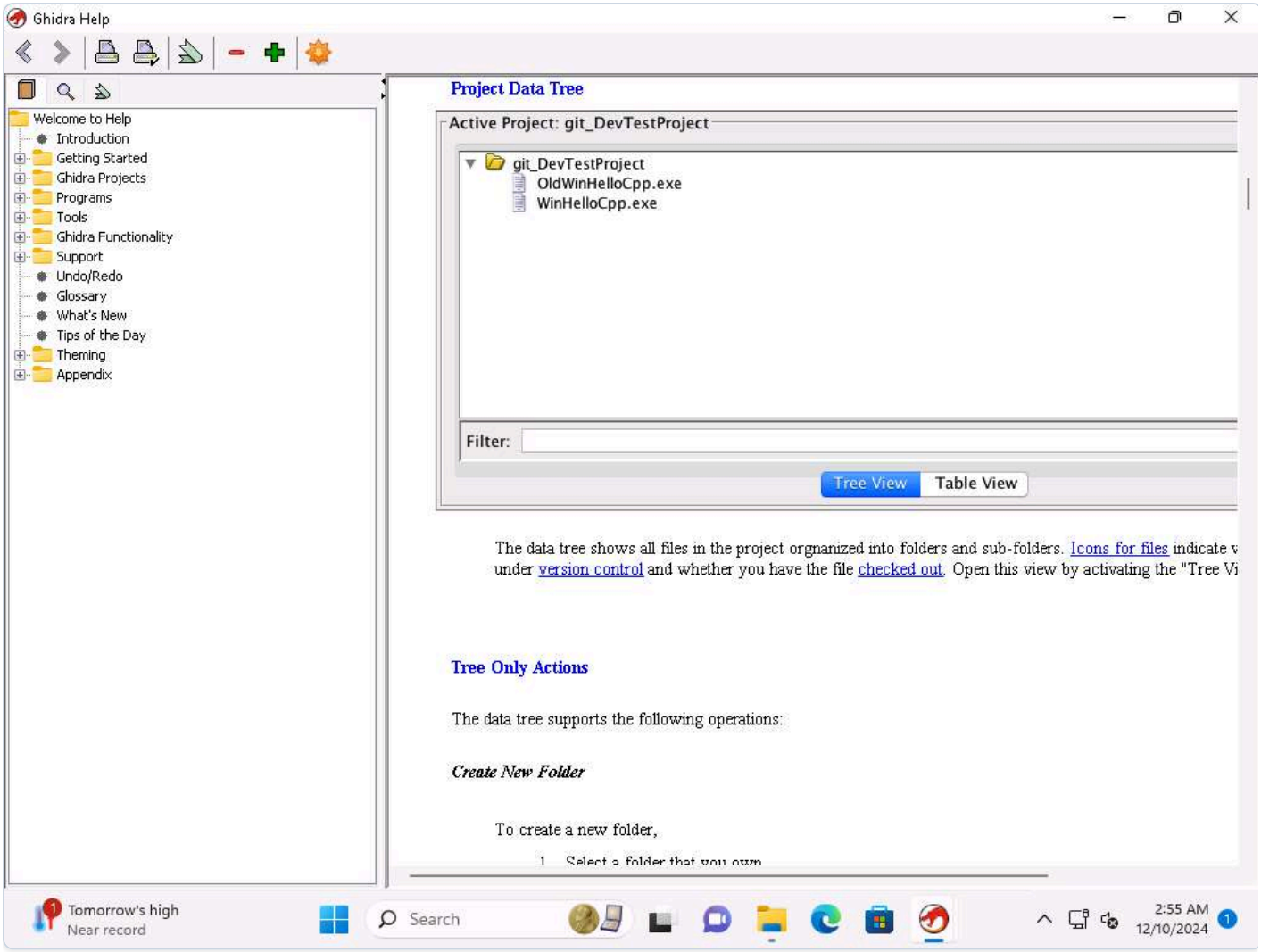

1. Click **Configure** under Ghidra Core to review active plugin list. Click **OK** and **Close**.

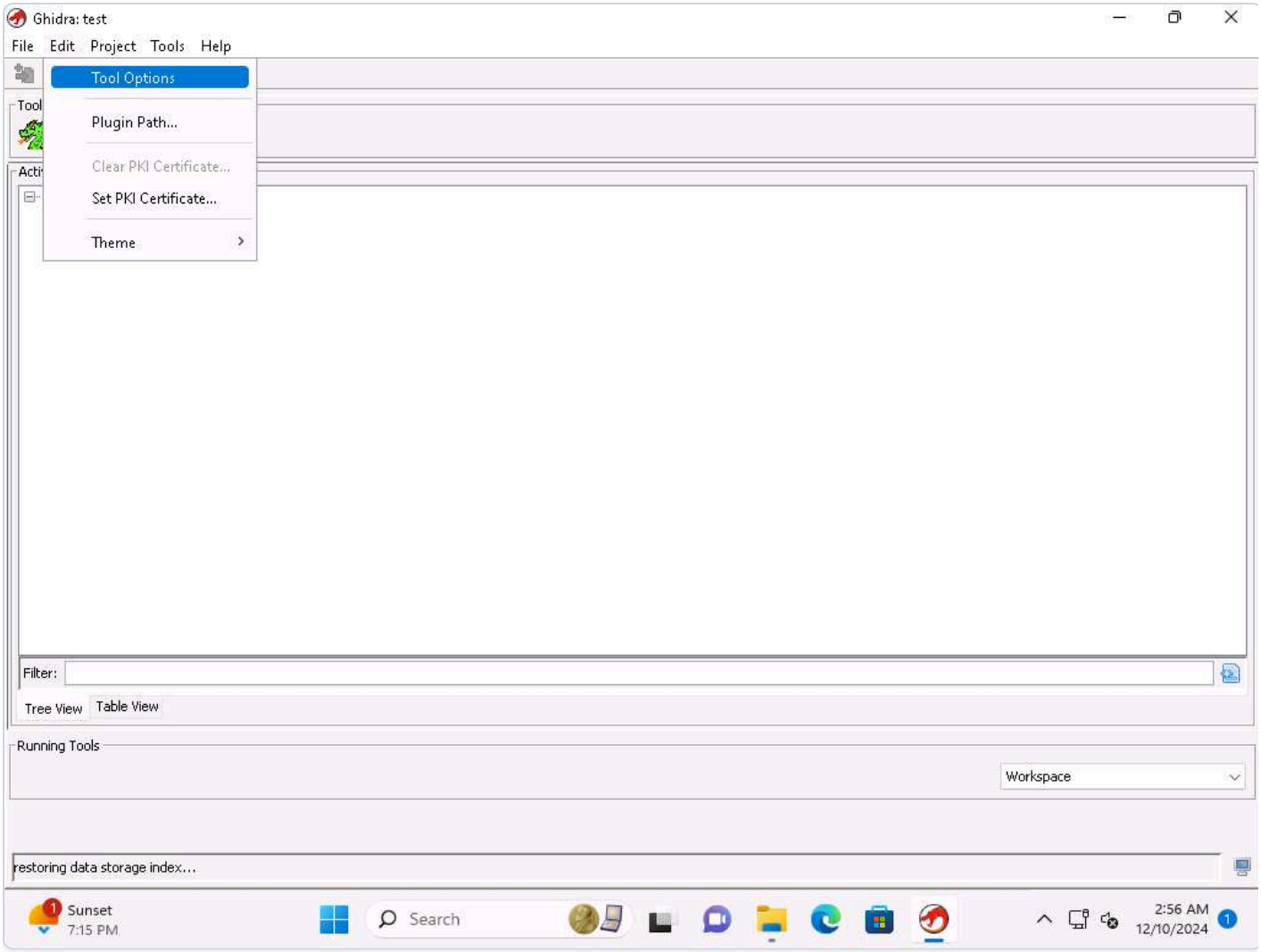



1. Press **F1** to open Ghidra Help documentation.

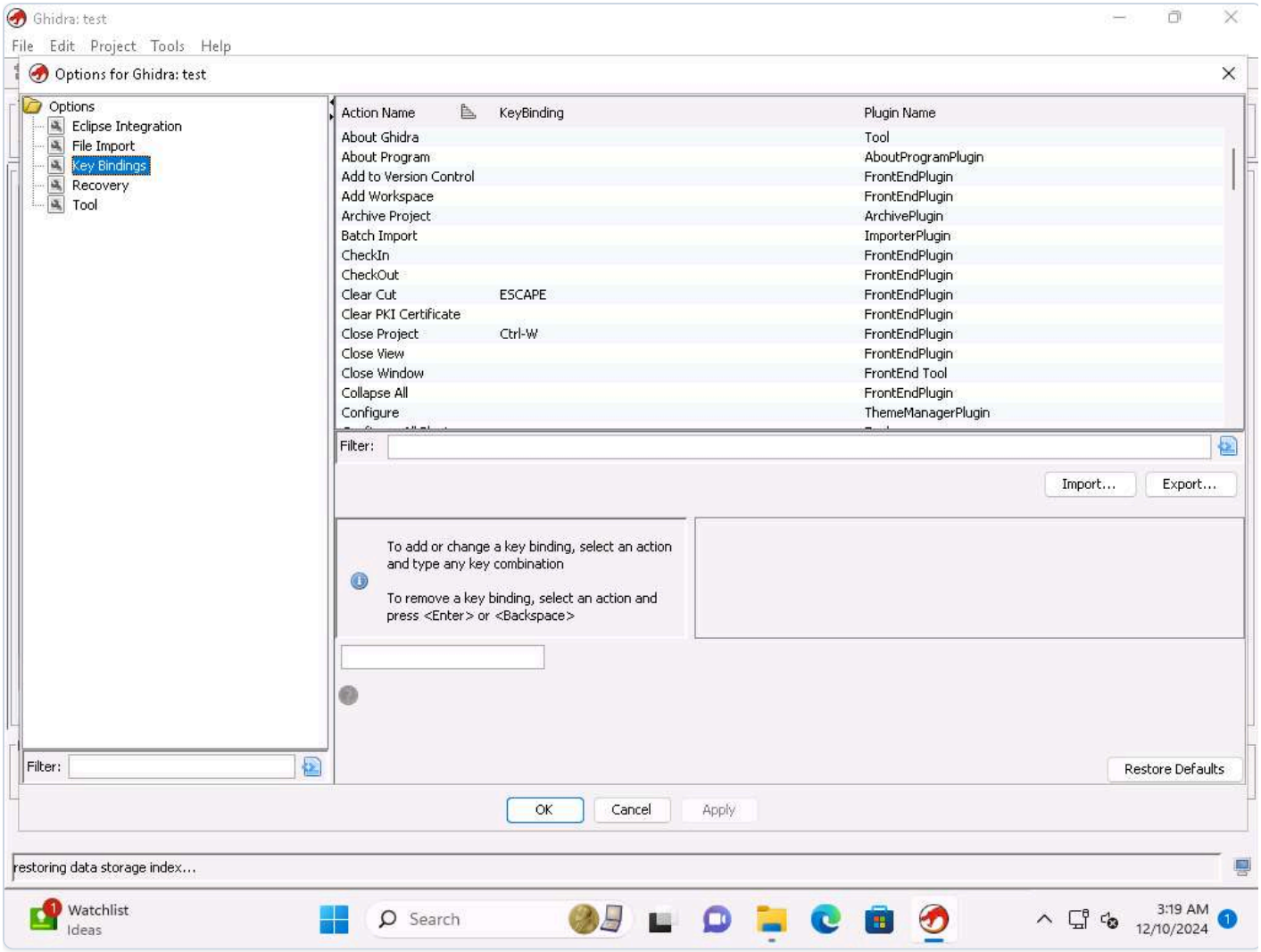


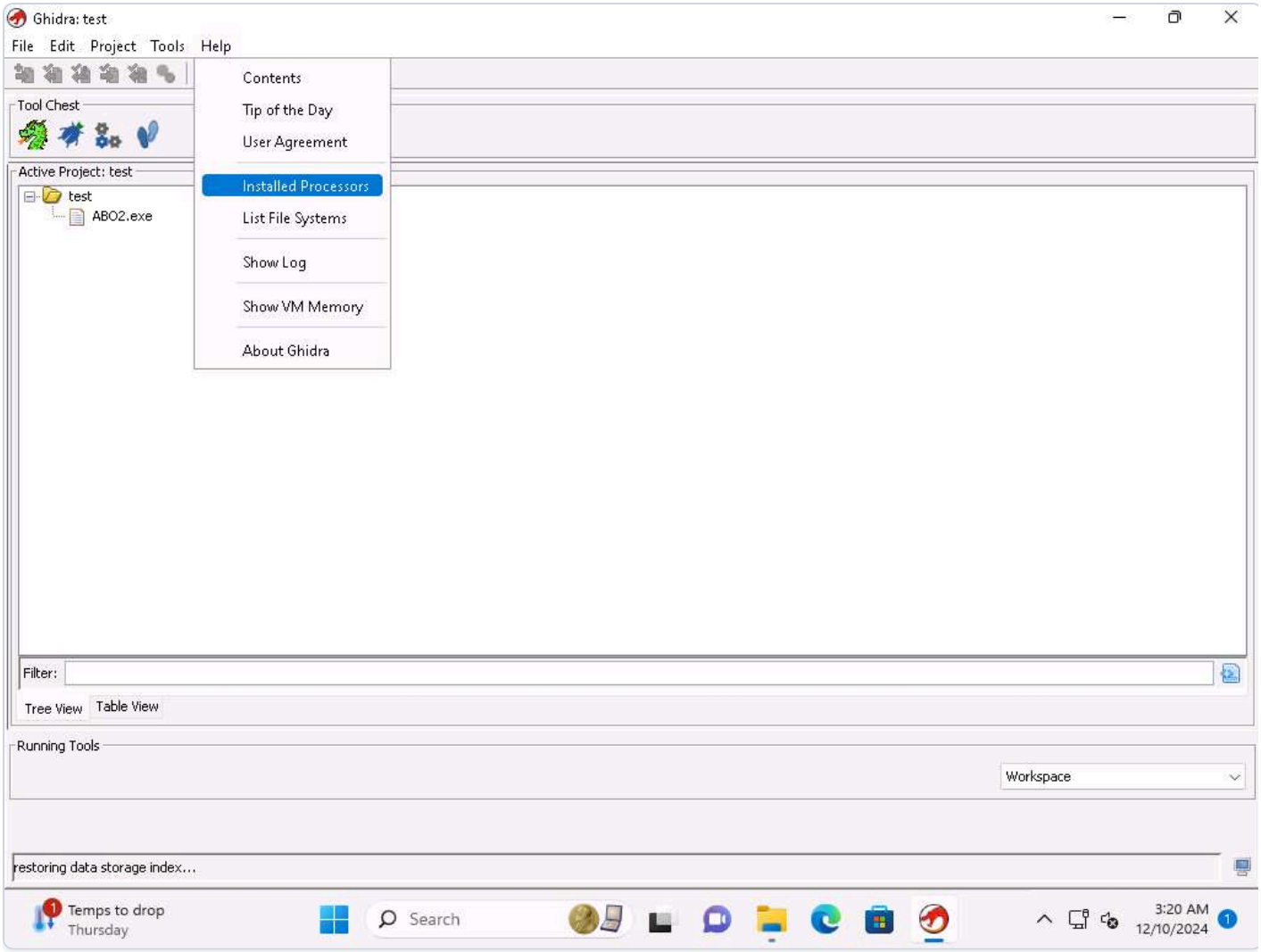

1. Click **Edit -> Tool Options**, select **Key Bindings**, and review hotkey assignments.


Ghidra Help
Welcome to Help
Introduction
Getting Started
Ghidra Projects
Programs
Tools
Ghidra Functionality
Support
Undo/Redo
Glossary
What's New
Tips of the Day
Theming
Appendix
Project Data Tree
Active Project: git_DevTestProject
git_DevTestProject
OldWinHelloCpp.exe
WinHelloCpp.exe
Filter:
Tree View
Table View
The data tree shows all files in the project orgnanized into folders and sub-folders. Icons for files indicate v
under version control and whether you have the file checked out. Open this view by activating the "Tree Vi
Tree Only Actions
The data tree supports the following operations:
Create New Folder
To create a new folder,
1. Select a folder that you own
Tomorrow's high
Near record
Search
2:55 AM
12/10/2024

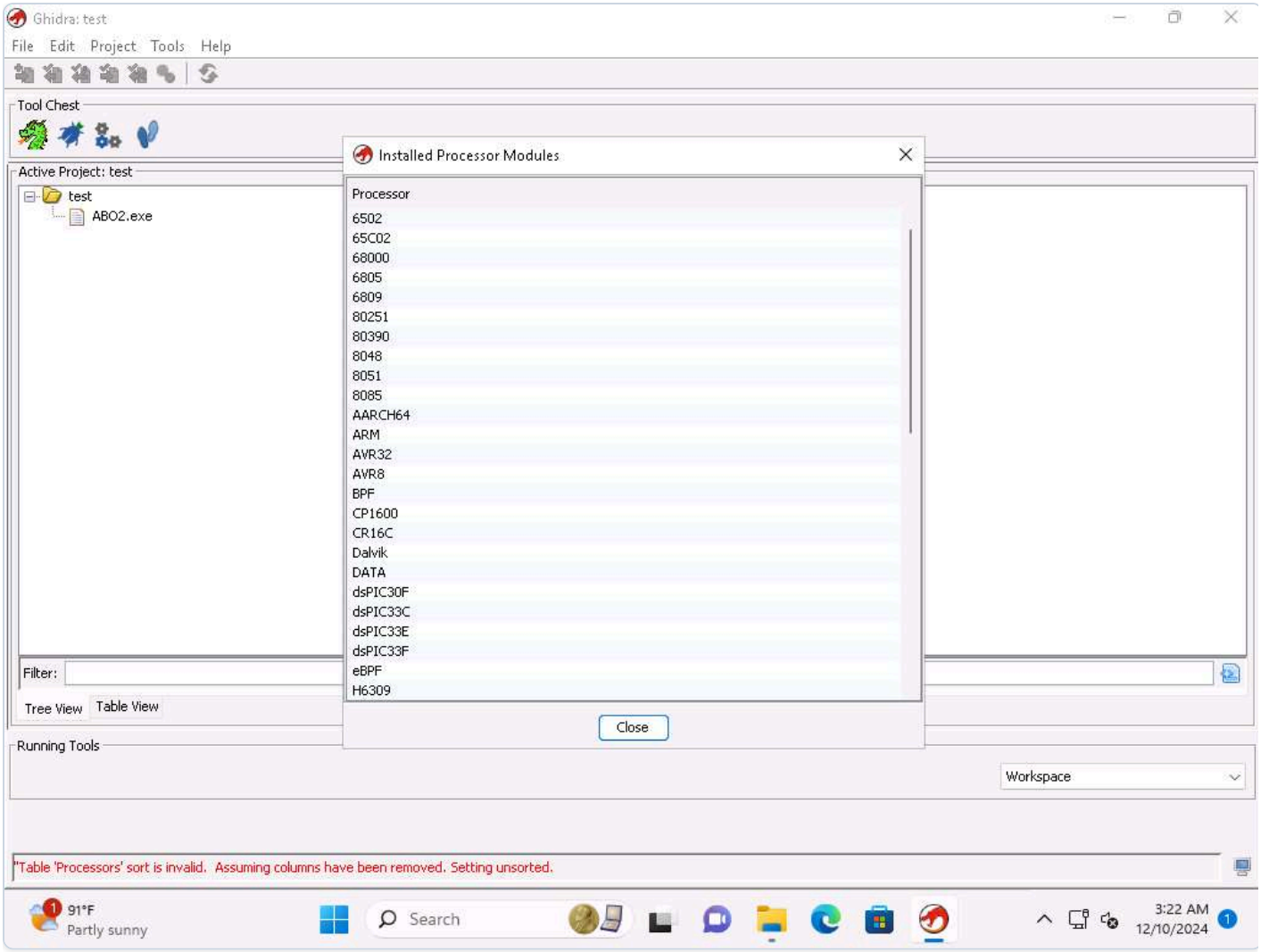



1. Click **Help -> Installed Processors** to review architecture modules.


Ghidra: test
File Edit Project Tools Help
Options for Ghidra: test
Options
Eclipse Integration
File Import
Key Bindings
Recovery
Tool
Action Name
KeyBinding
Plugin Name
About Ghidra
Tool
About Program
AboutProgramPlugin
Add to Version Control
FrontEndPlugin
Add Workspace
FrontEndPlugin
Archive Project
ArchivePlugin
Batch Import
ImporterPlugin
CheckIn
FrontEndPlugin
CheckOut
FrontEndPlugin
Clear Cut
ESCAPE
FrontEndPlugin
Clear PKI Certificate
FrontEndPlugin
Close Project
Ctrl-W
FrontEndPlugin
Close View
FrontEndPlugin
Close Window
FrontEnd Tool
Collapse All
FrontEndPlugin
Configure
ThemeManagerPlugin
Filter:
Import...
Export...
To add or change a key binding, select an action and type any key combination
To remove a key binding, select an action and press <Enter> or <Backspace>
Filter:
Restore Defaults
OK
Cancel
Apply
restoring data storage index...
Watchlist
Ideas
Search
3:19 AM
12/10/2024



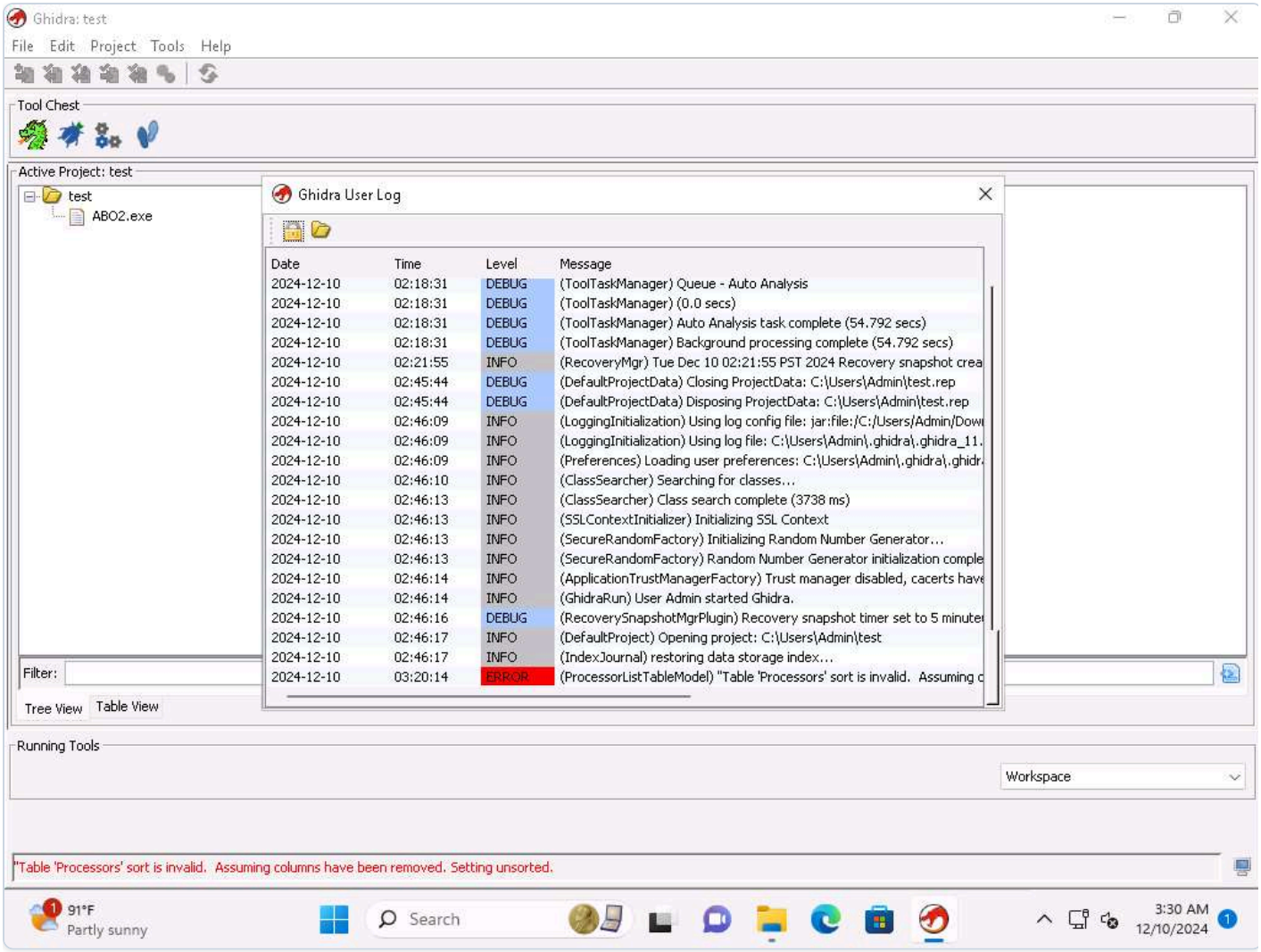

1. Click **Help -> Show Log** to inspect system logs.



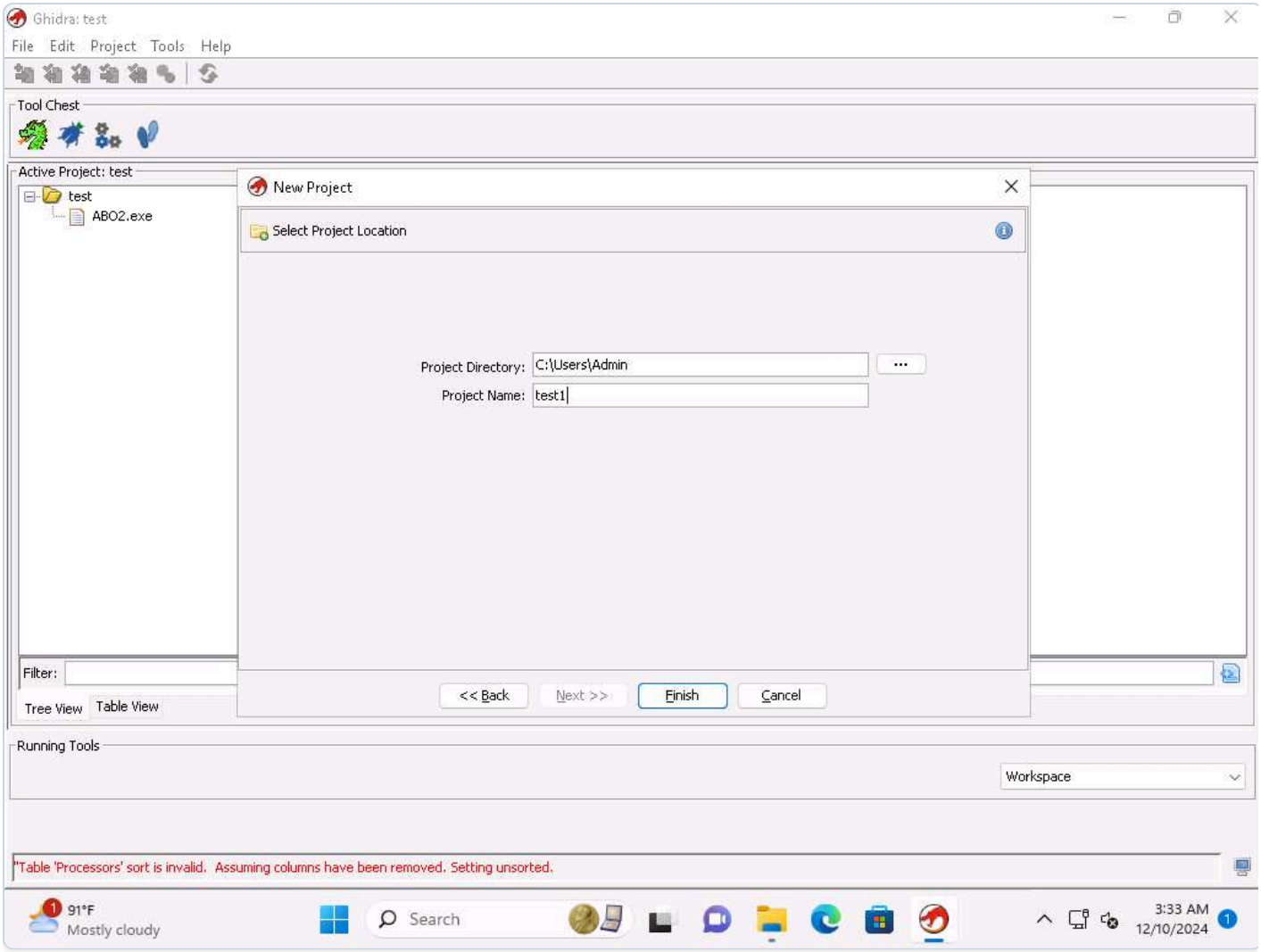

## Task 3: Reverse Engineer Crackme0x00.exe and Extract Hardcoded Password

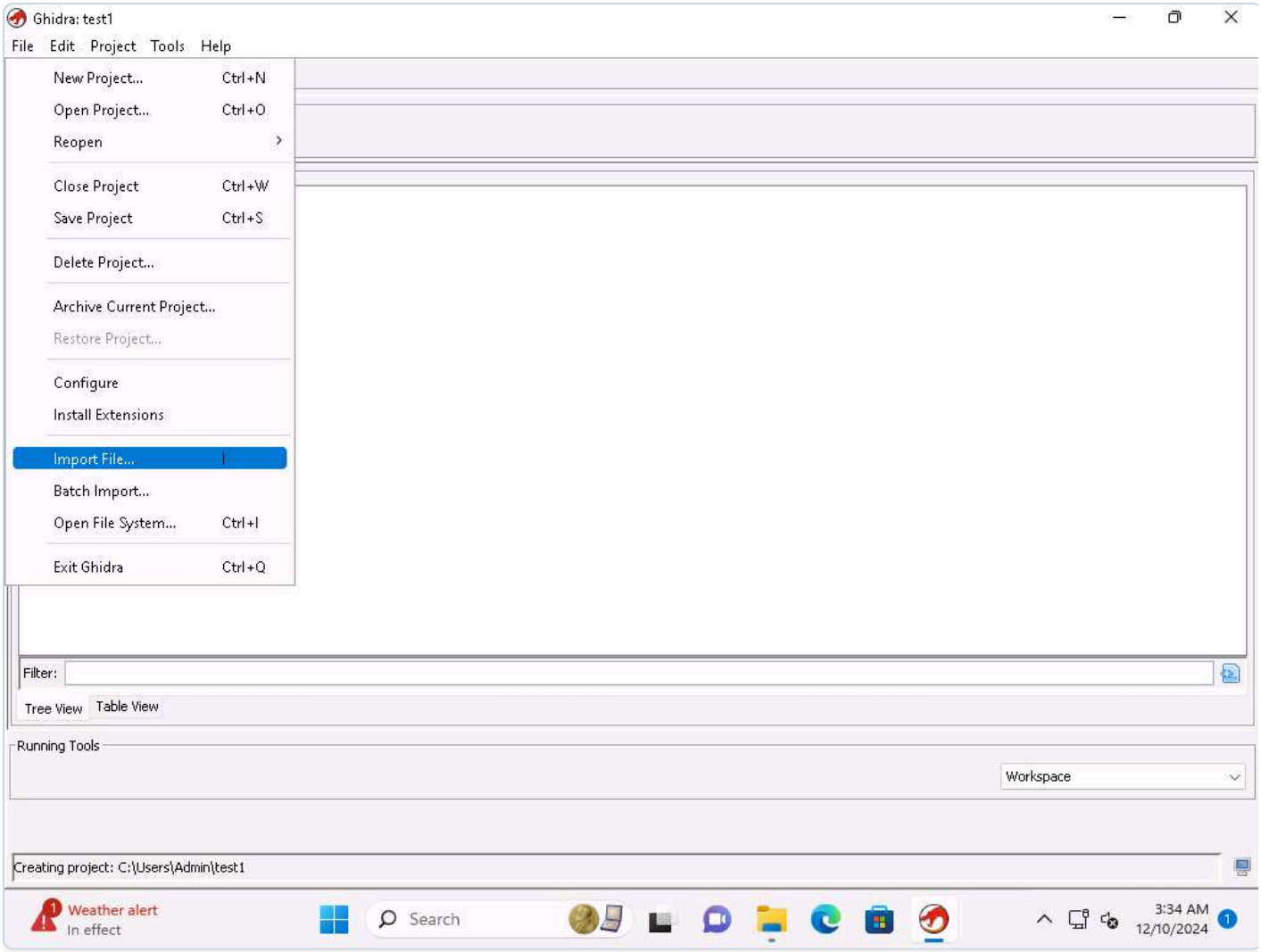
In this task, we create project `test1`, import `Crackme0x00.exe`, analyze stack frame setup (`PUSH EBP`, `MOV EBP, ESP`, `SUB ESP, 0x38`), inspect Defined Strings, set Equates, disassemble `main`, trace `strcmp` logic, recover password (**250382**), and verify execution in CMD.

1. Create project `test1` via **File -> New Project**.

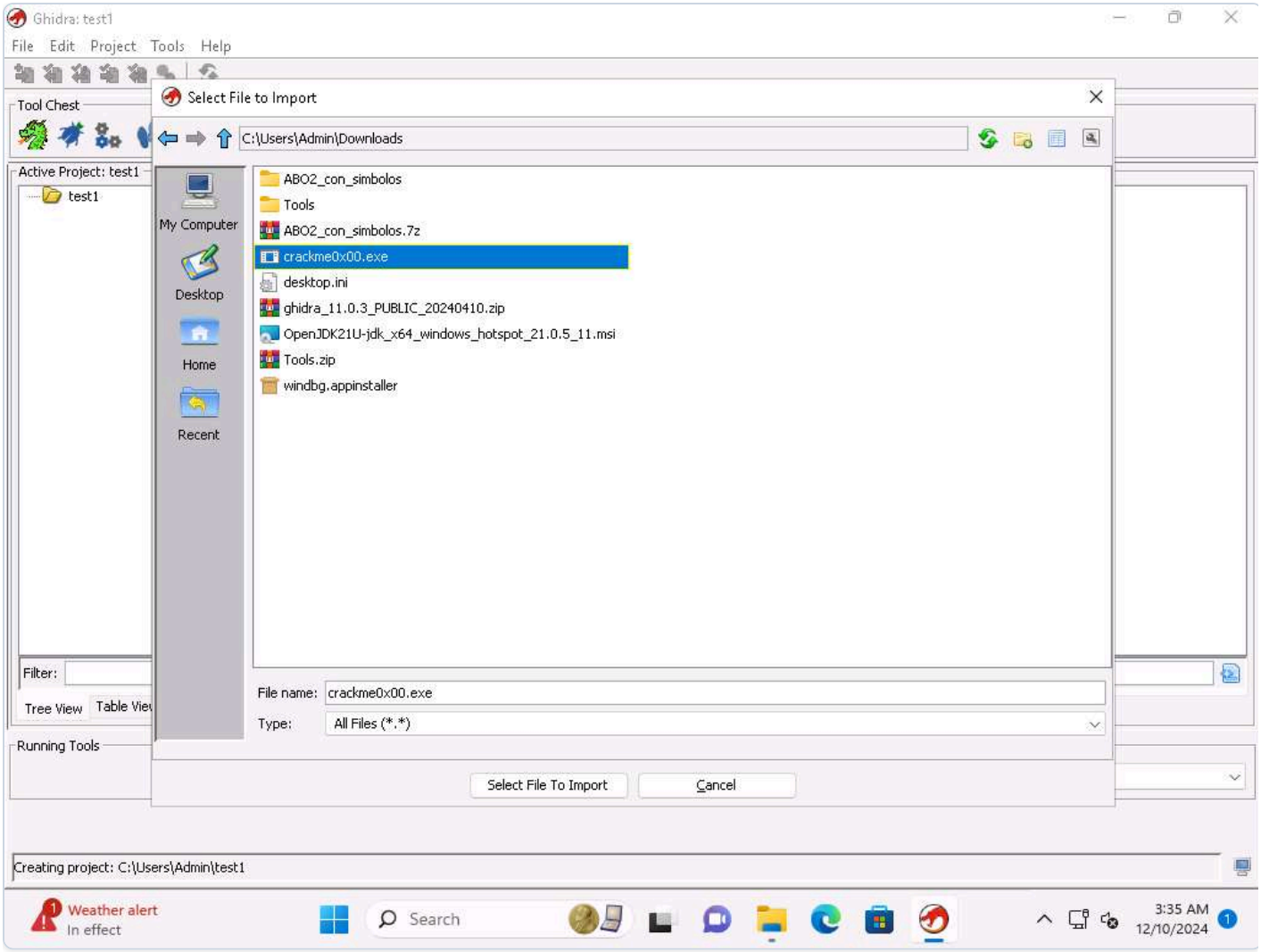



1. Import **Crackme0x00.exe** from `C:\Users\Admin\Downloads`.

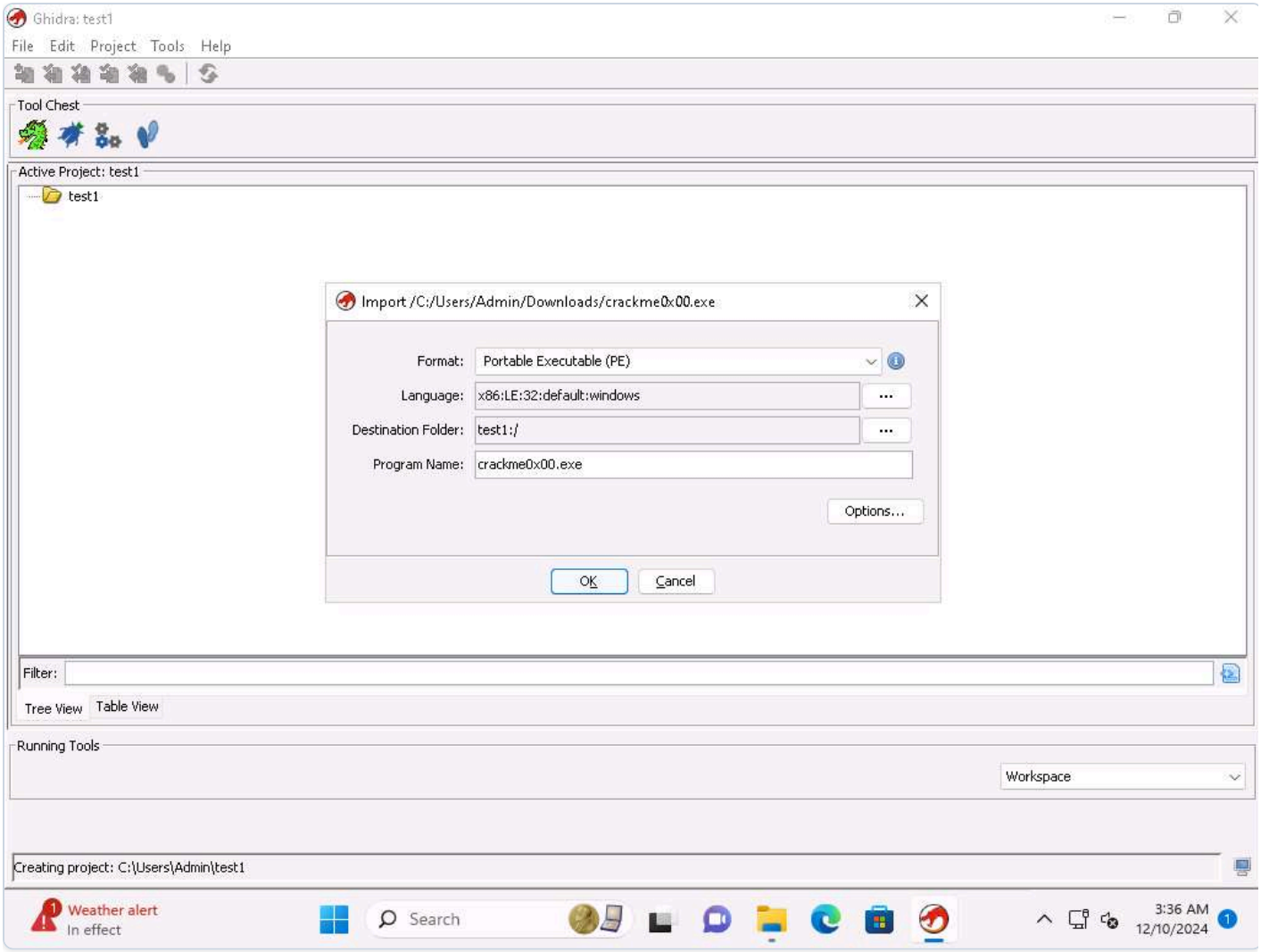

Ghidra: test
File Edit Project Tools Help
Tool Chest
Active Project: test
test
ABO2.exe
New Project
Select Project Location
Project Directory: C:\Users\Admin
Project Name: test1
<< Back
Next >>
Finish
Cancel
Filter:
Tree View
Table View
Running Tools
Workspace
'Table 'Processors' sort is invalid. Assuming columns have been removed. Setting unsorted.
91°F
Mostly cloudy
Search
3:33 AM
12/10/2024



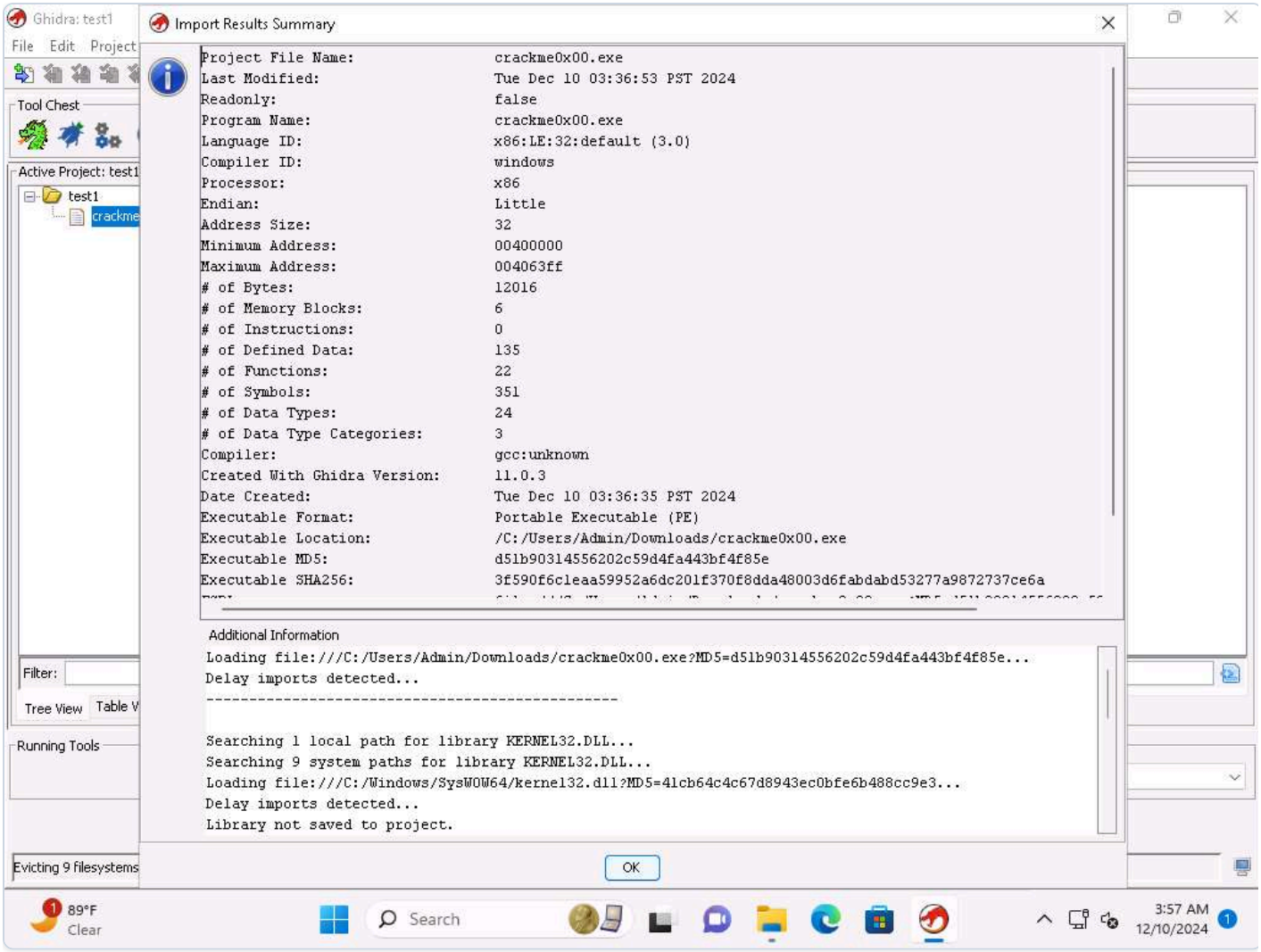

1. Review Portable Executable format metadata and click **OK**.

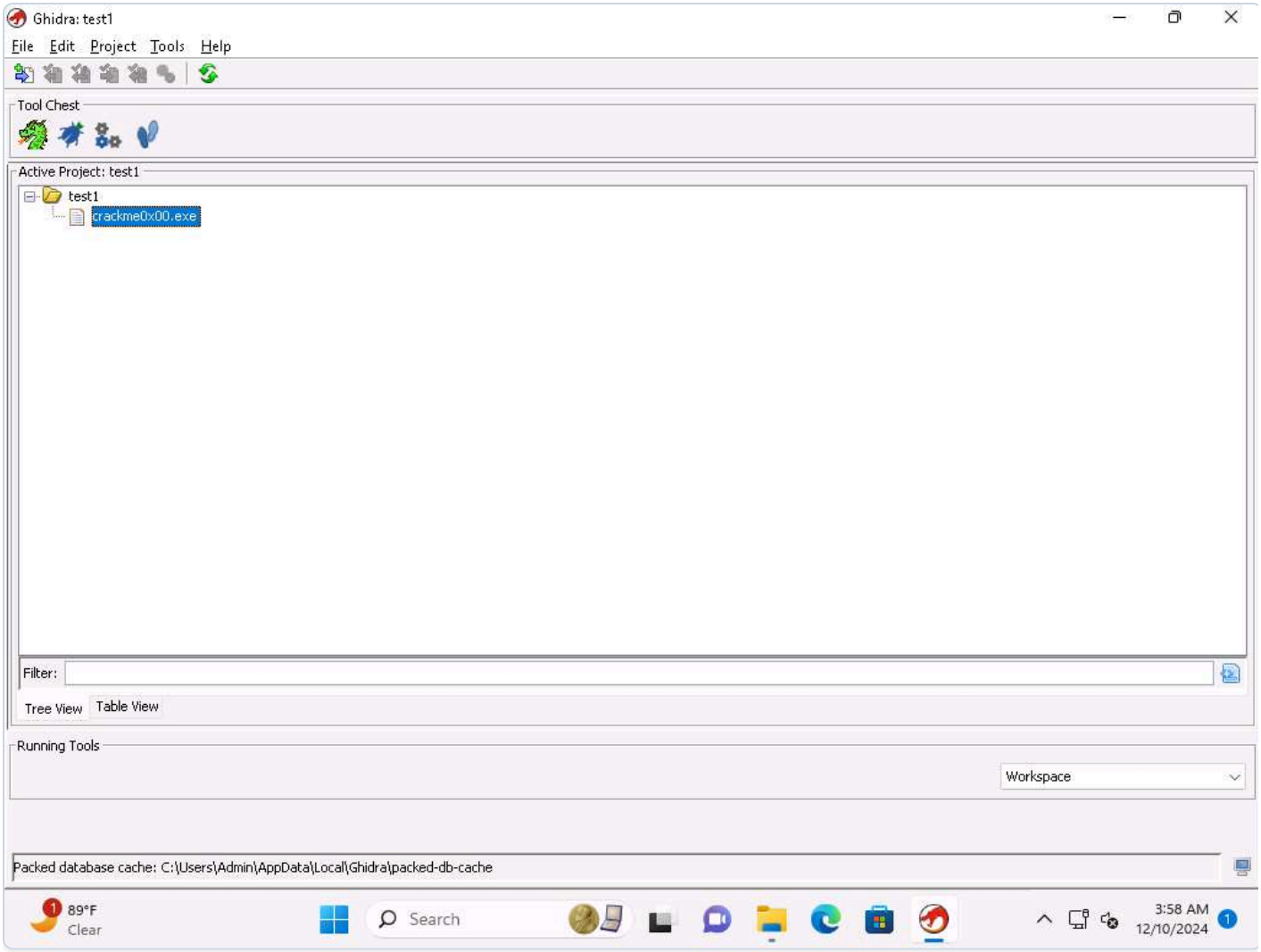


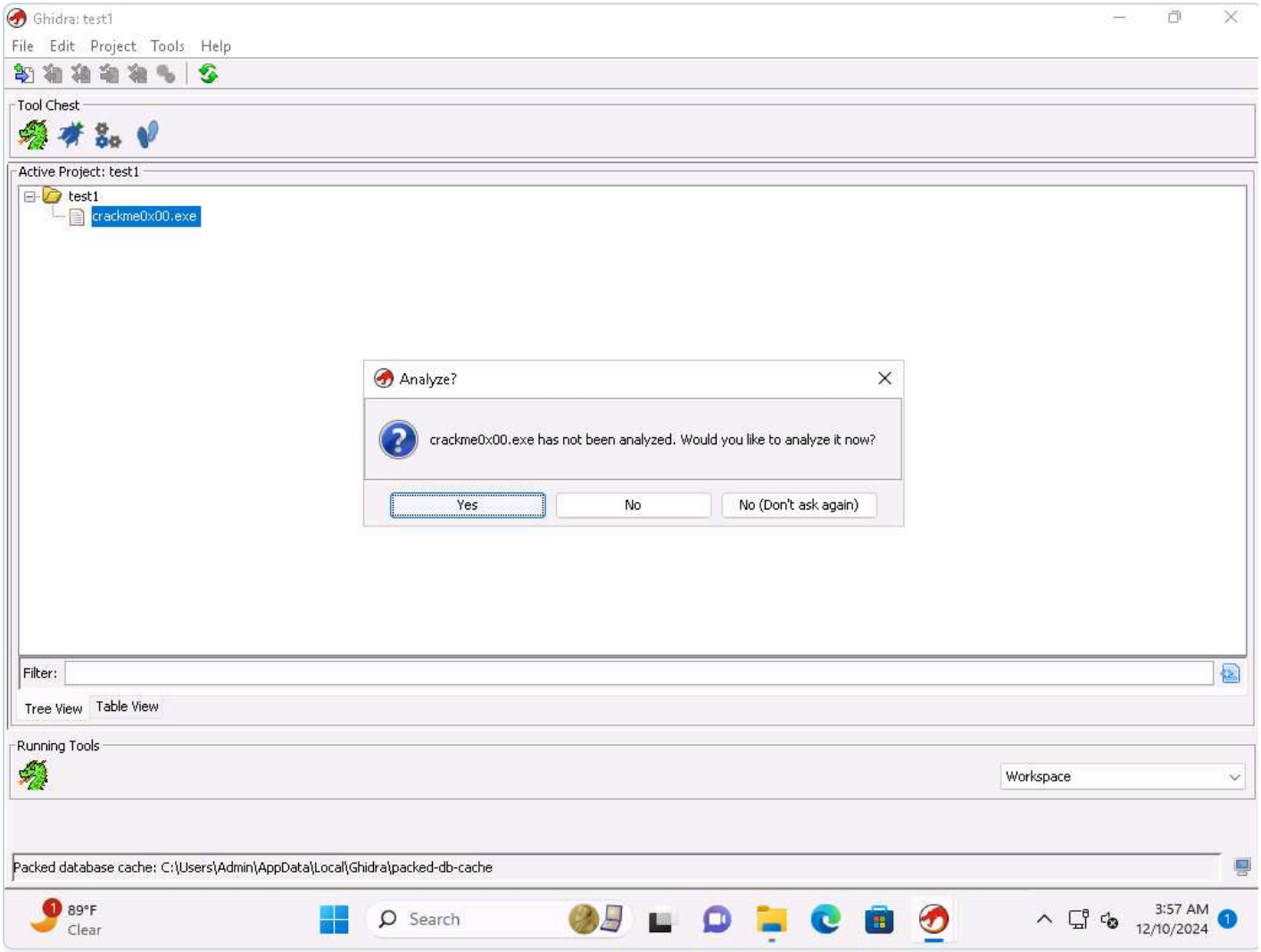

1. Confirm Windows Firewall alert and import results.



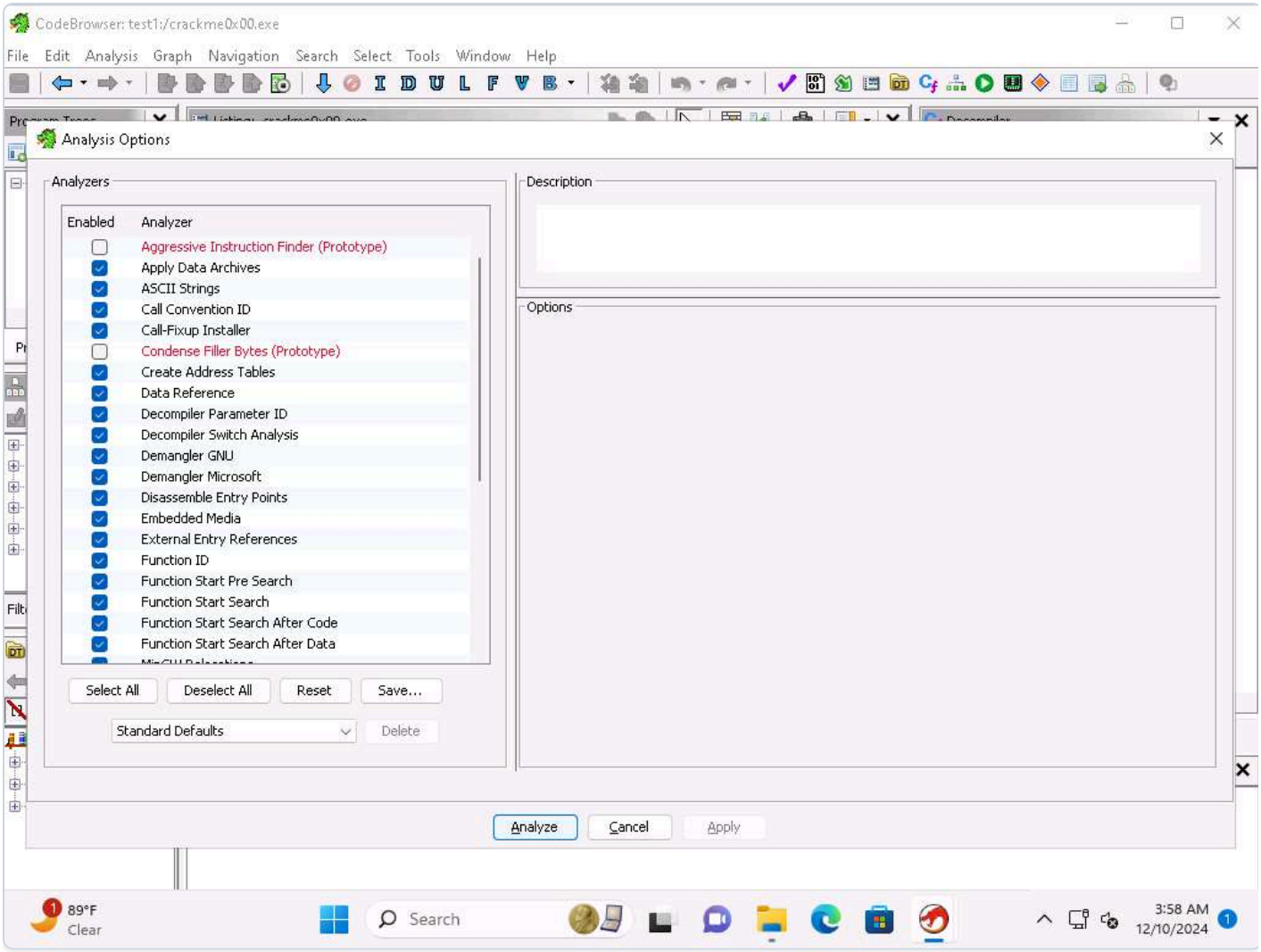

1. Double-click `Crackme0x00.exe`, click **Yes**, and run **Analyze** with `WindowsPE x86 Propagate External Parameters` checked.


Ghidra: test1
File Edit Project
Tool Chest
Active Project: test1
test1
crackme
Filter:
Tree View
Running Tools
Evicting 9 filesystems
Import Results Summary
Project File Name: crackme0x00.exe
Last Modified: Tue Dec 10 03:36:53 PST 2024
Readonly: false
Program Name: crackme0x00.exe
Language ID: x86:LE:32:default (3.0)
Compiler ID: windows
Processor: x86
Endian: Little
Address Size: 32
Minimum Address: 00400000
Maximum Address: 004063ff
# of Bytes: 12016
# of Memory Blocks: 6
# of Instructions: 0
# of Defined Data: 135
# of Functions: 22
# of Symbols: 351
# of Data Types: 24
# of Data Type Categories: 3
Compiler: gcc:unknown
Created With Ghidra Version: 11.0.3
Date Created: Tue Dec 10 03:36:35 PST 2024
Executable Format: Portable Executable (PE)
Executable Location: /C:/Users/Admin/Downloads/crackme0x00.exe
Executable MD5: d51b90314556202c59d4fa443bf4f85e
Executable SHA256: 3f590f6cleaa59952a6dc201f370f8dda48003d6fabdabd53277a9872737ce6a
Additional Information
Loading file:///C:/Users/Admin/Downloads/crackme0x00.exe?MD5=d51b90314556202c59d4fa443bf4f85e...
Delay imports detected...
Searching 1 local path for library KERNEL32.DLL...
Searching 9 system paths for library KERNEL32.DLL...
Loading file:///C:/Windows/SysWOW64/kernel32.dll?MD5=41cb64c4c67d8943ec0bfe6b488cc9e3...
Delay imports detected...
Library not saved to project.
OK
89°F
Clear
Search
3:57 AM
12/10/2024

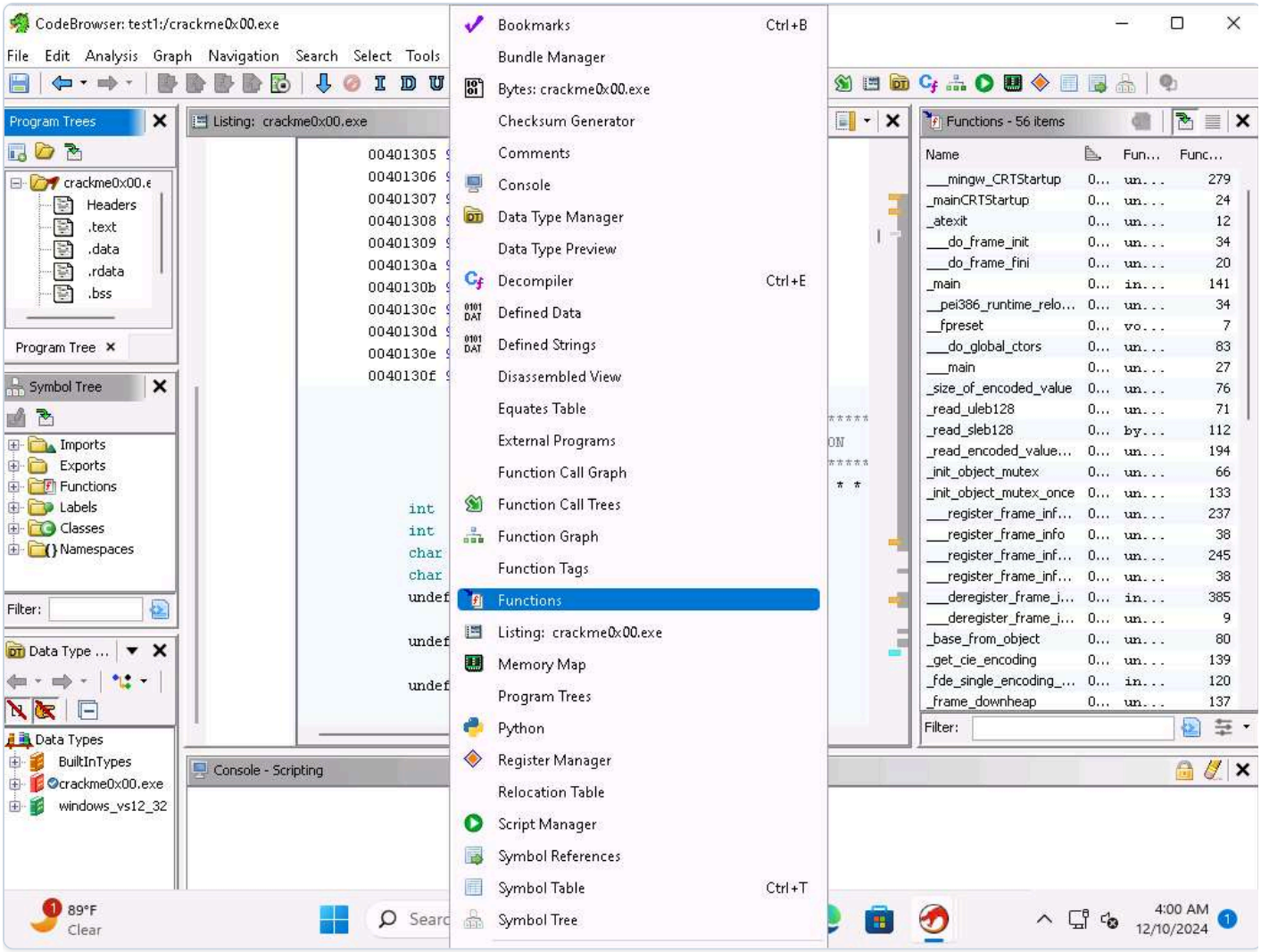

Ghidra: test1
File Edit Project Tools Help
Tool Chest
Active Project: test1
test1
crackme0x00.exe
Filter:
Tree View
Table View
Running Tools
Workspace
Packed database cache: C:\Users\Admin\AppData\Local\Ghidra\packed-db-cache
89°F
Clear
Search
3:58 AM
12/10/2024



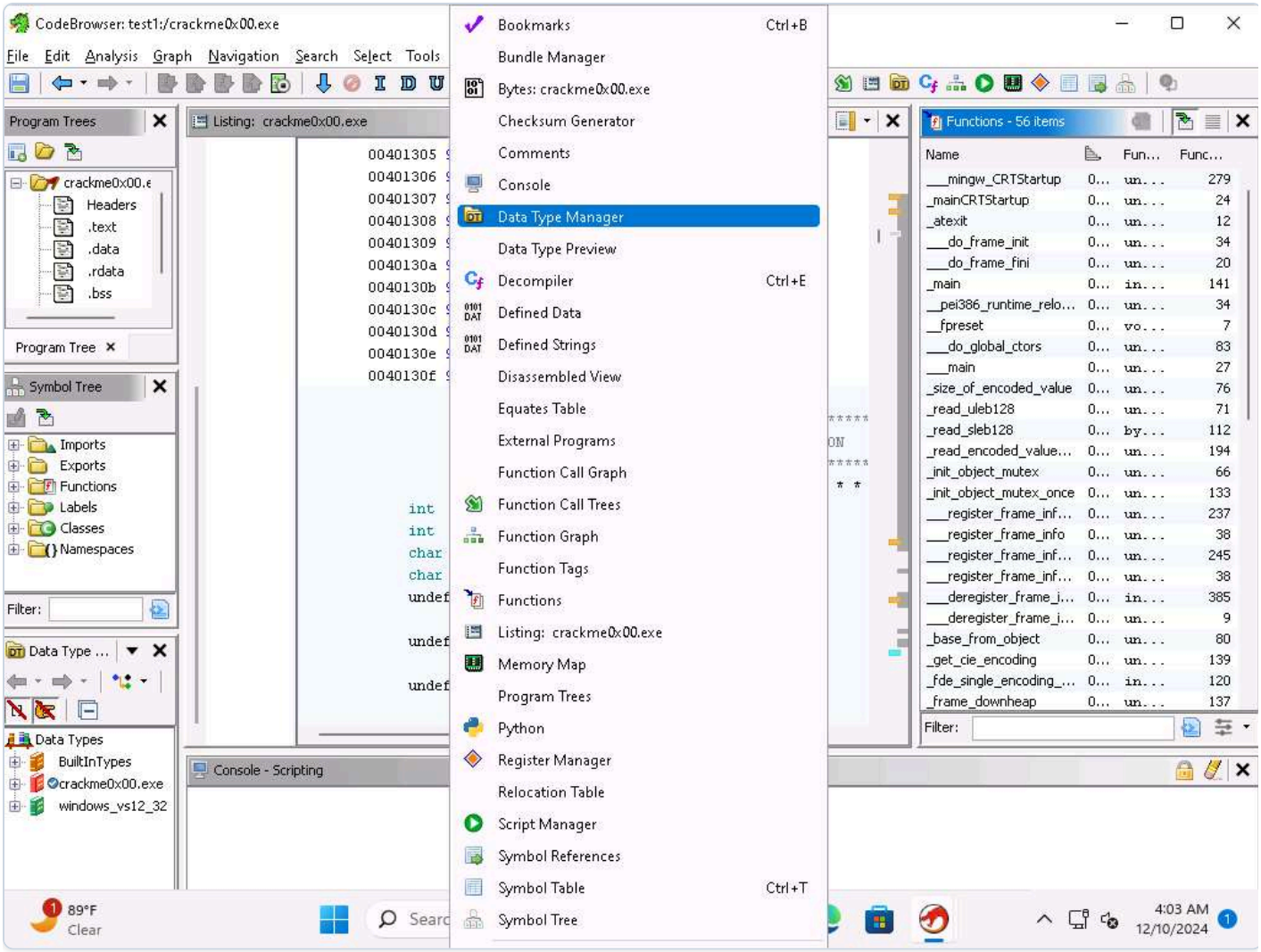

1. Click **Window -> Functions** to view binary imports.

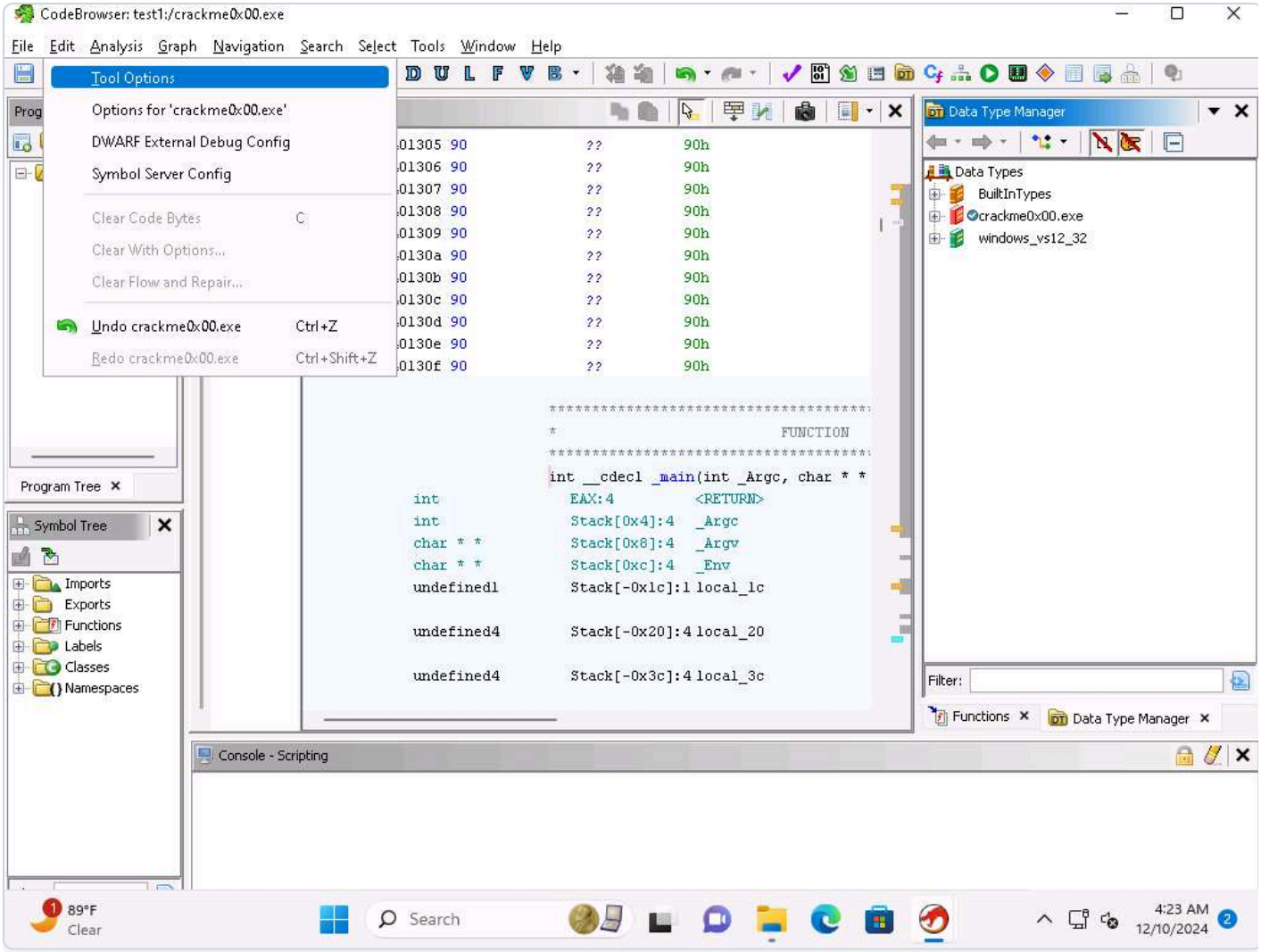



1. Click **Window -> Data Type Manager**.

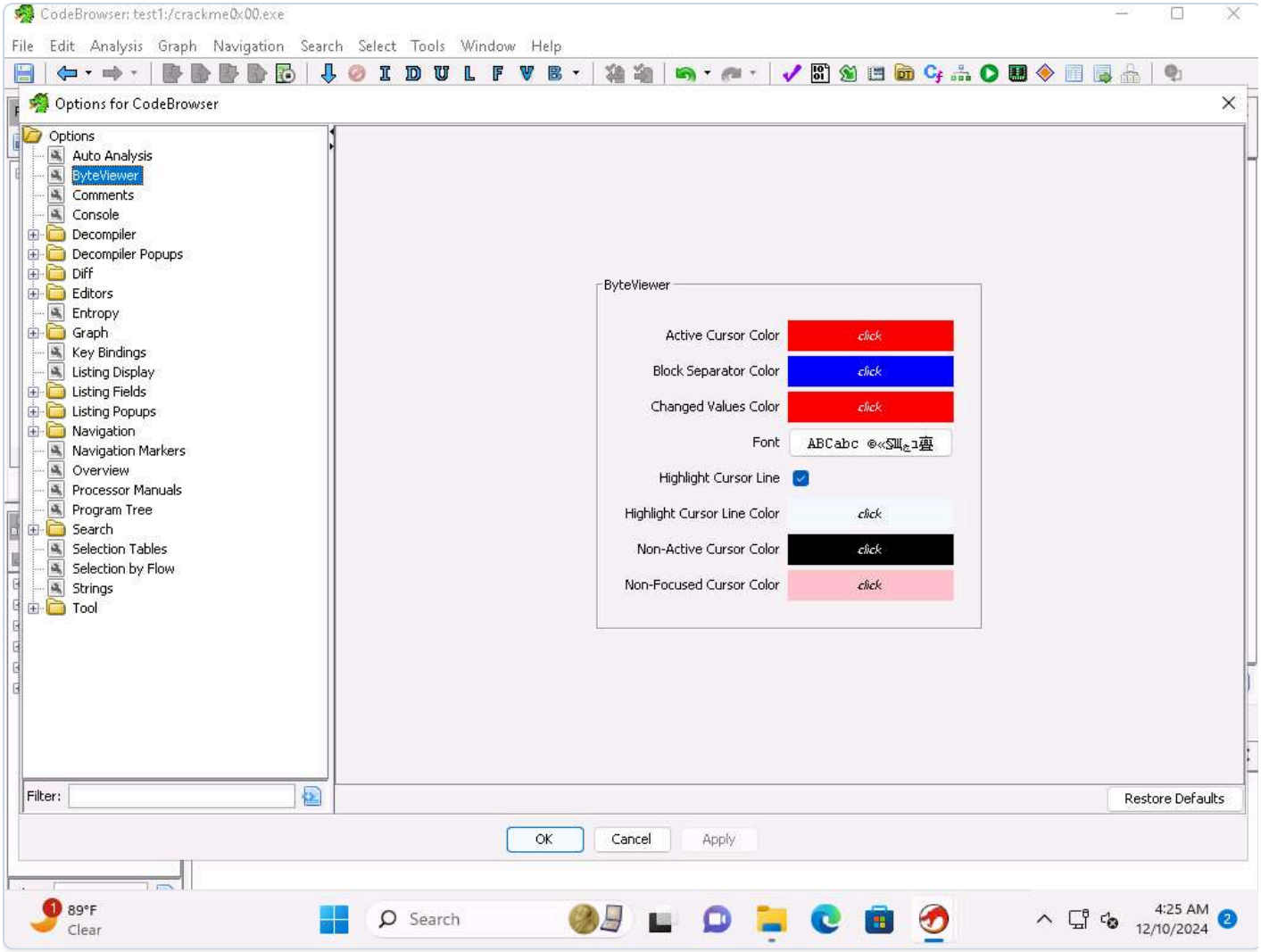



1. Explore ByteViewer options under **Edit -> Tool Options**.

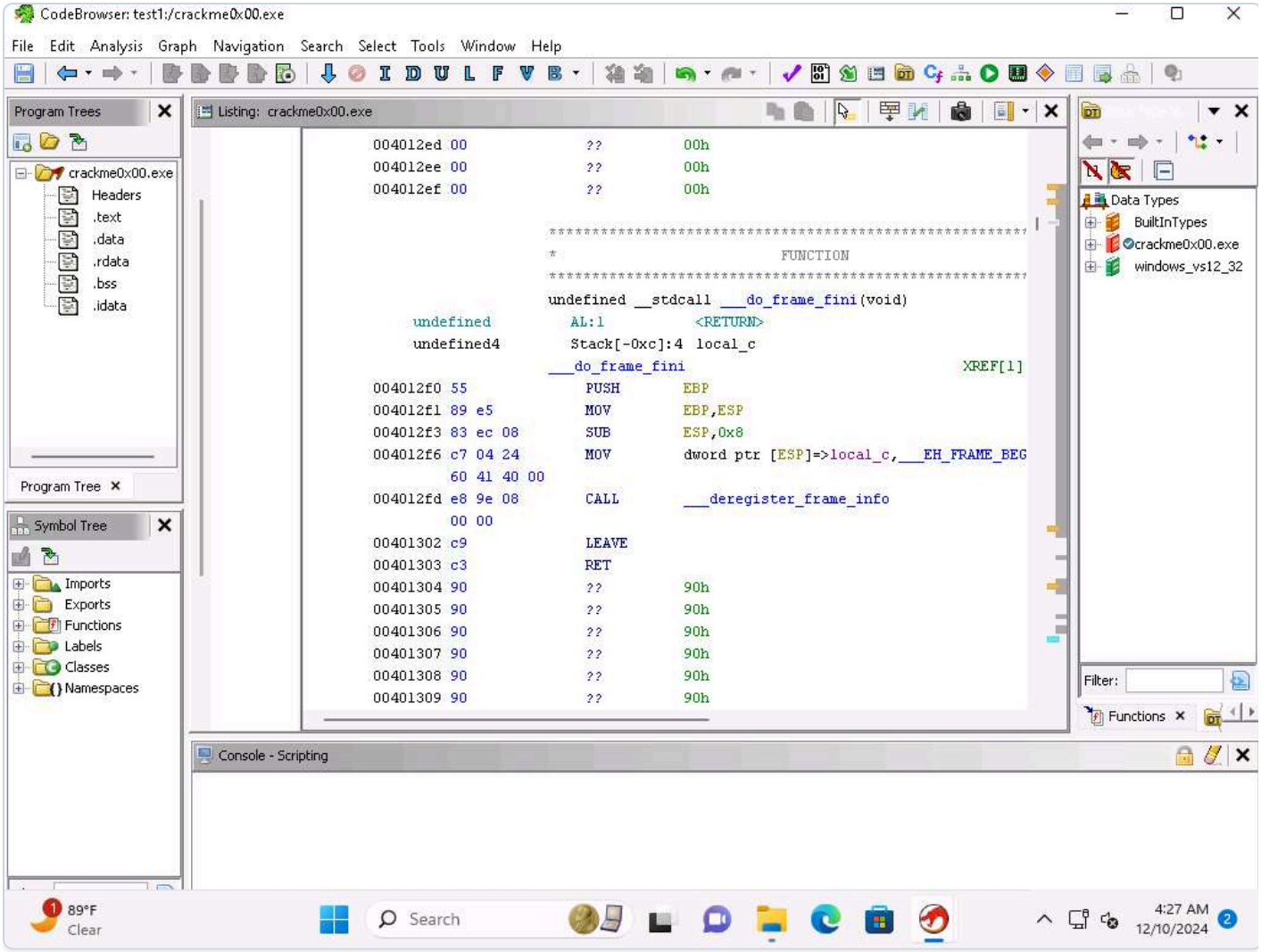

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
.bss
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Listing: crackme0x00.exe
00401305
00401306
00401307
00401308
00401309
0040130a
0040130b
0040130c
0040130d
0040130e
0040130f
int
int
char
char
undef
undef
undef
Console - Scripting
Bookmarks Ctrl+B
Bundle Manager
Bytes: crackme0x00.exe
Checksum Generator
Comments
Console
Data Type Manager
Data Type Preview
Decompiler Ctrl+E
Defined Data
Defined Strings
Disassembled View
Equates Table
External Programs
Function Call Graph
Function Call Trees
Function Graph
Function Tags
Functions
Listing: crackme0x00.exe
Memory Map
Program Trees
Python
Register Manager
Relocation Table
Script Manager
Symbol References
Symbol Table Ctrl+T
Symbol Tree
Functions - 56 items
Name
Fun...
Func...
___mingw_CRTStartup 279
_mainCRTStartup 24
_atexit 12
___do_frame_init 34
___do_frame_fini 20
_main 141
__pei386_runtime_relo... 34
__fpreset 7
___do_global_ctors 83
___main 27
_size_of_encoded_value 76
_read_uleb128 71
_read_sleb128 112
_read_encoded_value... 194
_init_object_mutex 66
_init_object_mutex_once 133
___register_frame_inf... 237
___register_frame_info 38
___register_frame_inf... 245
___register_frame_inf... 38
___deregister_frame_i... 385
___deregister_frame_i... 9
_base_from_object 80
_get_cie_encoding 139
_fde_single_encoding_... 120
_frame_downheap 137
Filter:
89°F
Clear
4:03 AM
12/10/2024



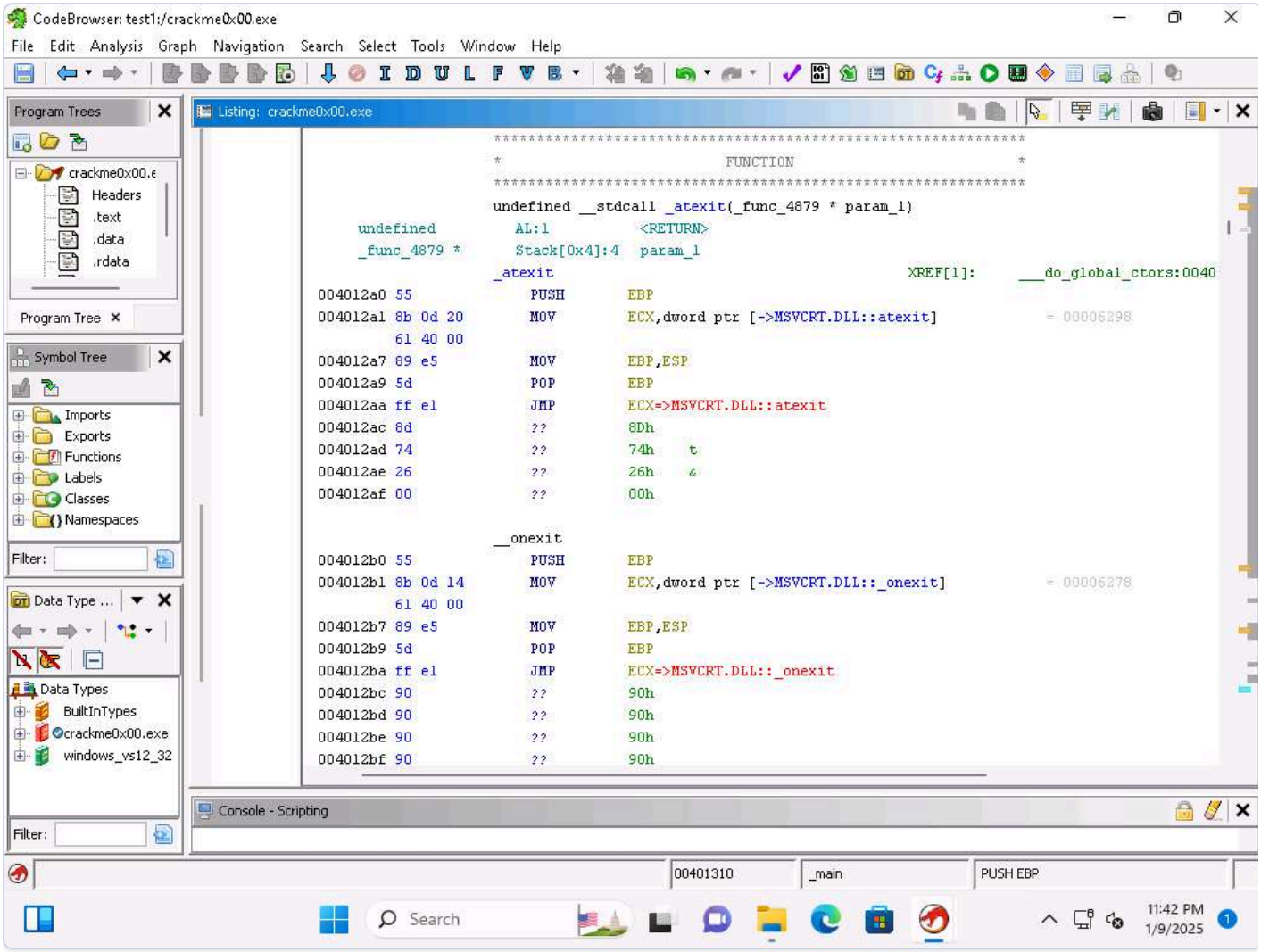

1. Inspect decompiled code in Listing window.

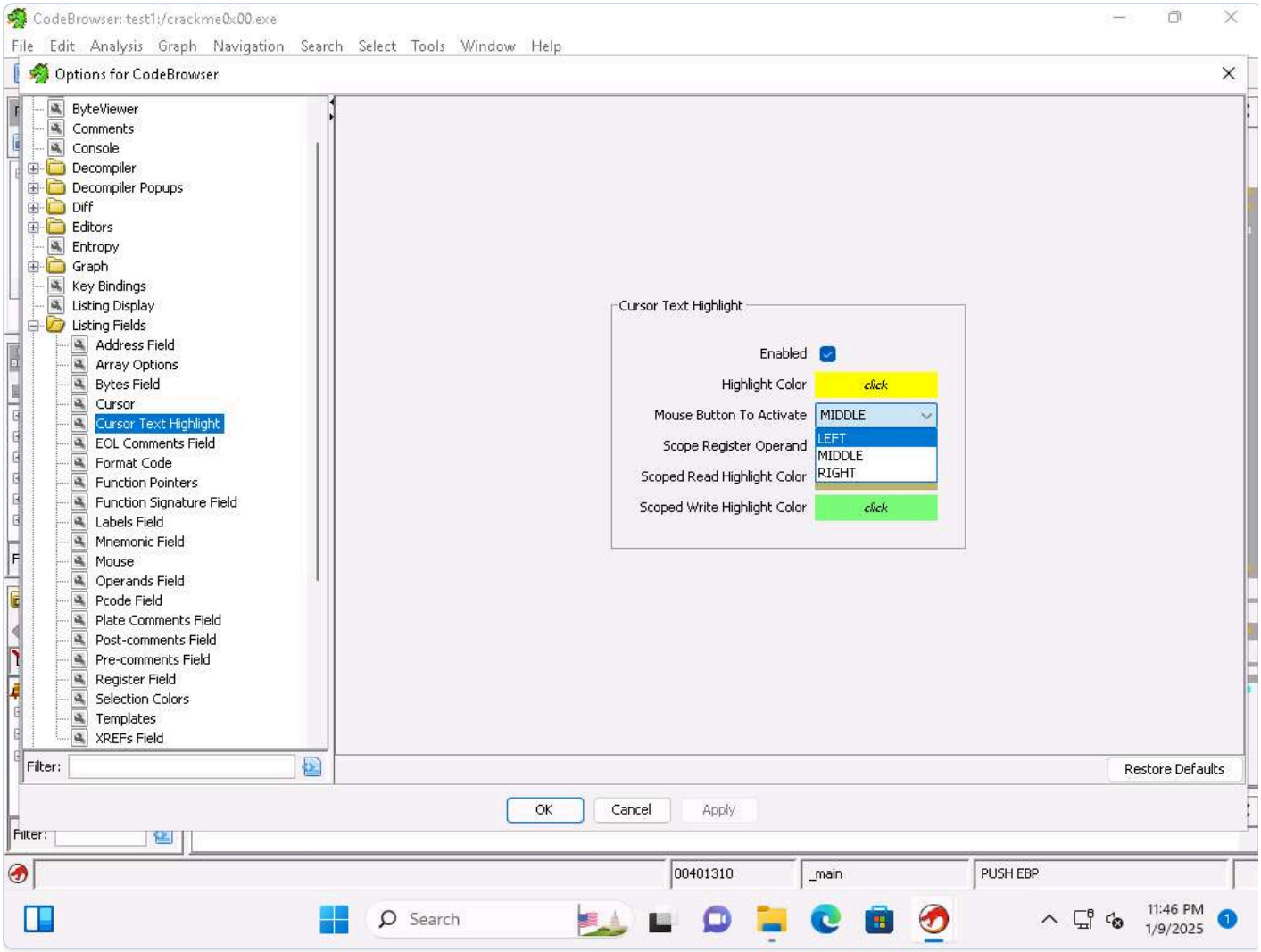



1. Locate stack initialization instruction `PUSH EBP`.

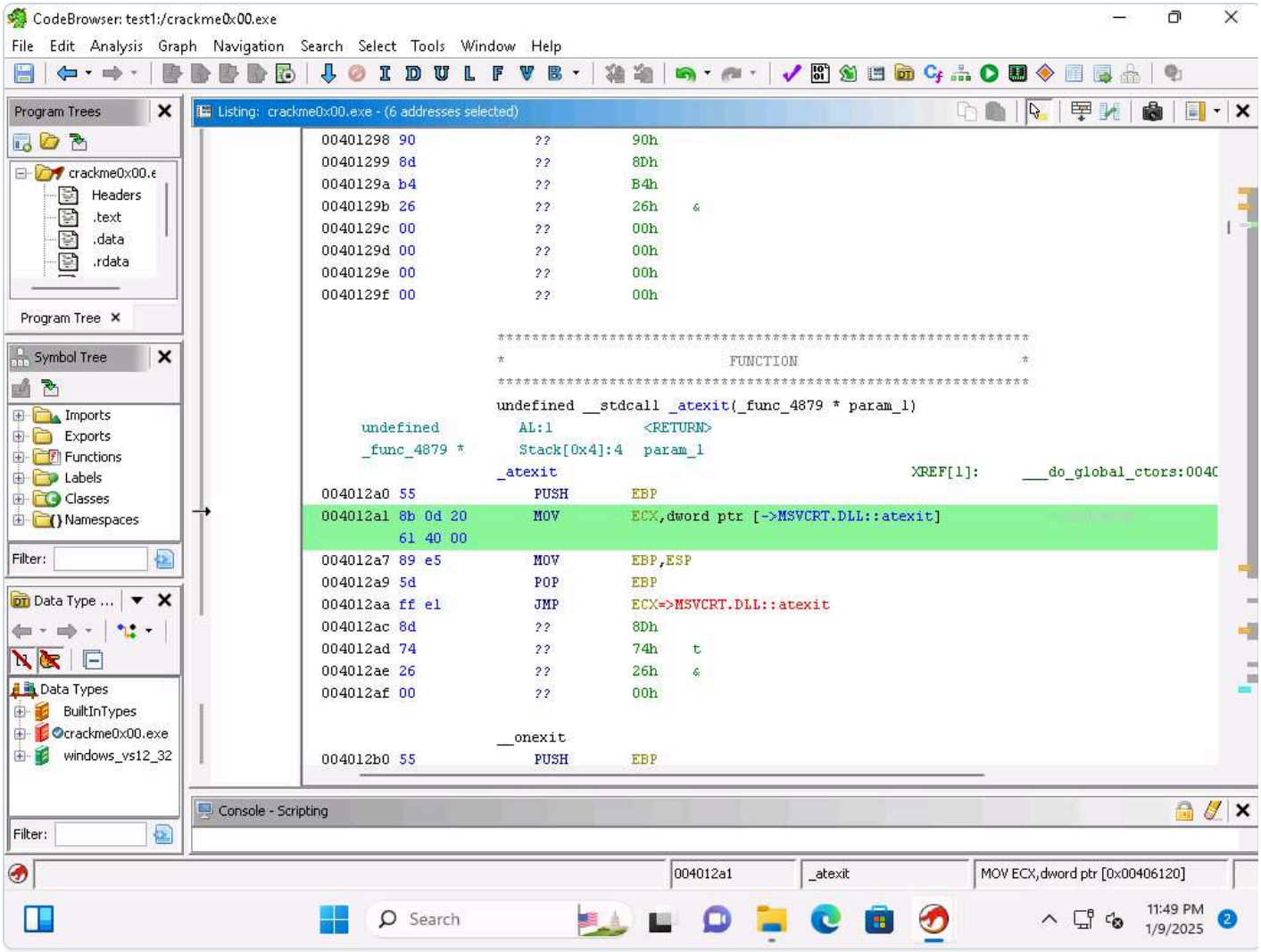



1. Configure Cursor TextHighlight under Options -> Listing Fields.

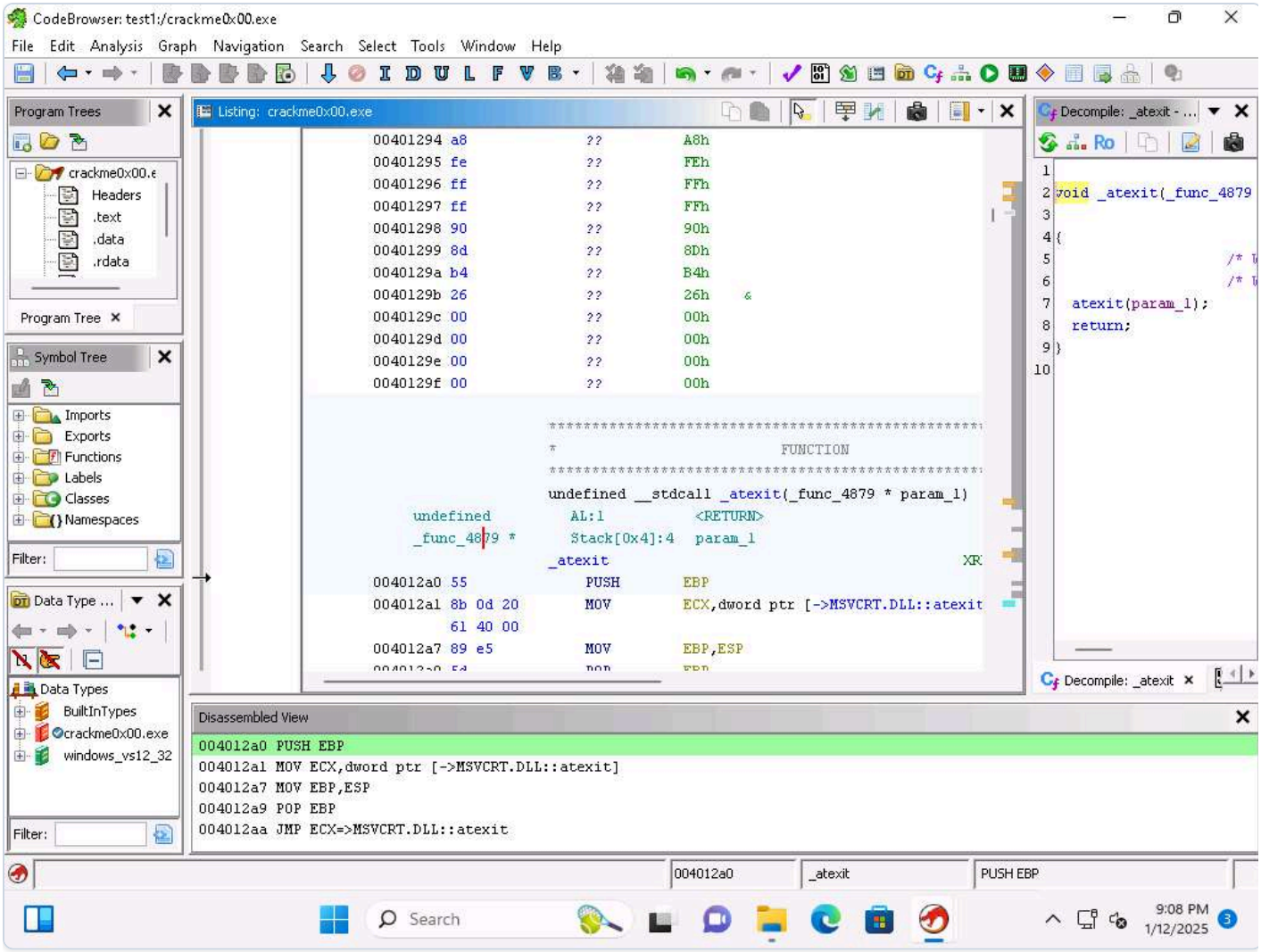

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
FUNCTION
undefined __stdcall _atexit(_func_4879 * param_1)
undefined AL:1 <RETURN>
_func_4879 * Stack[0x4]:4 param_1
_atexit XREF[1]: ___do_global_ctors:0040
004012a0 55 PUSH EBP
004012a1 8b 0d 20 61 40 00 MOV ECX,dword ptr [->MSVCRT.DLL::atexit] = 00006298
004012a7 89 e5 MOV EBP,ESP
004012a9 5d POP EBP
004012aa ff e1 JMP ECX=>MSVCRT.DLL::atexit
004012ac 8d ?? 8Dh
004012ad 74 ?? 74h t
004012ae 26 ?? 26h &
004012af 00 ?? 00h
__onexit
004012b0 55 PUSH EBP
004012b1 8b 0d 14 61 40 00 MOV ECX,dword ptr [->MSVCRT.DLL::_onexit] = 00006278
004012b7 89 e5 MOV EBP,ESP
004012b9 5d POP EBP
004012ba ff e1 JMP ECX=>MSVCRT.DLL::_onexit
004012bc 90 ?? 90h
004012bd 90 ?? 90h
004012be 90 ?? 90h
004012bf 90 ?? 90h
Console - Scripting
00401310
_main
PUSH EBP
Search
11:42 PM
1/9/2025



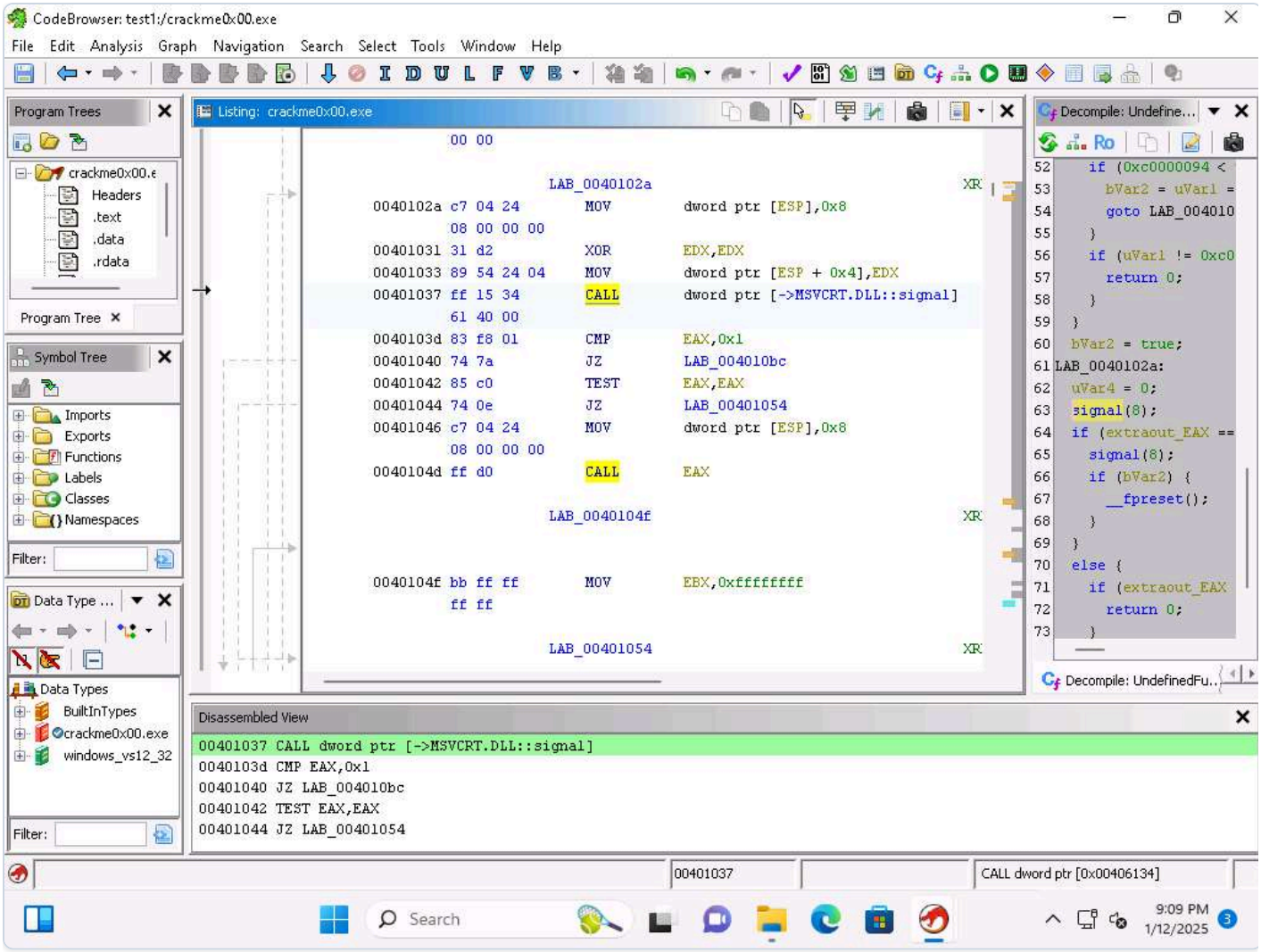

1. Click inside Listing window to open Decompiler view.



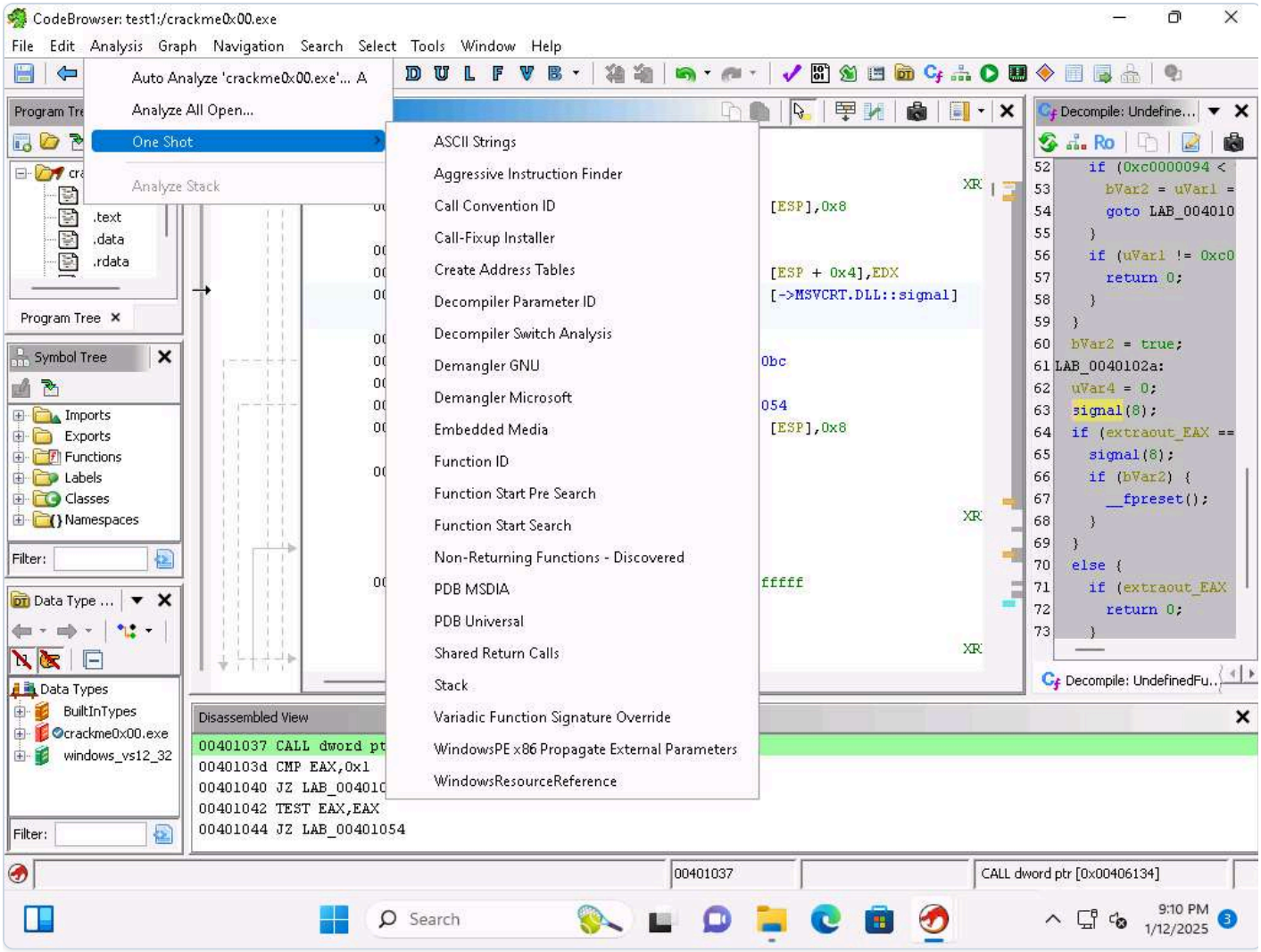

1. Observe statement highlighting on `CALL` instructions.

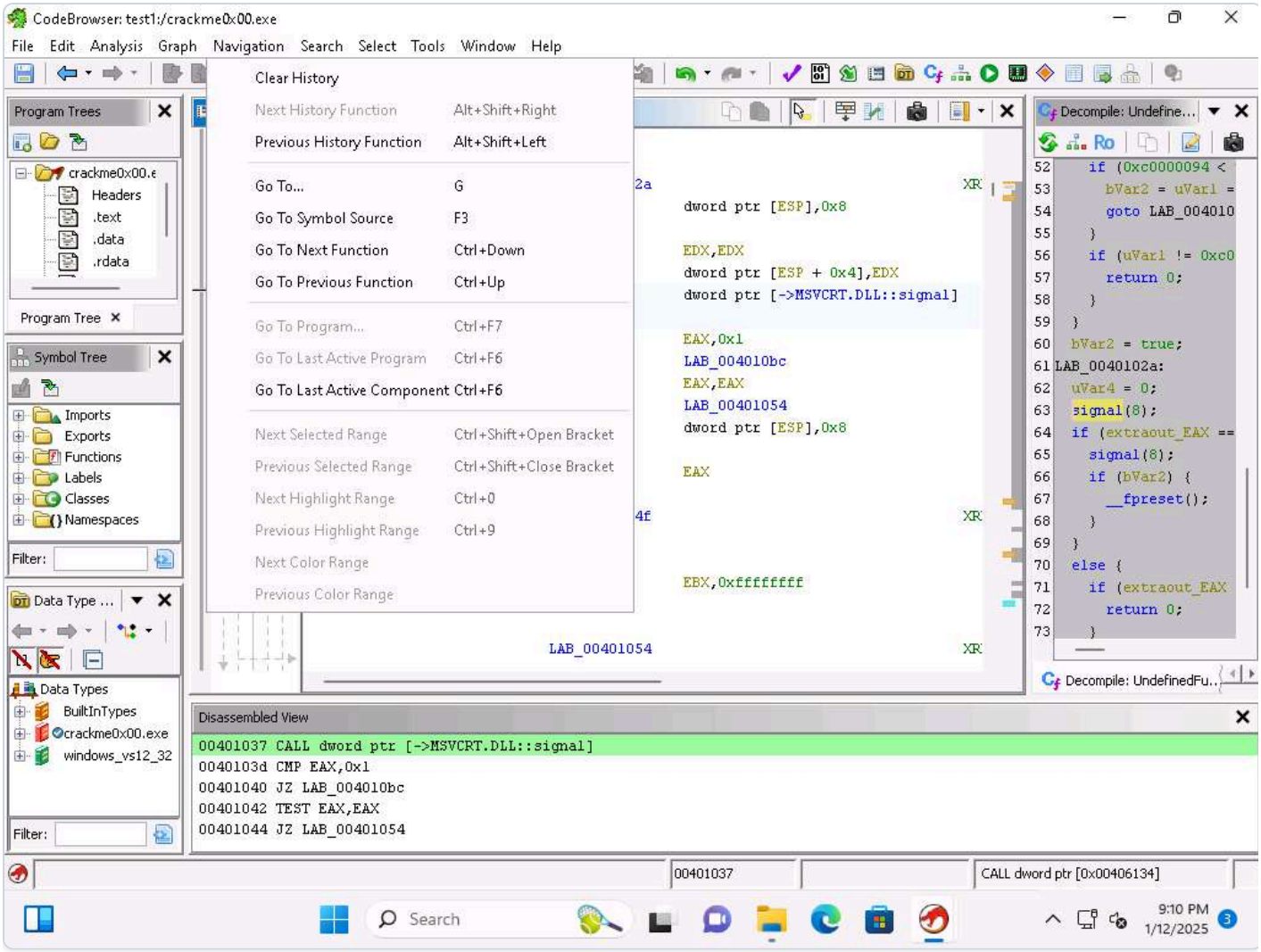



1. Access Analysis -> One Shot menu options.

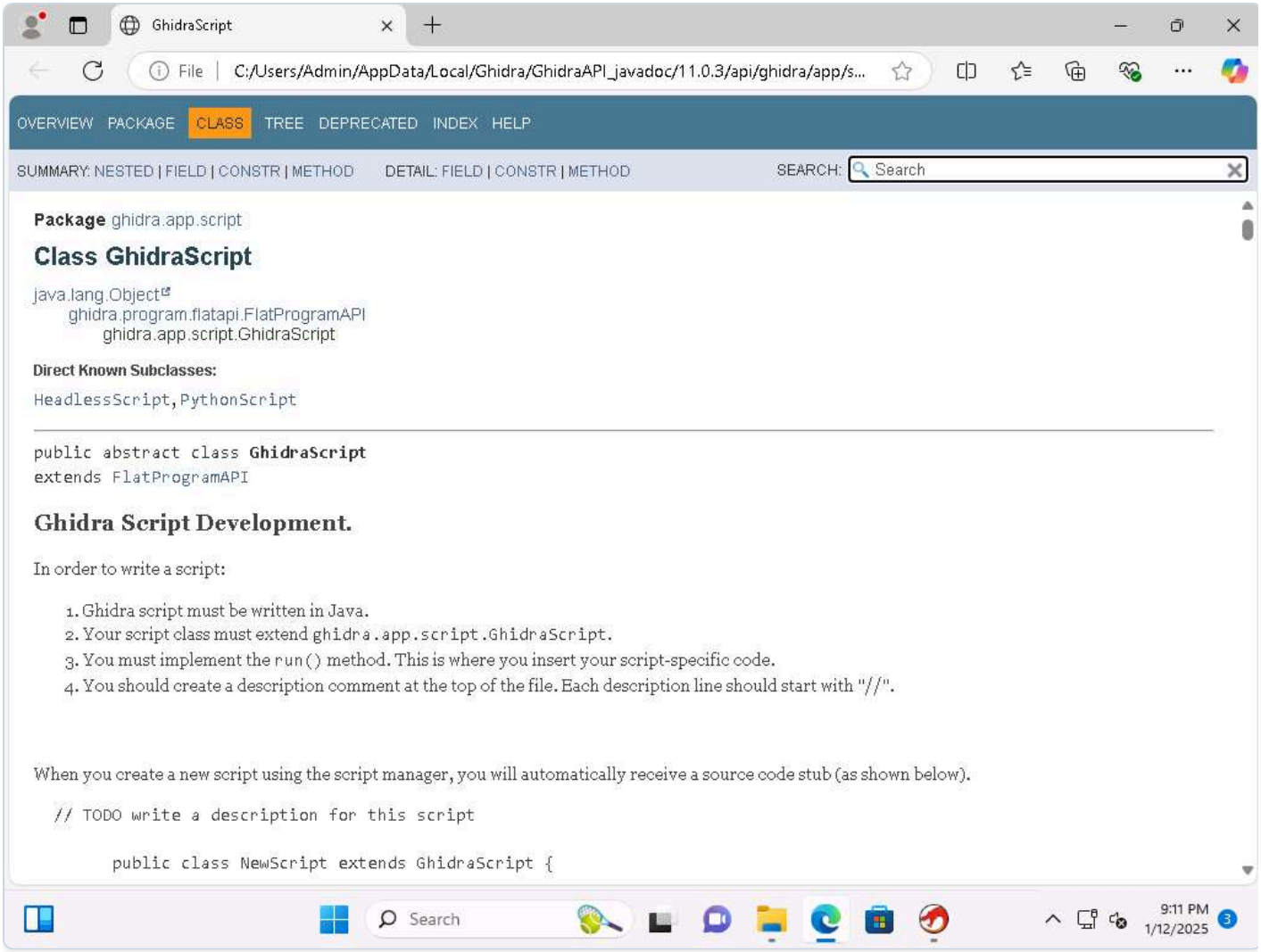


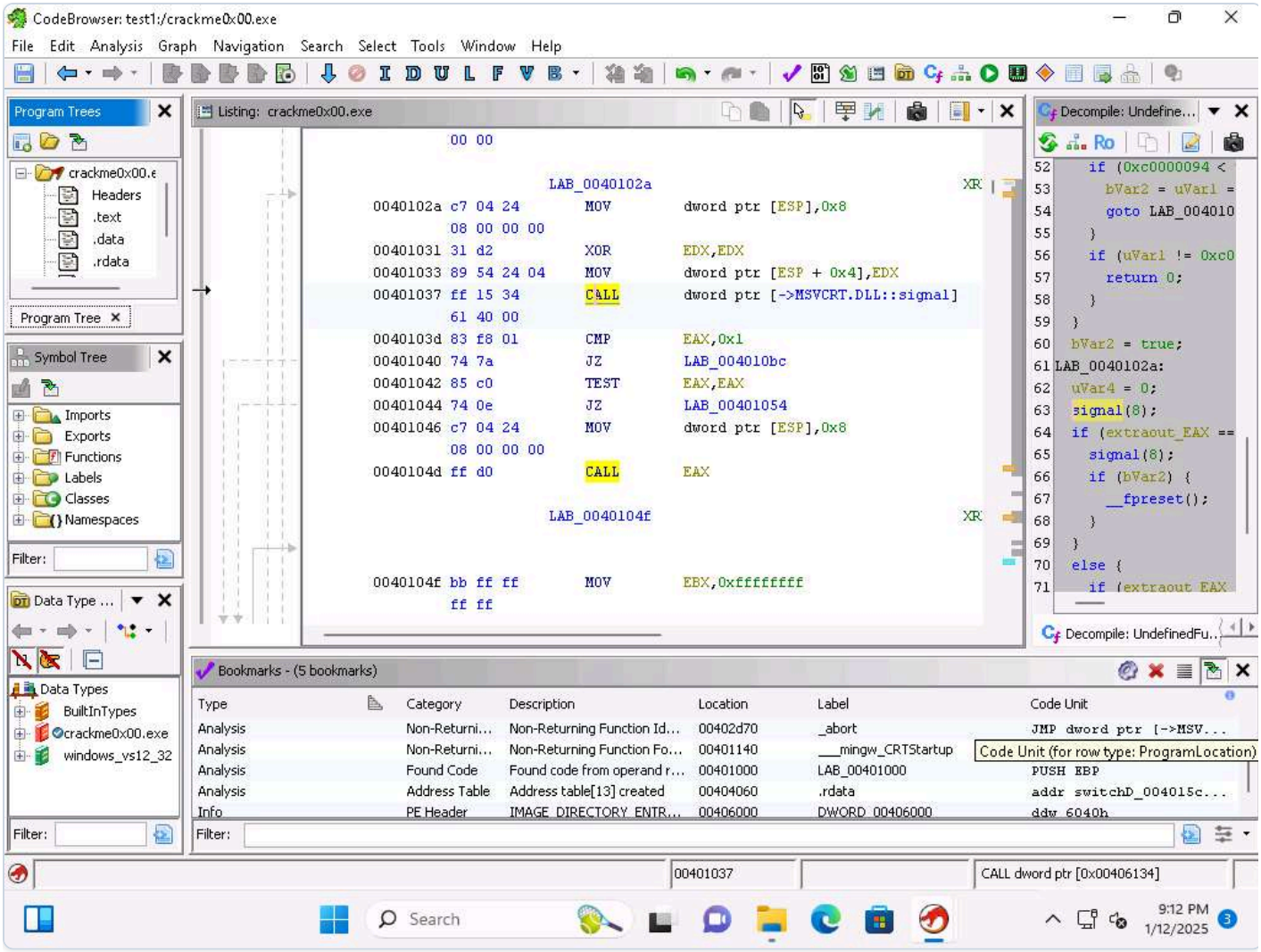

1. Open Ghidra API Help documentation.

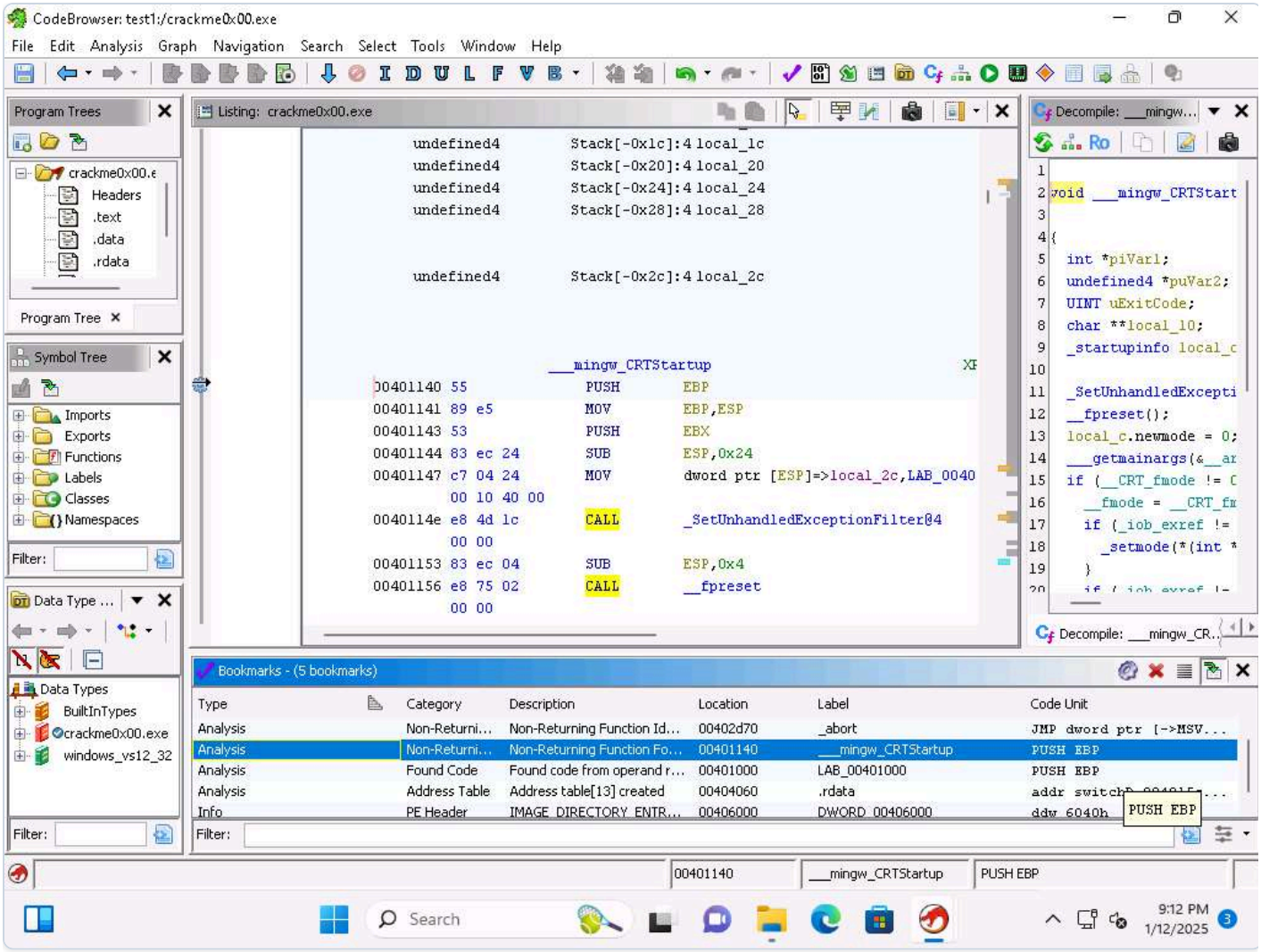

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Auto Analyze 'crackme0x00.exe'... A
Analyze All Open...
One Shot
Analyze Stack
ASCII Strings
Aggressive Instruction Finder
Call Convention ID
Call-Fixup Installer
Create Address Tables
Decompiler Parameter ID
Decompiler Switch Analysis
Demangler GNU
Demangler Microsoft
Embedded Media
Function ID
Function Start Pre Search
Function Start Search
Non-Returning Functions - Discovered
PDB MSDIA
PDB Universal
Shared Return Calls
Stack
Variadic Function Signature Override
WindowsPE x86 Propagate External Parameters
WindowsResourceReference
Program Tree
.text
.data
.rdata
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Disassembled View
00401037 CALL dword pt
0040103d CMP EAX,0x1
00401040 JZ LAB_004010
00401042 TEST EAX,EAX
00401044 JZ LAB_00401054
Decompile: Undefine...
if (0xc0000094 <
bVar2 = uVar1 =
goto LAB_004010
if (uVar1 != 0xc0
return 0;
bVar2 = true;
LAB_0040102a:
uVar4 = 0;
signal(8);
if (extraout_EAX ==
signal(8);
if (bVar2) {
__fpreset();
else {
if (extraout_EAX
return 0;
Decompile: UndefinedFu...
00401037
CALL dword ptr [0x00406134]
Search
9:10 PM
1/12/2025



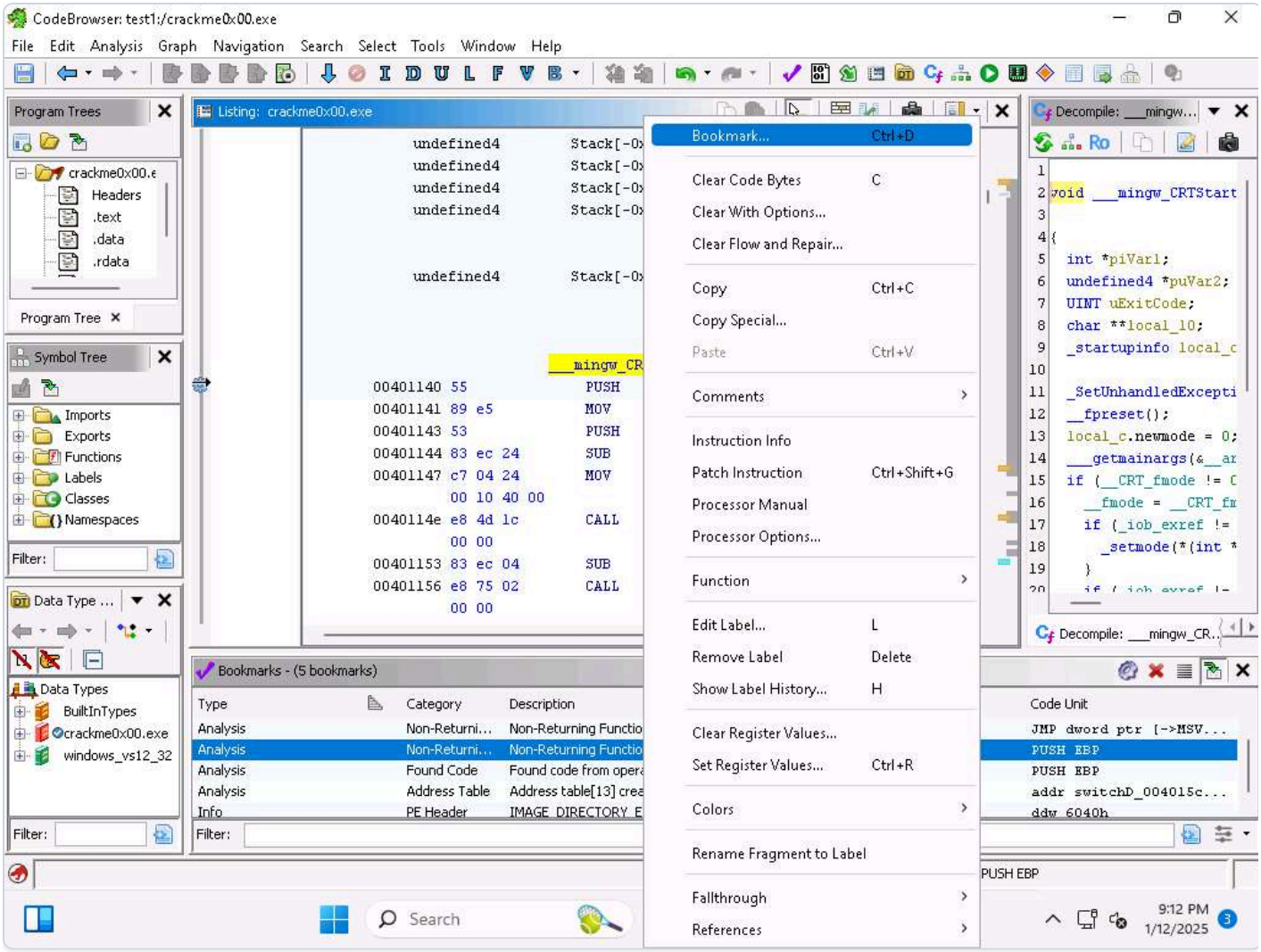

1. Open Bookmarks window to view code bookmarks.

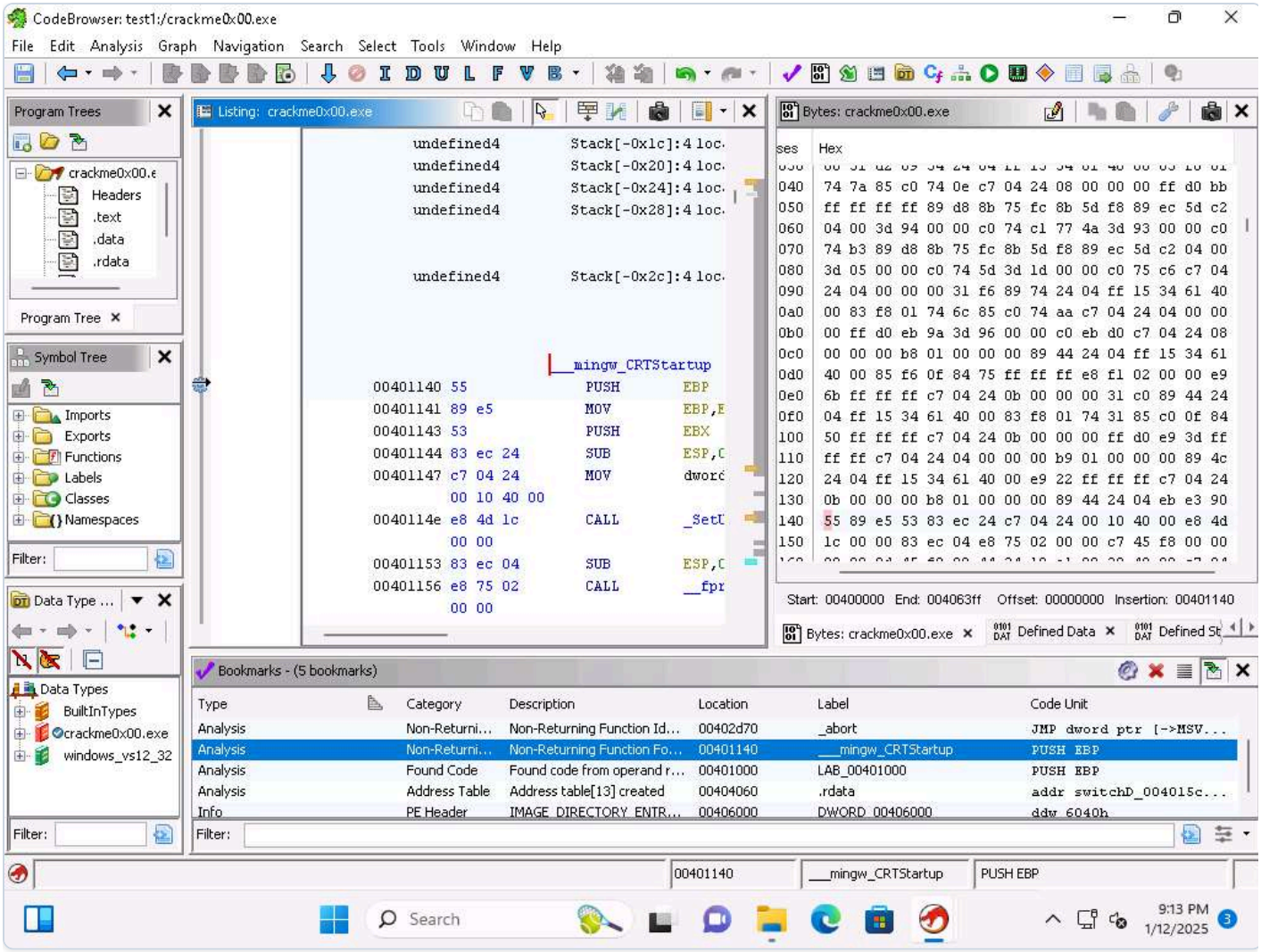

GhidraScript
File | C:/Users/Admin/AppData/Local/Ghidra/GhidraAPI_javadoc/11.0.3/api/ghidra/app/s...
OVERVIEW PACKAGE CLASS TREE DEPRECATED INDEX HELP
SUMMARY: NESTED | FIELD | CONSTR | METHOD DETAIL: FIELD | CONSTR | METHOD SEARCH: Search
Package ghidra.app.script
Class GhidraScript
java.lang.Object
ghidra.program.flatapi.FlatProgramAPI
ghidra.app.script.GhidraScript
Direct Known Subclasses:
HeadlessScript, PythonScript
public abstract class GhidraScript
extends FlatProgramAPI
Ghidra Script Development.
In order to write a script:
1. Ghidra script must be written in Java.
2. Your script class must extend ghidra.app.script.GhidraScript.
3. You must implement the run() method. This is where you insert your script-specific code.
4. You should create a description comment at the top of the file. Each description line should start with "//".
When you create a new script using the script manager, you will automatically receive a source code stub (as shown below).
// TODO write a description for this script
public class NewScript extends GhidraScript {
Search
9:11 PM
1/12/2025

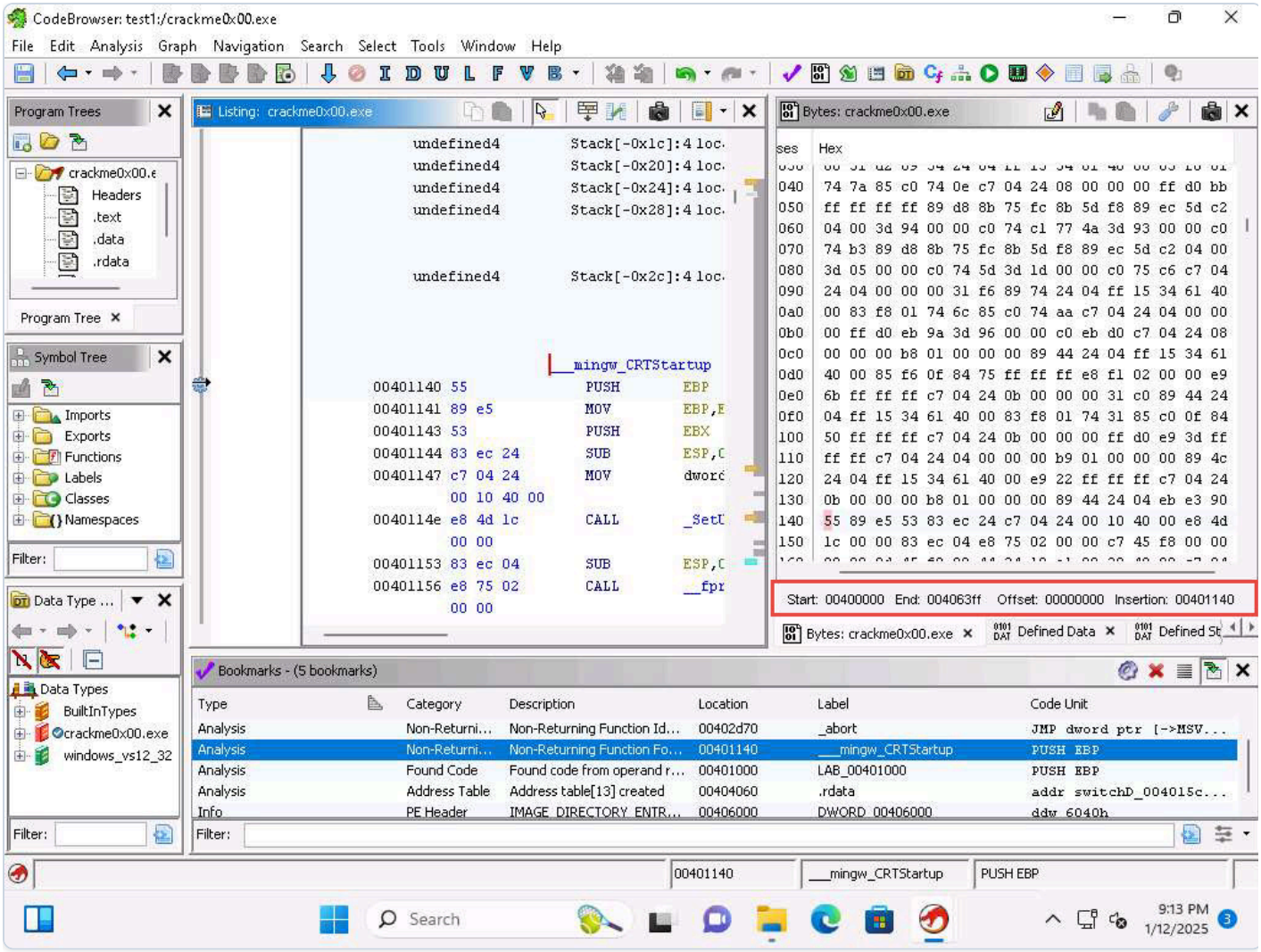

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
00 00
LAB_0040102a XR
0040102a c7 04 24 08 00 00 00 MOV dword ptr [ESP],0x8
00401031 31 d2 XOR EDX,EDX
00401033 89 54 24 04 MOV dword ptr [ESP + 0x4],EDX
00401037 ff 15 34 61 40 00 CALL dword ptr [->MSVCRT.DLL::signal]
0040103d 83 f8 01 CMP EAX,0x1
00401040 74 7a JZ LAB_004010bc
00401042 85 c0 TEST EAX,EAX
00401044 74 0e JZ LAB_00401054
00401046 c7 04 24 08 00 00 00 MOV dword ptr [ESP],0x8
0040104d ff d0 CALL EAX
LAB_0040104f XR
0040104f bb ff ff ff ff MOV EBX,0xffffffff
Decompile: Undefine...
52 if (0xc0000094 <
53 bVar2 = uVar1 =
54 goto LAB_004010
55 }
56 if (uVar1 != 0xc0
57 return 0;
58 }
59 }
60 bVar2 = true;
61 LAB_0040102a:
62 uVar4 = 0;
63 signal(8);
64 if (extraout_EAX ==
65 signal(8);
66 if (bVar2) {
67 __fpreset();
68 }
69 }
70 else {
71 if (extraout_EAX
Decompile: UndefinedFu...
Bookmarks - (5 bookmarks)
Type Category Description Location Label Code Unit
Analysis Non-Returni... Non-Returning Function Id... 00402d70 _abort JMP dword ptr [->MSV...
Analysis Non-Returni... Non-Returning Function Fo... 00401140 ___mingw_CRTStartup
Analysis Found Code Found code from operand r... 00401000 LAB_00401000 PUSH EBP
Analysis Address Table Address table[13] created 00404060 .rdata addr switchD_004015c...
Info PE Header IMAGE_DIRECTORY_ENTR... 00406000 DWORD_00406000 ddw 6040h
Code Unit (for row type: ProgramLocation)
Filter:
00401037
CALL dword ptr [0x00406134]
Search
9:12 PM
1/12/2025



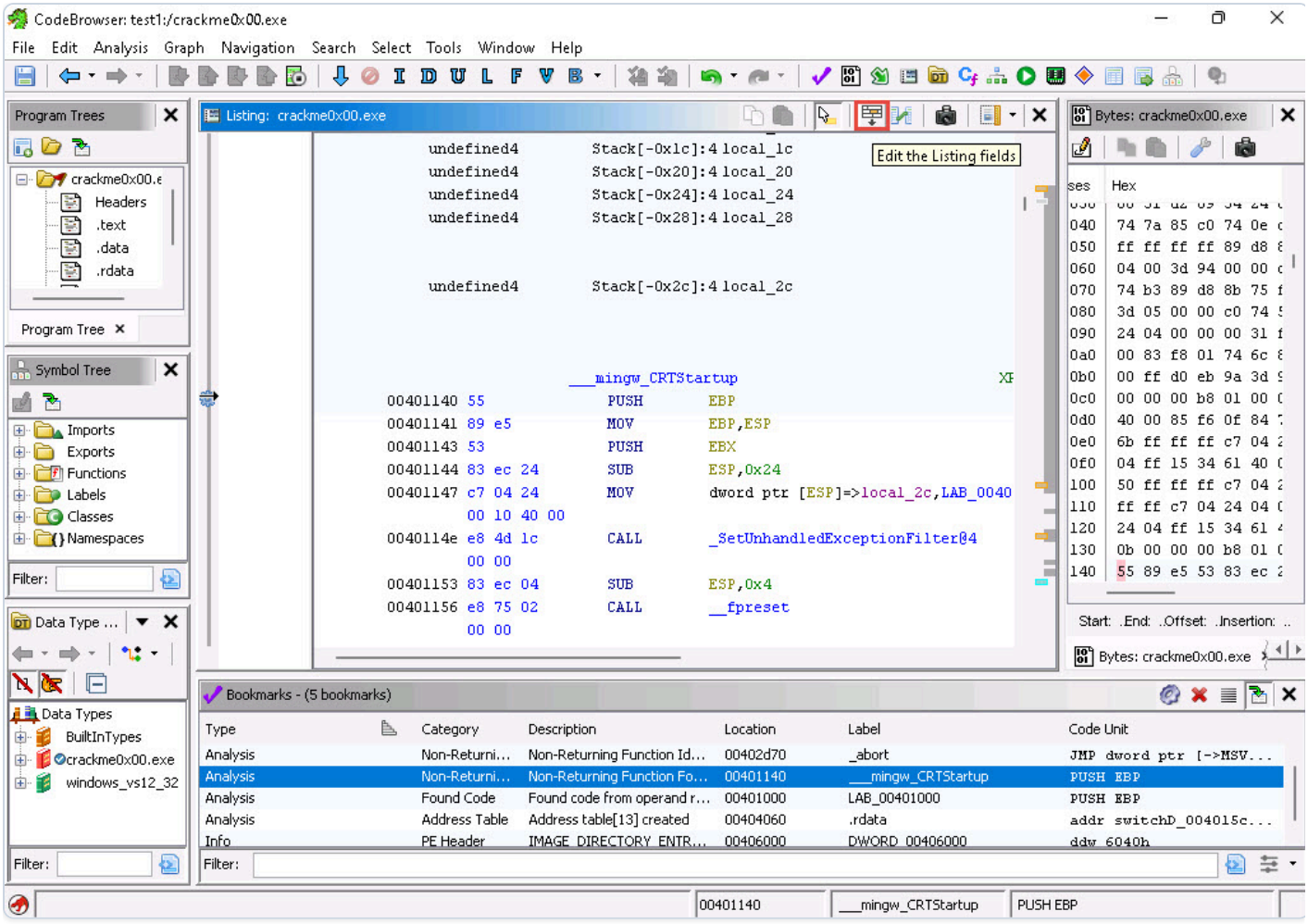

1. Open Bytes window (**Window -> Bytes**) for hex view.

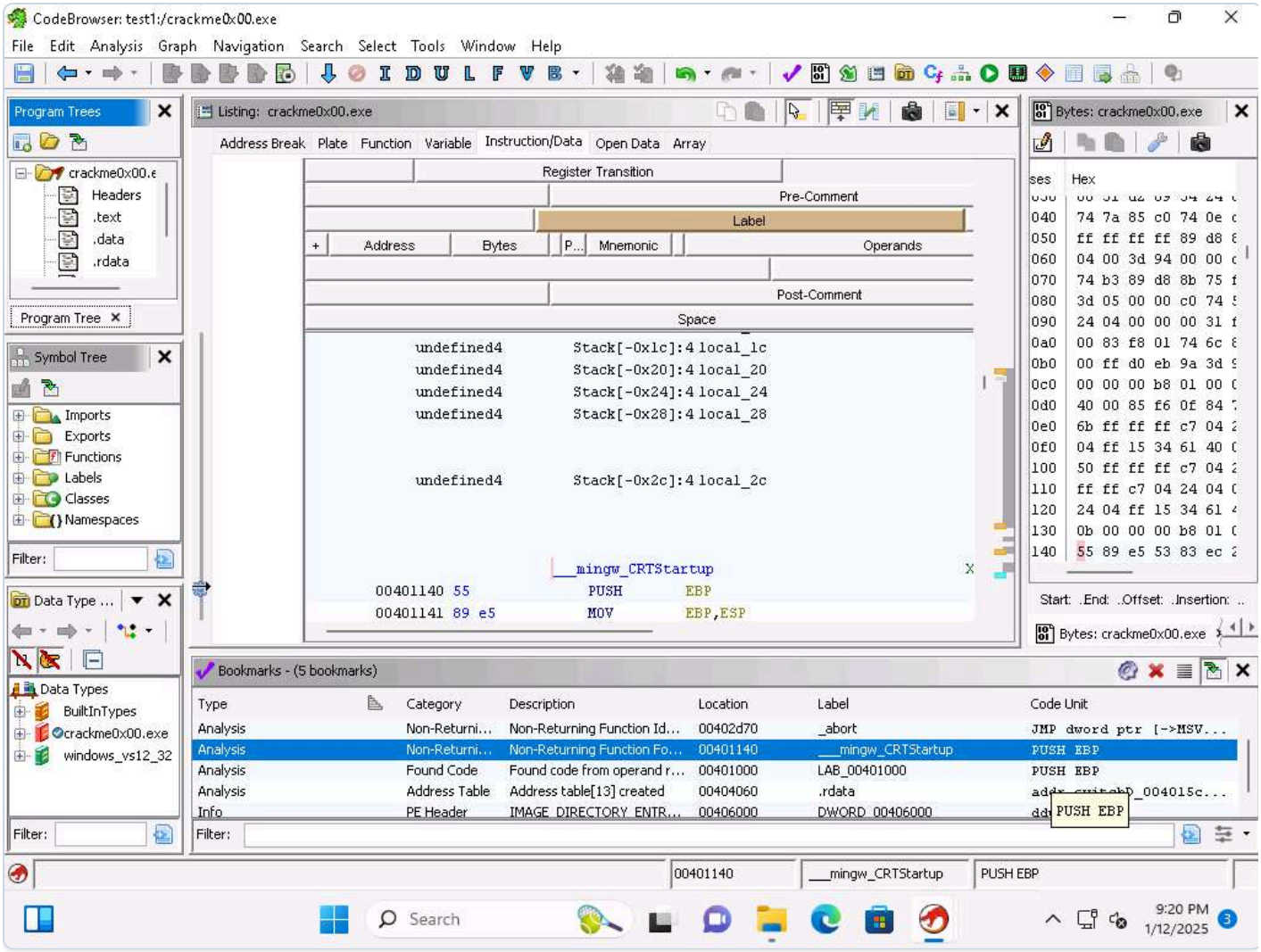

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
undefined4 Stack[-0x
undefined4 Stack[-0x
undefined4 Stack[-0x
undefined4 Stack[-0x
undefined4 Stack[-0x
__mingw_CR
00401140 55 PUSH
00401141 89 e5 MOV
00401143 53 PUSH
00401144 83 ec 24 SUB
00401147 c7 04 24 MOV
00 10 40 00
0040114e e8 4d 1c CALL
00 00
00401153 83 ec 04 SUB
00401156 e8 75 02 CALL
00 00
Bookmarks - (5 bookmarks)
Type Category Description
Analysis Non-Returni... Non-Returning Functio
Analysis Non-Returni... Non-Returning Functio
Analysis Found Code Found code from oper
Analysis Address Table Address table[13] crea
Info PE Header IMAGE_DIRECTORY_E
Filter:
Bookmark... Ctrl+D
Clear Code Bytes C
Clear With Options...
Clear Flow and Repair...
Copy Ctrl+C
Copy Special...
Paste Ctrl+V
Comments
Instruction Info
Patch Instruction Ctrl+Shift+G
Processor Manual
Processor Options...
Function
Edit Label... L
Remove Label Delete
Show Label History... H
Clear Register Values...
Set Register Values... Ctrl+R
Colors
Rename Fragment to Label
Fallthrough
References
Decompile: ___mingw...
void ___mingw_CRTStart
{
int *piVar1;
undefined4 *puVar2;
UINT uExitCode;
char **local_10;
_startupinfo local_c
_SetUnhandledExcepti
__fpreset();
local_c.newmode = 0;
___getmainargs(&__ar
if (__CRT_fmode != C
__fmode = __CRT_fm
if (_iob_exref !=
_setmode(*(int *
}
Decompile: ___mingw_CR...
Code Unit
JMP dword ptr [->MSV...
PUSH EBP
PUSH EBP
addr switchD_004015c...
ddw 6040h
PUSH EBP
Search
9:12 PM
1/12/2025


CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
undefined4 Stack[-0x1c]:4 loc.
undefined4 Stack[-0x20]:4 loc.
undefined4 Stack[-0x24]:4 loc.
undefined4 Stack[-0x28]:4 loc.
undefined4 Stack[-0x2c]:4 loc.
__mingw_CRTStartup
00401140 55 PUSH EBP
00401141 89 e5 MOV EBP,E
00401143 53 PUSH EBX
00401144 83 ec 24 SUB ESP,0
00401147 c7 04 24 00 10 40 00 MOV dword
0040114e e8 4d 1c 00 00 CALL _SetU
00401153 83 ec 04 SUB ESP,0
00401156 e8 75 02 00 00 CALL __fpr
Bytes: crackme0x00.exe
ses Hex
040 74 7a 85 c0 74 0e c7 04 24 08 00 00 00 ff d0 bb
050 ff ff ff ff 89 d8 8b 75 fc 8b 5d f8 89 ec 5d c2
060 04 00 3d 94 00 00 c0 74 c1 77 4a 3d 93 00 00 c0
070 74 b3 89 d8 8b 75 fc 8b 5d f8 89 ec 5d c2 04 00
080 3d 05 00 00 c0 74 5d 3d 1d 00 00 c0 75 c6 c7 04
090 24 04 00 00 00 31 f6 89 74 24 04 ff 15 34 61 40
0a0 00 83 f8 01 74 6c 85 c0 74 aa c7 04 24 04 00 00
0b0 00 ff d0 eb 9a 3d 96 00 00 c0 eb d0 c7 04 24 08
0c0 00 00 00 b8 01 00 00 00 89 44 24 04 ff 15 34 61
0d0 40 00 85 f6 0f 84 75 ff ff ff e8 f1 02 00 00 e9
0e0 6b ff ff ff c7 04 24 0b 00 00 00 31 c0 89 44 24
0f0 04 ff 15 34 61 40 00 83 f8 01 74 31 85 c0 0f 84
100 50 ff ff ff c7 04 24 0b 00 00 00 ff d0 e9 3d ff
110 ff ff c7 04 24 04 00 00 00 b9 01 00 00 00 89 4c
120 24 04 ff 15 34 61 40 00 e9 22 ff ff ff c7 04 24
130 0b 00 00 00 b8 01 00 00 00 89 44 24 04 eb e3 90
140 55 89 e5 53 83 ec 24 c7 04 24 00 10 40 00 e8 4d
150 1c 00 00 83 ec 04 e8 75 02 00 00 c7 45 f8 00 00
Start: 00400000 End: 004063ff Offset: 00000000 Insertion: 00401140
Bytes: crackme0x00.exe
Defined Data
Bookmarks - (5 bookmarks)
Type Category Description Location Label Code Unit
Analysis Non-Returni... Non-Returning Function Id... 00402d70 _abort JMP dword ptr [->MSV...
Analysis Non-Returni... Non-Returning Function Fo... 00401140 __mingw_CRTStartup PUSH EBP
Analysis Found Code Found code from operand r... 00401000 LAB_00401000 PUSH EBP
Analysis Address Table Address table[13] created 00404060 .rdata addr switchD_004015c...
Info PE Header IMAGE_DIRECTORY_ENTR... 00406000 DWORD_00406000 ddw 6040h
Filter:
00401140
__mingw_CRTStartup
PUSH EBP
Search
9:13 PM
1/12/2025

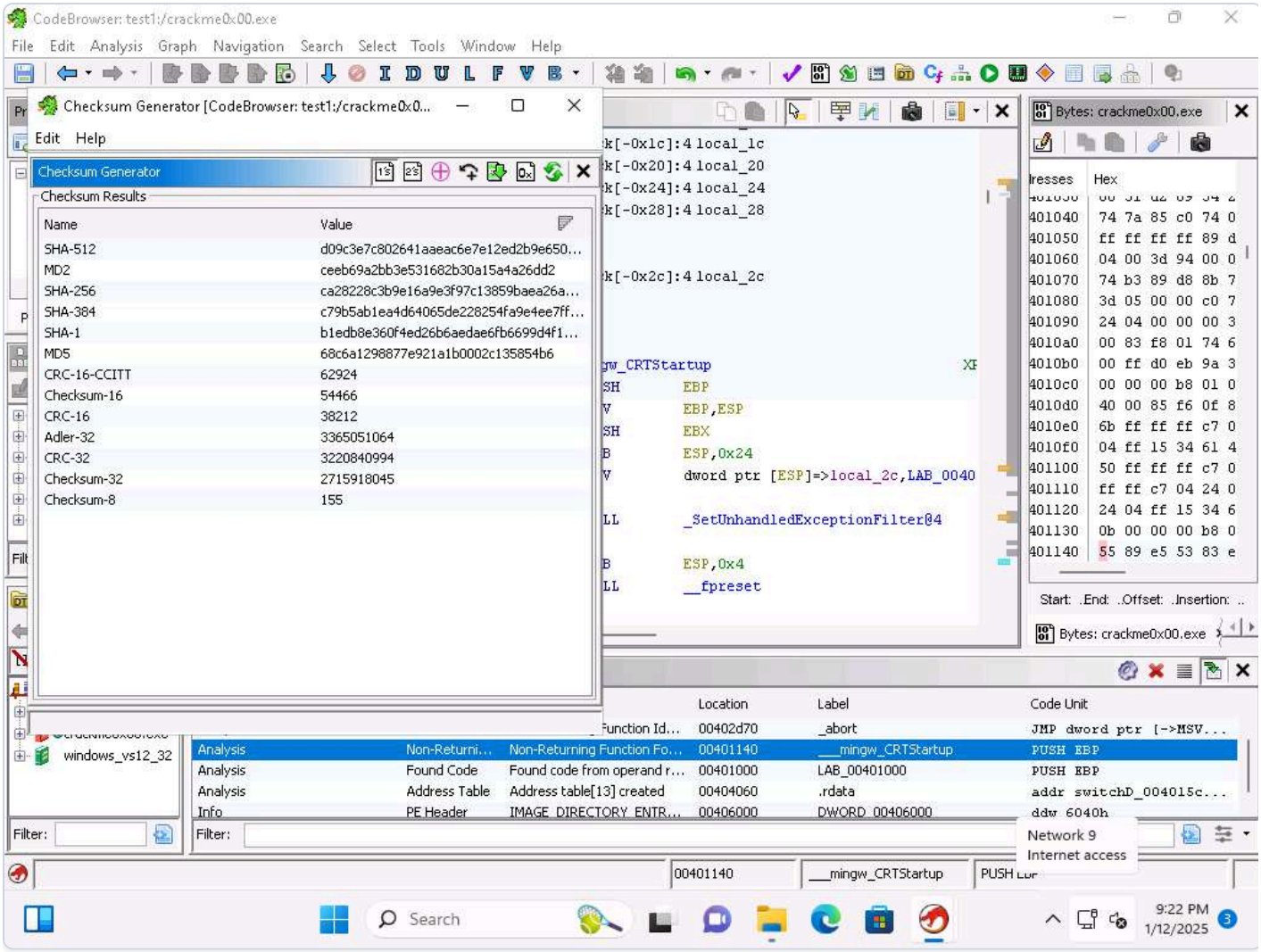



1. Toggle Listing fields editor.

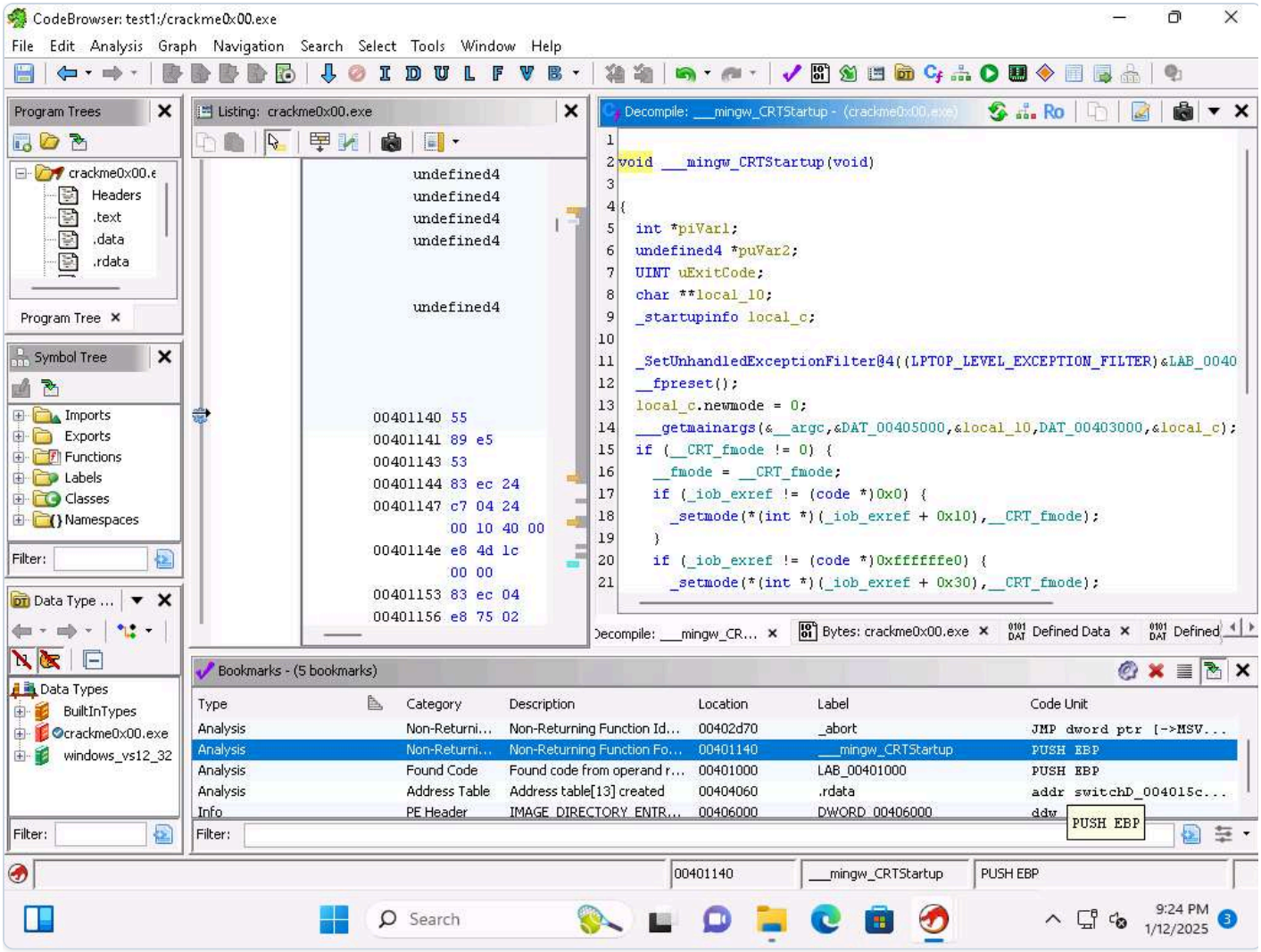

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
Edit the Listing fields
undefined4 Stack[-0x1c]:4 local_1c
undefined4 Stack[-0x20]:4 local_20
undefined4 Stack[-0x24]:4 local_24
undefined4 Stack[-0x28]:4 local_28
undefined4 Stack[-0x2c]:4 local_2c
___mingw_CRTStartup
00401140 55 PUSH EBP
00401141 89 e5 MOV EBP,ESP
00401143 53 PUSH EBX
00401144 83 ec 24 SUB ESP,0x24
00401147 c7 04 24 00 10 40 00 MOV dword ptr [ESP]=>local_2c,LAB_00401000
0040114e e8 4d 1c 00 00 CALL _SetUnhandledExceptionFilter@4
00401153 83 ec 04 SUB ESP,0x4
00401156 e8 75 02 00 00 CALL __fpreset
Bytes: crackme0x00.exe
ses Hex
040 74 7a 85 c0 74 0e
050 ff ff ff ff 89 d8
060 04 00 3d 94 00 00
070 74 b3 89 d8 8b 75
080 3d 05 00 00 c0 74
090 24 04 00 00 00 31
0a0 00 83 f8 01 74 6c
0b0 00 ff d0 eb 9a 3d
0c0 00 00 00 b8 01 00
0d0 40 00 85 f6 0f 84
0e0 6b ff ff ff c7 04
0f0 04 ff 15 34 61 40
100 50 ff ff ff c7 04
110 ff ff c7 04 24 04
120 24 04 ff 15 34 61
130 0b 00 00 00 b8 01
140 55 89 e5 53 83 ec
Start: .End: .Offset: .Insertion: ..
Bytes: crackme0x00.exe
Bookmarks - (5 bookmarks)
Type Category Description Location Label Code Unit
Analysis Non-Returni... Non-Returning Function Id... 00402d70 _abort JMP dword ptr [->MSV...
Analysis Non-Returni... Non-Returning Function Fo... 00401140 ___mingw_CRTStartup PUSH EBP
Analysis Found Code Found code from operand r... 00401000 LAB_00401000 PUSH EBP
Analysis Address Table Address table[13] created 00404060 .rdata addr switchD_004015c...
Info PE Header IMAGE_DIRECTORY_ENTR... 00406000 DWORD_00406000 ddw 6040h
Filter:
00401140
___mingw_CRTStartup
PUSH EBP

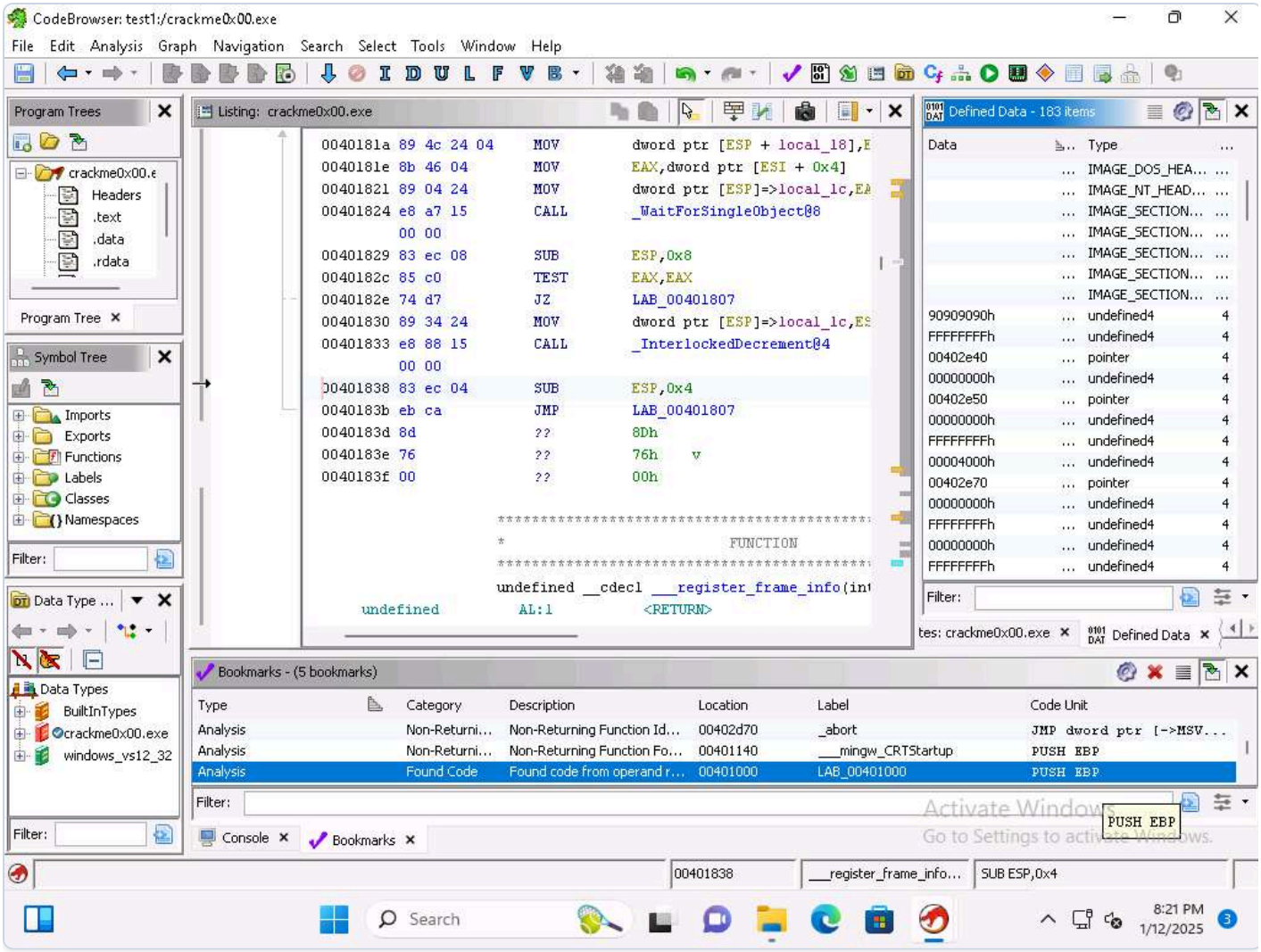



1. Open Checksum Generator tool (**Window -> Checksum Generator**).

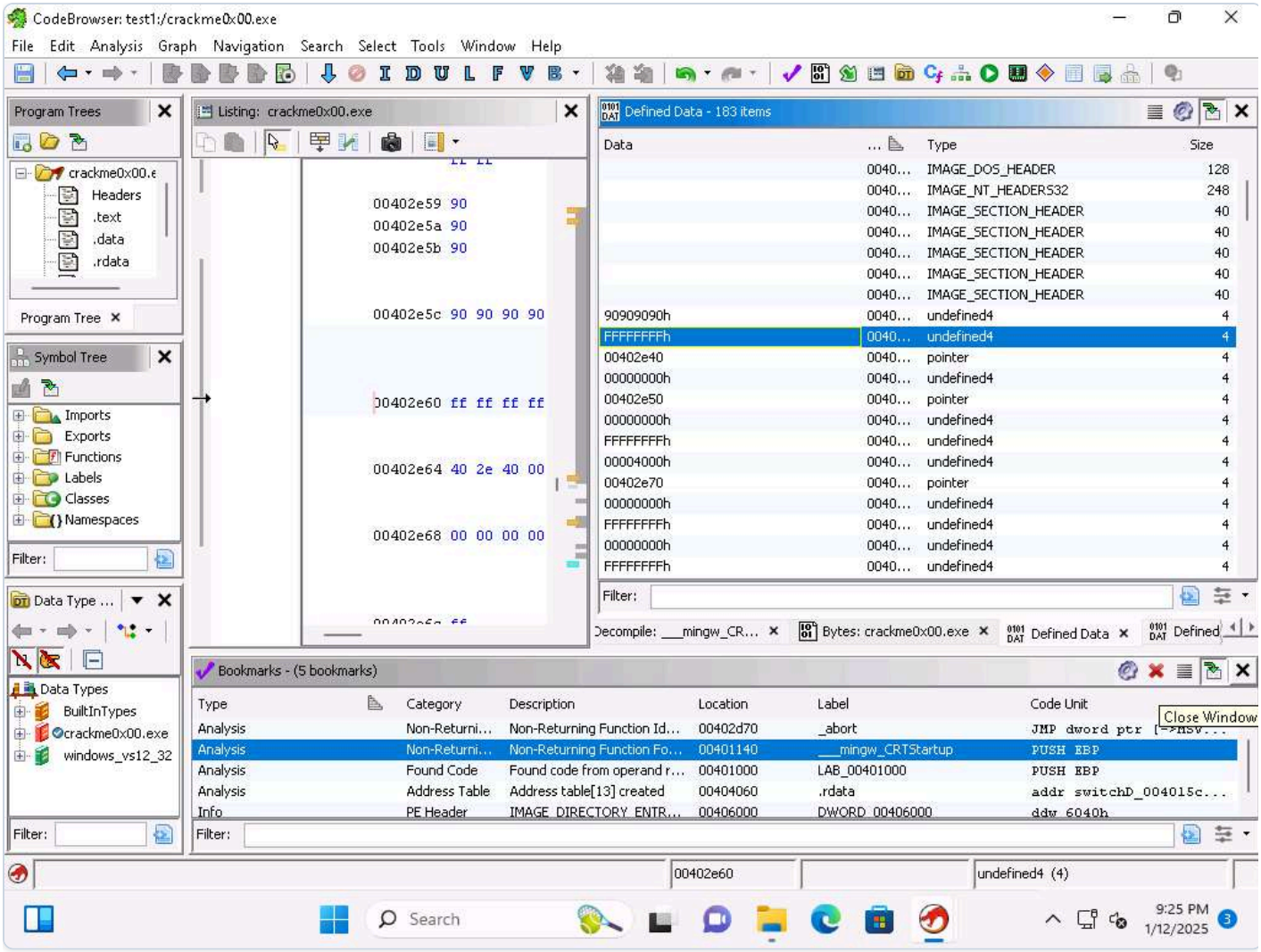



1. Re-open Decompiler and Defined Data views.

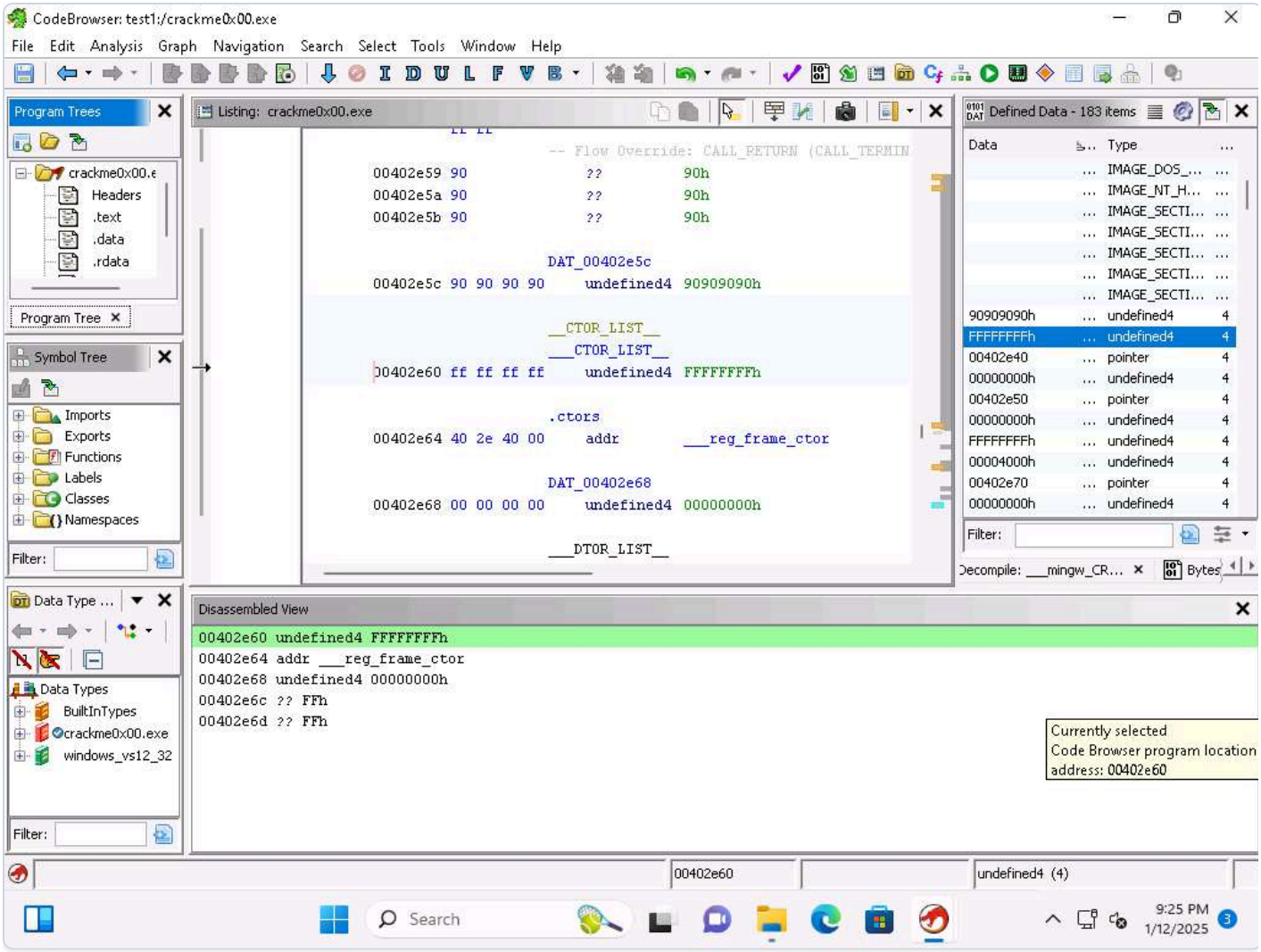

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
undefined4
undefined4
undefined4
undefined4
undefined4
00401140 55
00401141 89 e5
00401143 53
00401144 83 ec 24
00401147 c7 04 24
00 10 40 00
0040114e e8 4d 1c
00 00
00401153 83 ec 04
00401156 e8 75 02
Decompile: ___mingw_CRTStartup - (crackme0x00.exe)
void ___mingw_CRTStartup(void)
{
int *piVar1;
undefined4 *puVar2;
UINT uExitCode;
char **local_10;
_startupinfo local_c;
_SetUnhandledExceptionFilter@4((LPTOP_LEVEL_EXCEPTION_FILTER)&LAB_0040
__fpreset();
local_c.newmode = 0;
___getmainargs(&__argc,&DAT_00405000,&local_10,DAT_00403000,&local_c);
if (__CRT_fmode != 0) {
__fmode = __CRT_fmode;
if (_iob_exref != (code *)0x0) {
_setmode(*(int *)(_iob_exref + 0x10),__CRT_fmode);
}
if (_iob_exref != (code *)0xfffffffe0) {
_setmode(*(int *)(_iob_exref + 0x30),__CRT_fmode);
Decompile: ___mingw_CR...
Bytes: crackme0x00.exe
Defined Data
Defined
Bookmarks - (5 bookmarks)
Type Category Description Location Label Code Unit
Analysis Non-Returni... Non-Returning Function Id... 00402d70 _abort JMP dword ptr [->MSV...
Analysis Non-Returni... Non-Returning Function Fo... 00401140 ___mingw_CRTStartup PUSH EBP
Analysis Found Code Found code from operand r... 00401000 LAB_00401000 PUSH EBP
Analysis Address Table Address table[13] created 00404060 .rdata addr switchD_004015c...
Info PE Header IMAGE_DIRECTORY_ENTR... 00406000 DWORD_00406000 ddw
Filter:
PUSH EBP
00401140
___mingw_CRTStartup
PUSH EBP
Search
9:24 PM
1/12/2025



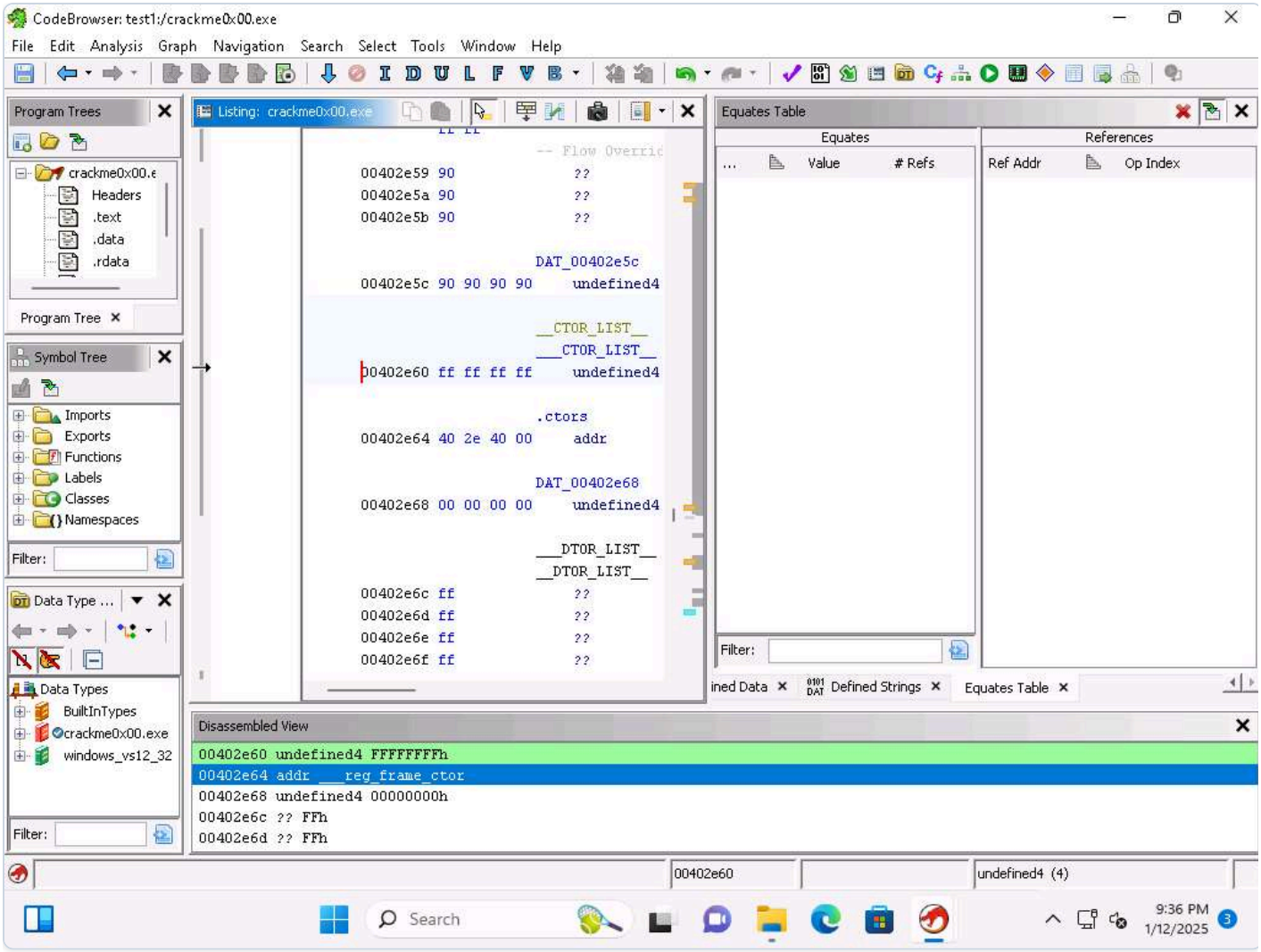

1. Open Defined Strings (**Window -> Defined Strings**) and review embedded literals.

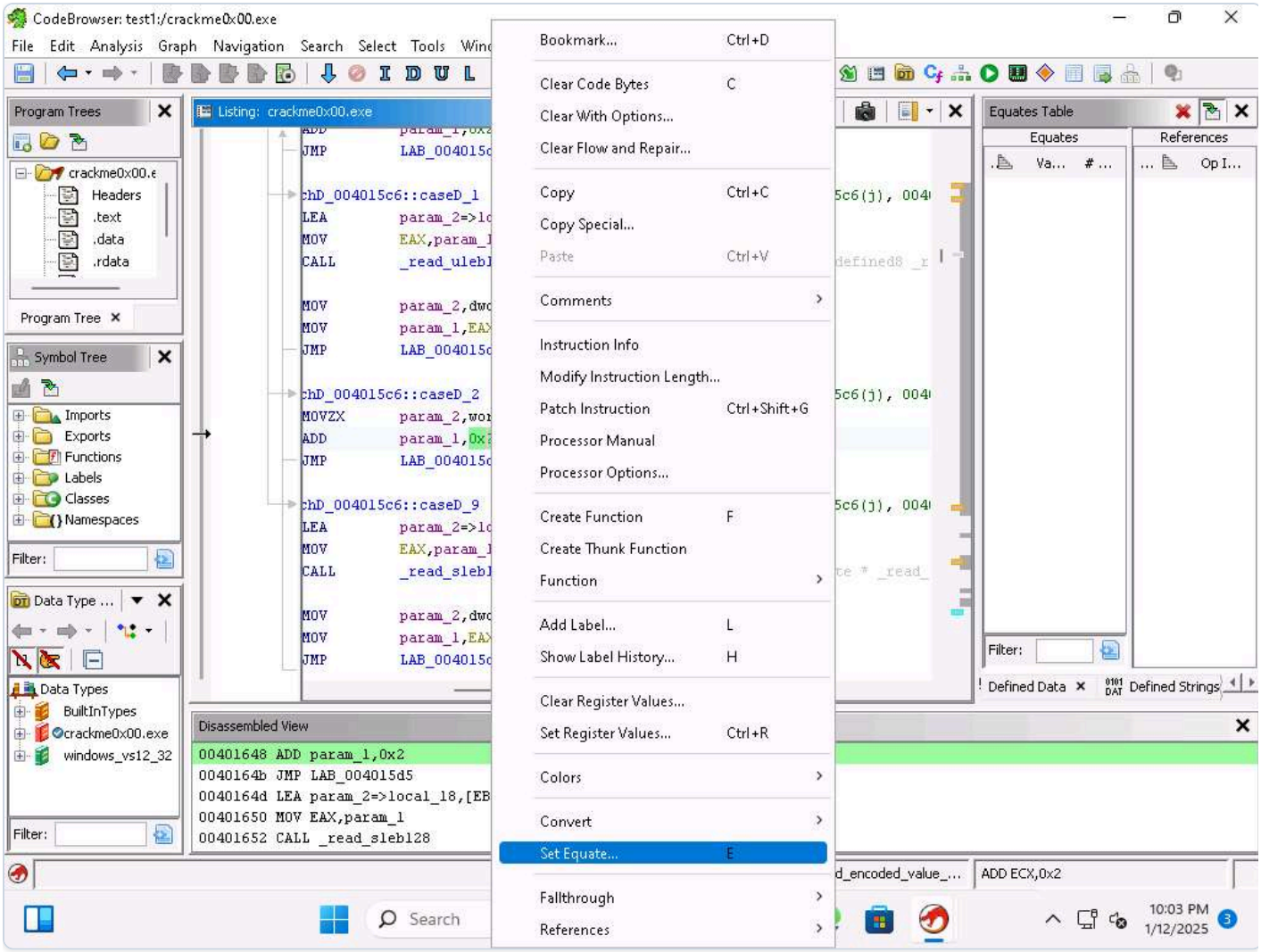


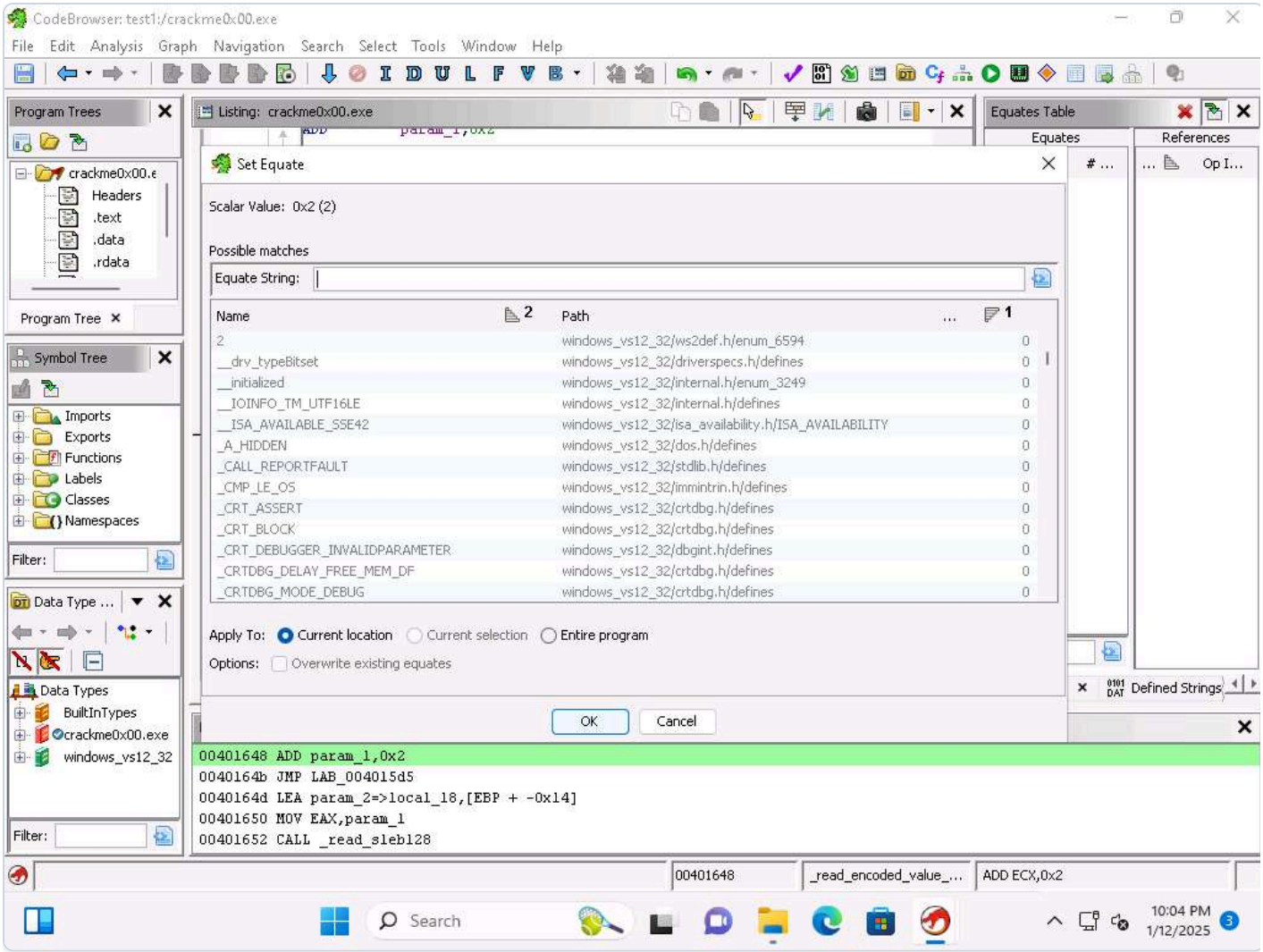

1. Open Disassembled View (**Window -> Disassembled View**).

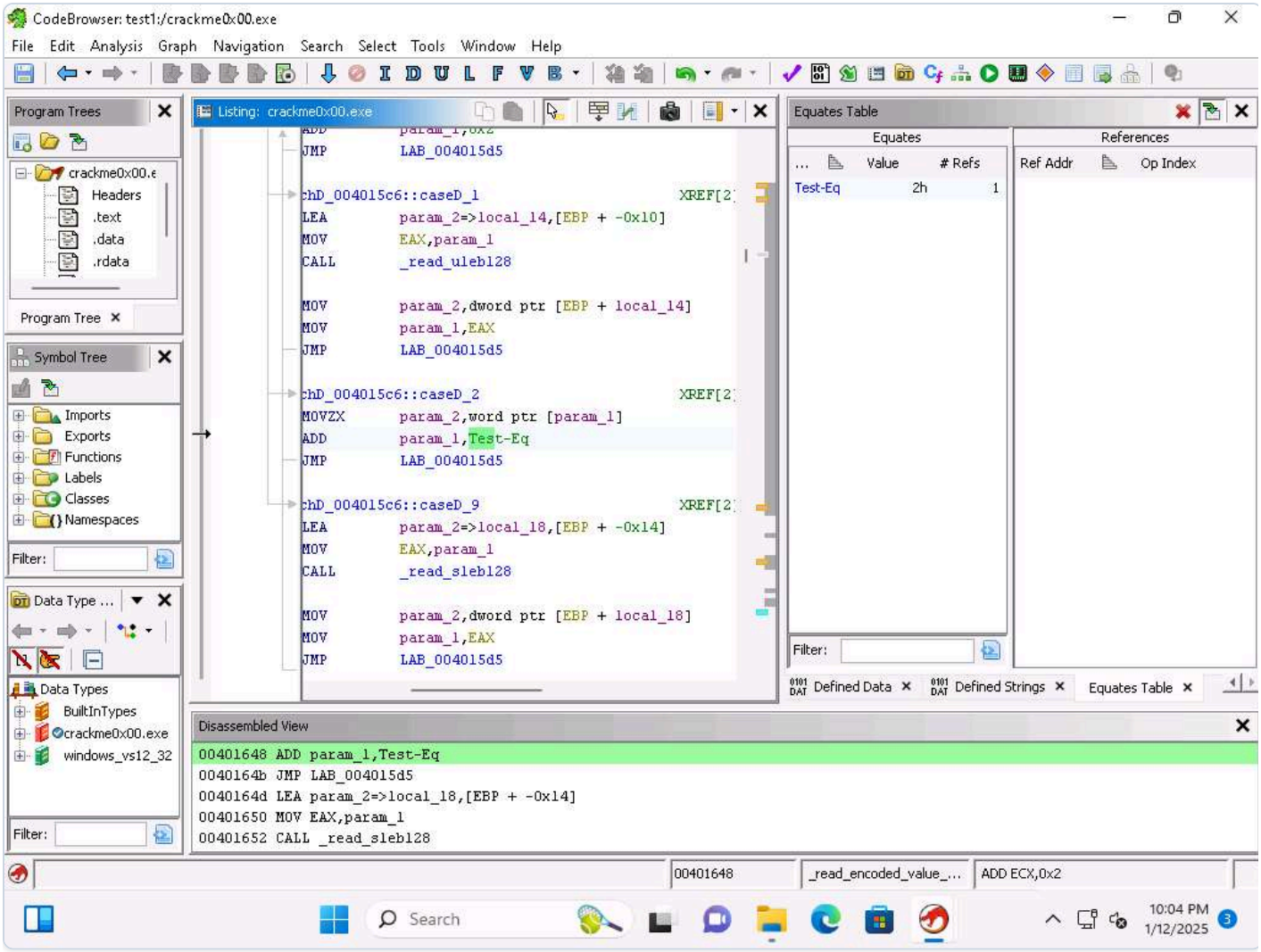



1. Open Equates Table (**Window -> Equates Table**), navigate to address `00402340`, right-click `0x2`, and set equate to `Test-Eq`.

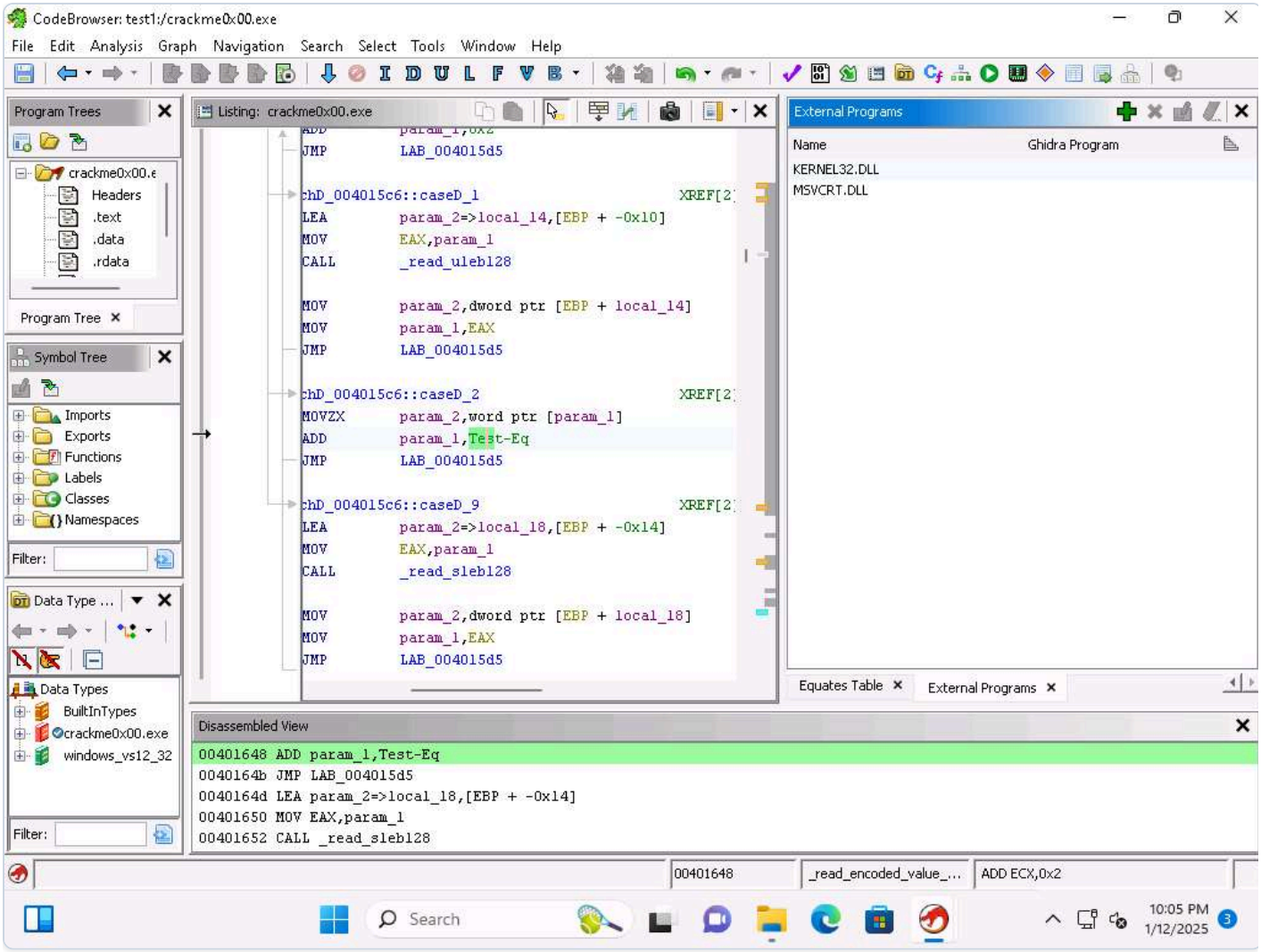

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
-- Flow Overric
00402e59 90 ??
00402e5a 90 ??
00402e5b 90 ??
DAT_00402e5c
00402e5c 90 90 90 90 undefined4
__CTOR_LIST__
___CTOR_LIST__
00402e60 ff ff ff ff undefined4
.ctors
00402e64 40 2e 40 00 addr
DAT_00402e68
00402e68 00 00 00 00 undefined4
___DTOR_LIST__
__DTOR_LIST__
00402e6c ff ??
00402e6d ff ??
00402e6e ff ??
00402e6f ff ??
Equates Table
Equates
... Value # Refs
References
Ref Addr Op Index
Filter:
ined Data
Defined Strings
Equates Table
Disassembled View
00402e60 undefined4 FFFFFFFFh
00402e64 addr ___reg_frame_ctor
00402e68 undefined4 00000000h
00402e6c ?? FFh
00402e6d ?? FFh
00402e60
undefined4 (4)
Search
9:36 PM
1/12/2025


CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
JMP LAB_004015c
chD_004015c6::caseD_1
LEA param_2=>lc
MOV EAX,param_1
CALL _read_ulebl
MOV param_2,dwc
MOV param_1,EAX
JMP LAB_004015c
chD_004015c6::caseD_2
MOVZX param_2,wor
ADD param_1,0x2
JMP LAB_004015c
chD_004015c6::caseD_9
LEA param_2=>lc
MOV EAX,param_1
CALL _read_slebl
MOV param_2,dwc
MOV param_1,EAX
JMP LAB_004015c
Disassembled View
00401648 ADD param_1,0x2
0040164b JMP LAB_004015d5
0040164d LEA param_2=>local_18,[EB
00401650 MOV EAX,param_1
00401652 CALL _read_sleb128
Bookmark... Ctrl+D
Clear Code Bytes C
Clear With Options...
Clear Flow and Repair...
Copy Ctrl+C
Copy Special...
Paste Ctrl+V
Comments
Instruction Info
Modify Instruction Length...
Patch Instruction Ctrl+Shift+G
Processor Manual
Processor Options...
Create Function F
Create Thunk Function
Function
Add Label... L
Show Label History... H
Clear Register Values...
Set Register Values... Ctrl+R
Colors
Convert
Set Equate... E
Fallthrough
References
Equates Table
Equates
References
Va... # ...
Op I...
Filter:
Defined Data
Defined Strings
d_encoded_value_...
ADD ECX,0x2
Search
10:03 PM
1/12/2025

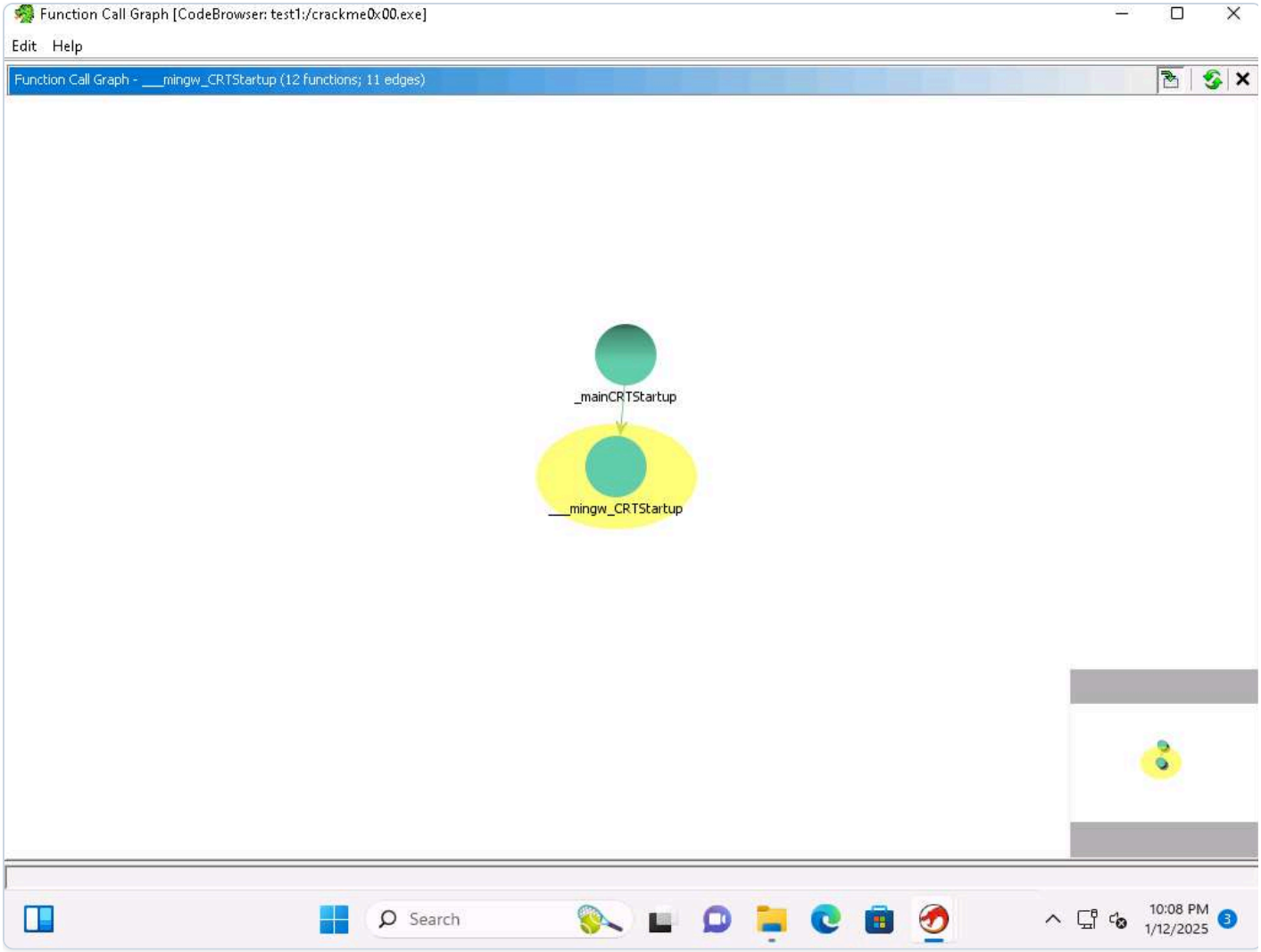

CodeBrowser: test1:/crackme0x00.exe
File Edit Analysis Graph Navigation Search Select Tools Window Help
Program Trees
crackme0x00.e
Headers
.text
.data
.rdata
Program Tree
Symbol Tree
Imports
Exports
Functions
Labels
Classes
Namespaces
Filter:
Data Type ...
Data Types
BuiltInTypes
crackme0x00.exe
windows_vs12_32
Filter:
Listing: crackme0x00.exe
Set Equate
Scalar Value: 0x2 (2)
Possible matches
Equate String:
Name
Path
2
windows_vs12_32/ws2def.h/enum_6594
__drv_typeBitset
windows_vs12_32/driverspecs.h/defines
__initialized
windows_vs12_32/internal.h/enum_3249
__IOINFO_TM_UTF16LE
windows_vs12_32/internal.h/defines
__ISA_AVAILABLE_SSE42
windows_vs12_32/isa_availability.h/ISA_AVAILABILITY
_A_HIDDEN
windows_vs12_32/dos.h/defines
_CALL_REPORTFAULT
windows_vs12_32/stdlib.h/defines
_CMP_LE_OS
windows_vs12_32/immintrin.h/defines
_CRT_ASSERT
windows_vs12_32/crtdbg.h/defines
_CRT_BLOCK
windows_vs12_32/crtdbg.h/defines
_CRT_DEBUGGER_INVALIDPARAMETER
windows_vs12_32/dbgint.h/defines
_CRTDBG_DELAY_FREE_MEM_DF
windows_vs12_32/crtdbg.h/defines
_CRTDBG_MODE_DEBUG
windows_vs12_32/crtdbg.h/defines
Apply To: Current location Current selection Entire program
Options: Overwrite existing equates
OK
Cancel
Equates Table
Equates
References
Defined Strings
00401648 ADD param_1,0x2
0040164b JMP LAB_004015d5
0040164d LEA param_2=>local_18,[EBP + -0x14]
00401650 MOV EAX,param_1
00401652 CALL _read_sleb128
00401648
_read_encoded_value_...
ADD ECX,0x2
Search
10:04 PM
1/12/2025

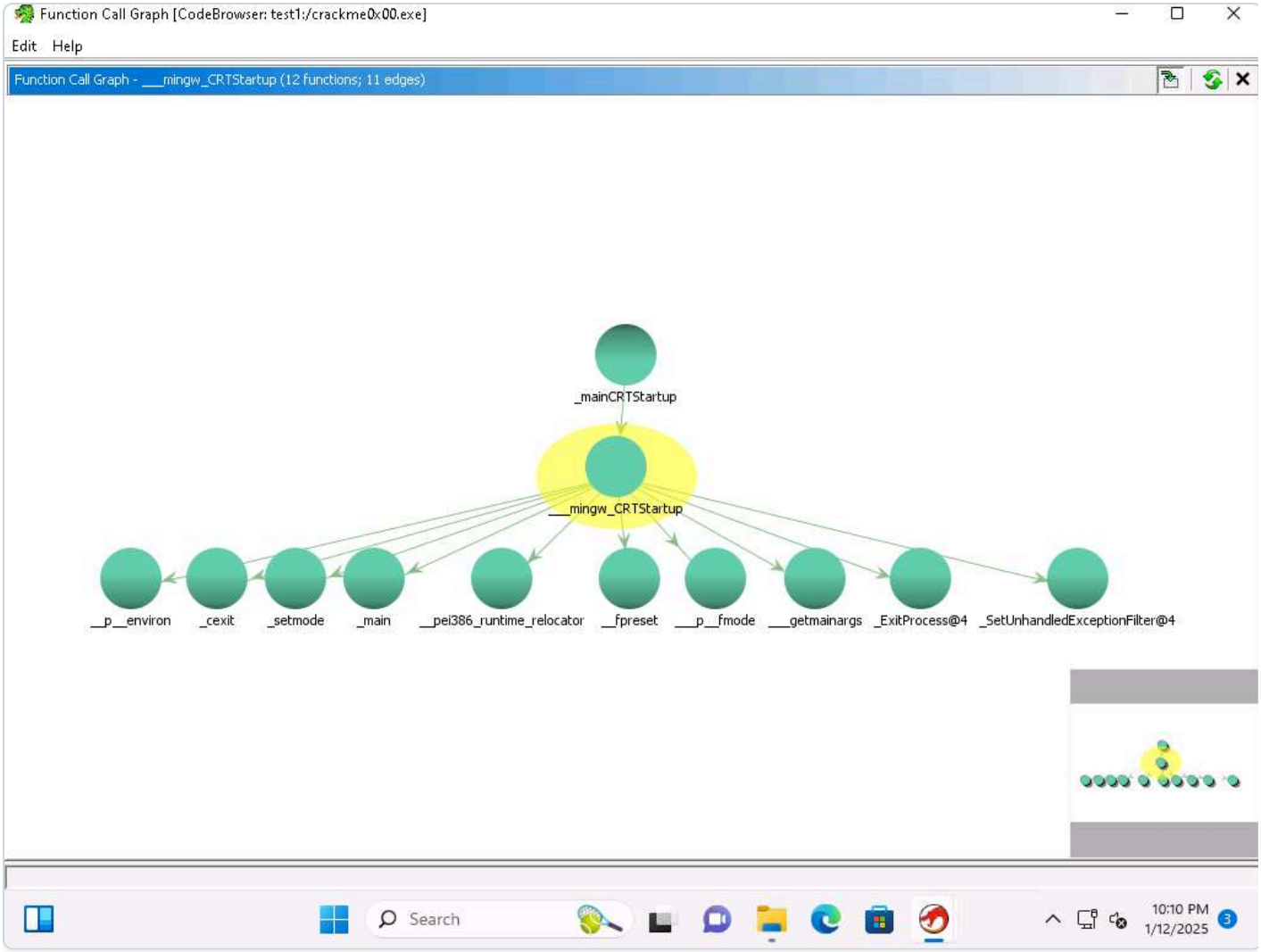



1. View External Programs window (**Window -> External Programs**).

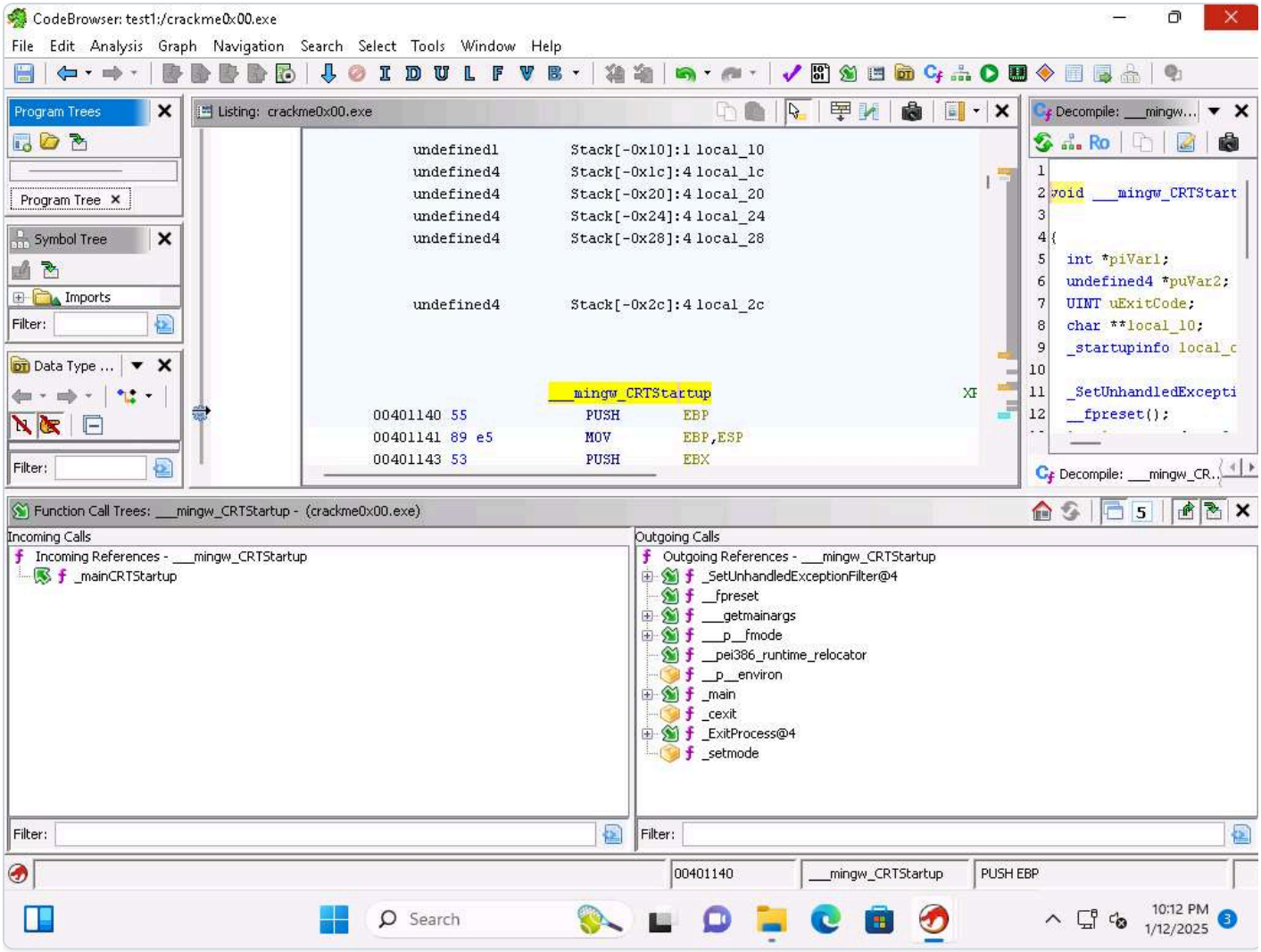



1. Inspect Function Call Graph (**Window -> Function Call Graph**).

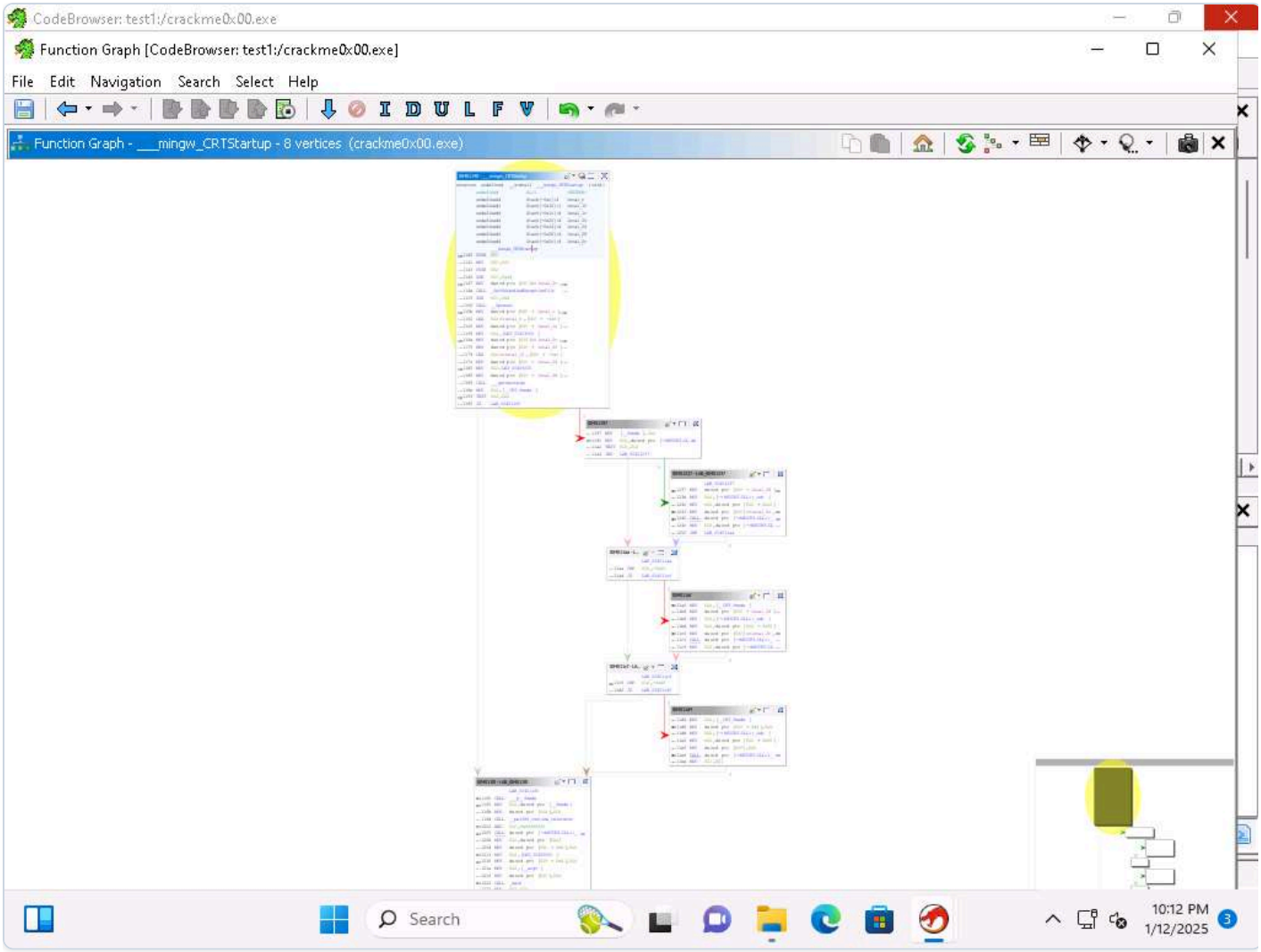

Function Call Graph [CodeBrowser: test1:/crackme0x00.exe]
Edit Help
Function Call Graph - ___mingw_CRTStartup (12 functions; 11 edges)
_mainCRTStartup
___mingw_CRTStartup
Search
10:08 PM
1/12/2025

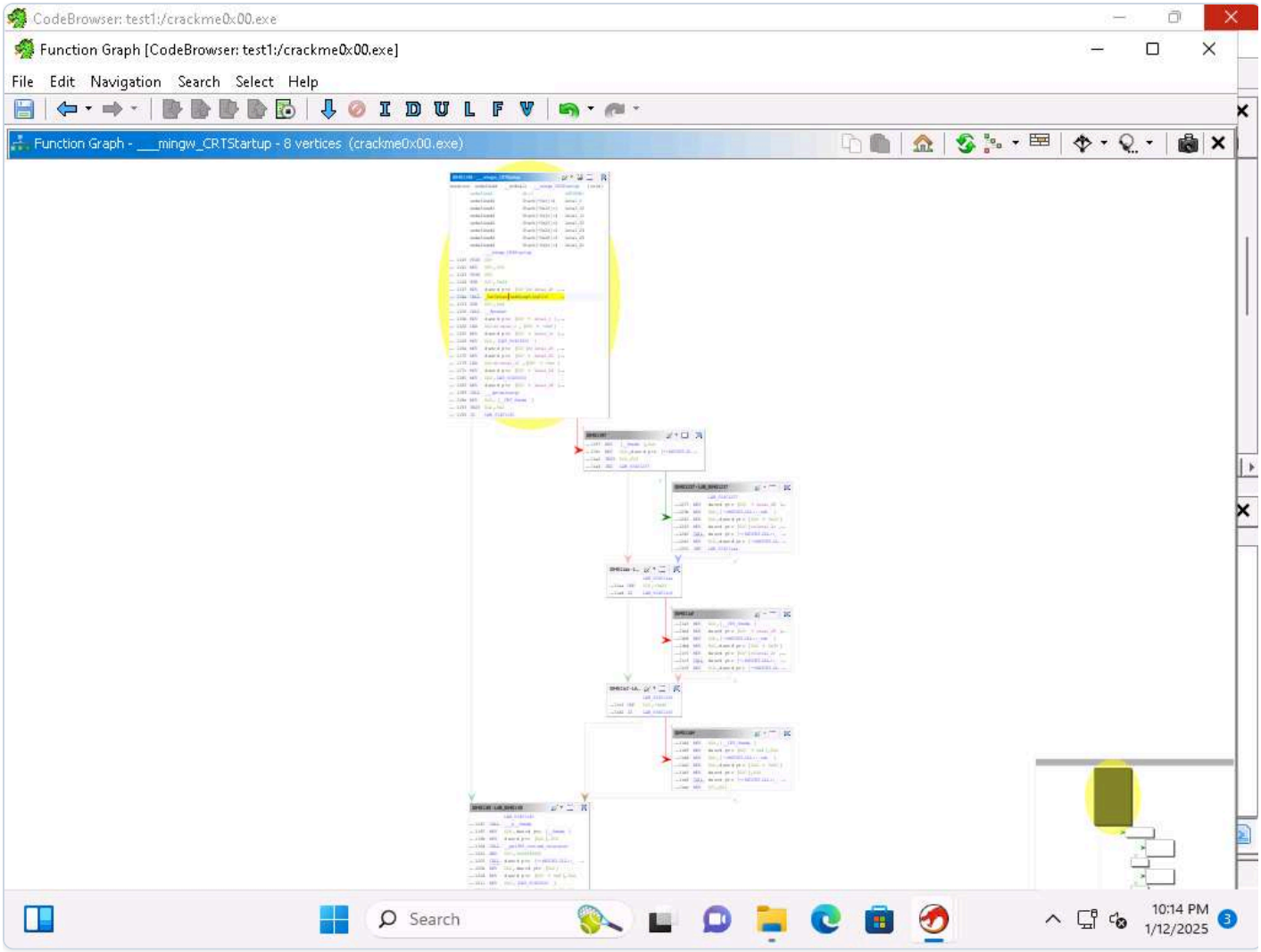



1. Inspect Function Call Tree (**Window -> Function Call Tree**).

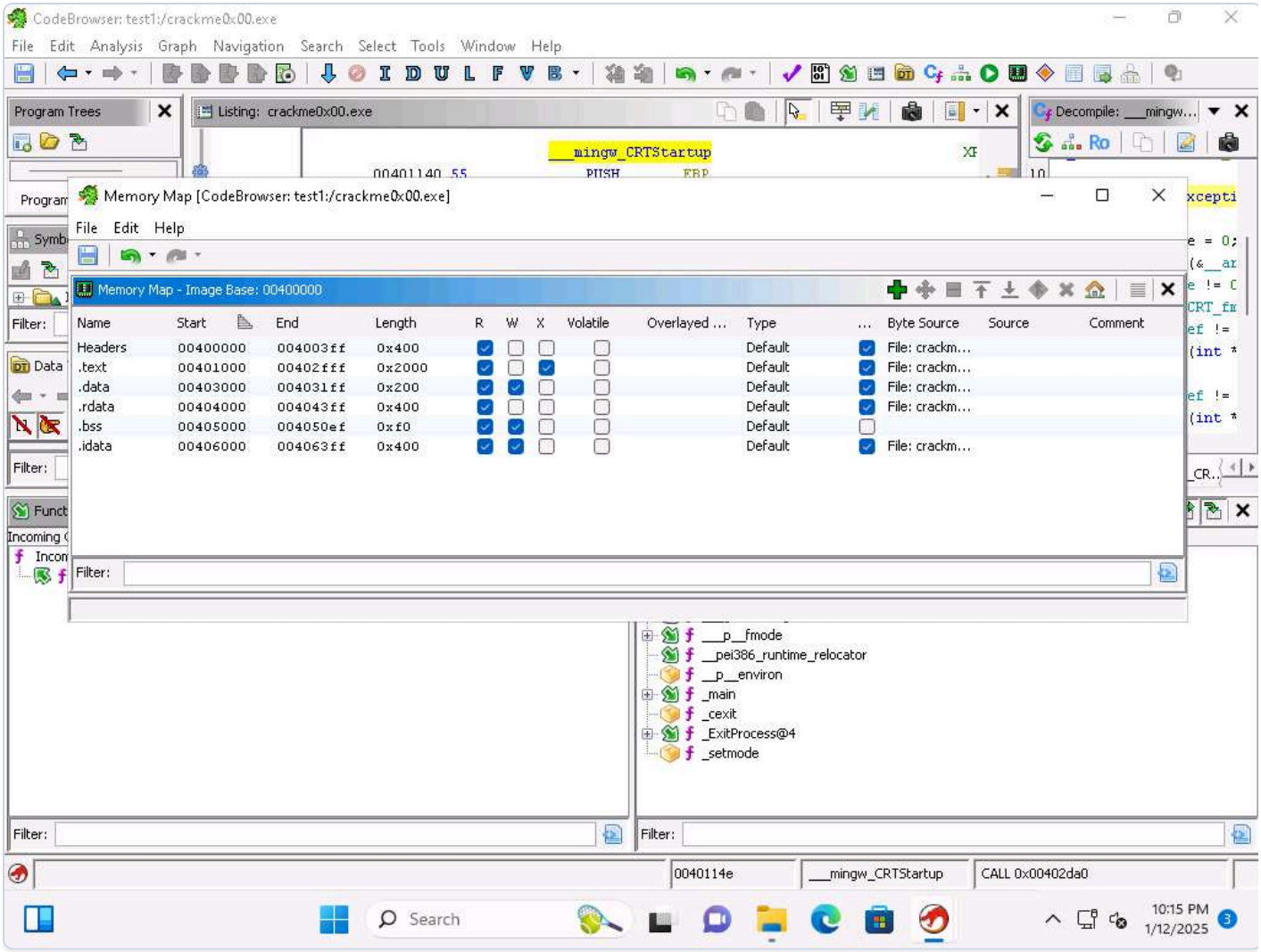



1. Open basic block Function Graph.

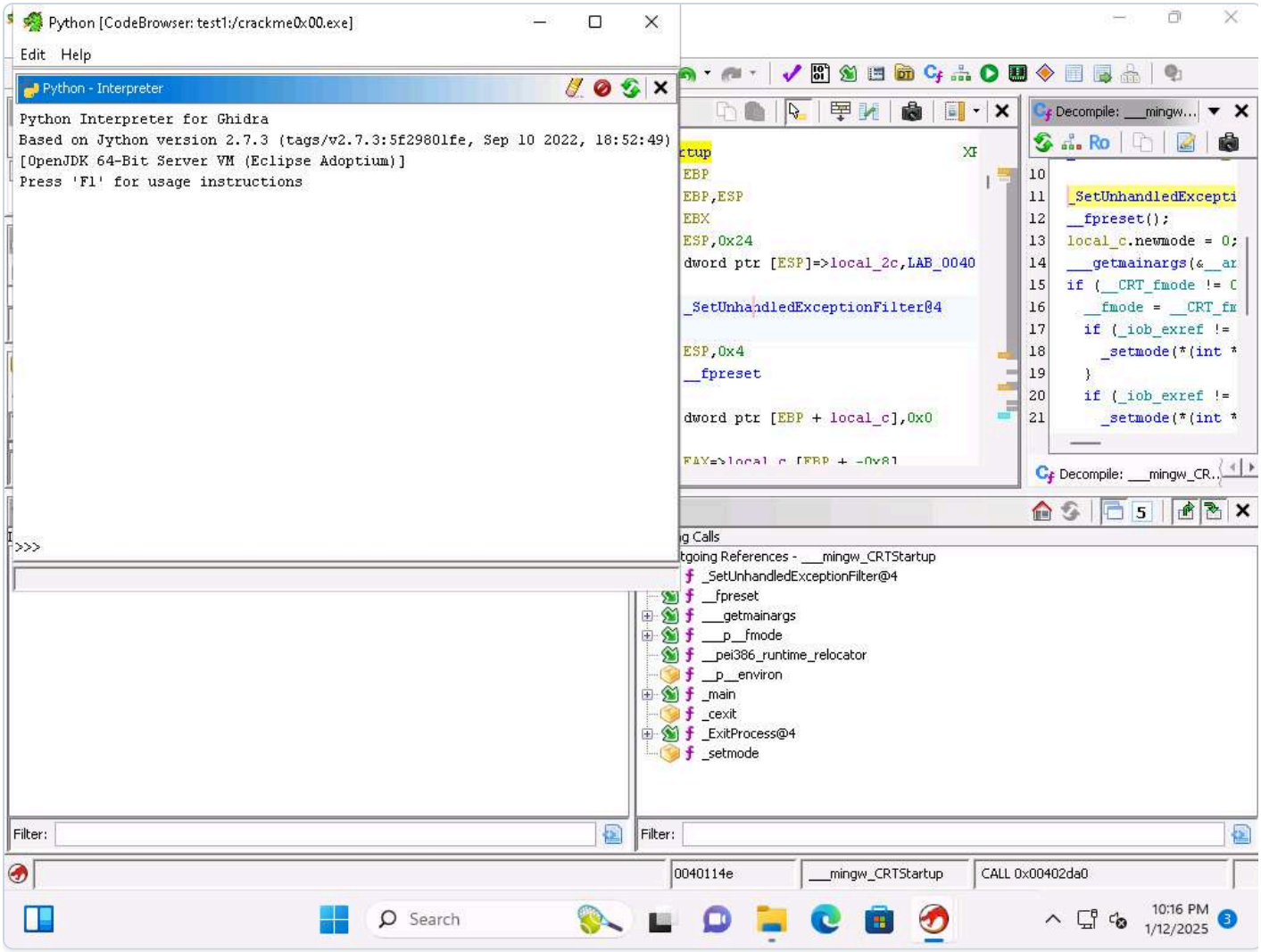

CodeBrowser: test1:/crackme0x00.exe
Function Graph [CodeBrowser: test1:/crackme0x00.exe]
File Edit Navigation Search Select Help
Function Graph - ___mingw_CRTStartup - 8 vertices (crackme0x00.exe)
Search
10:12 PM
1/12/2025



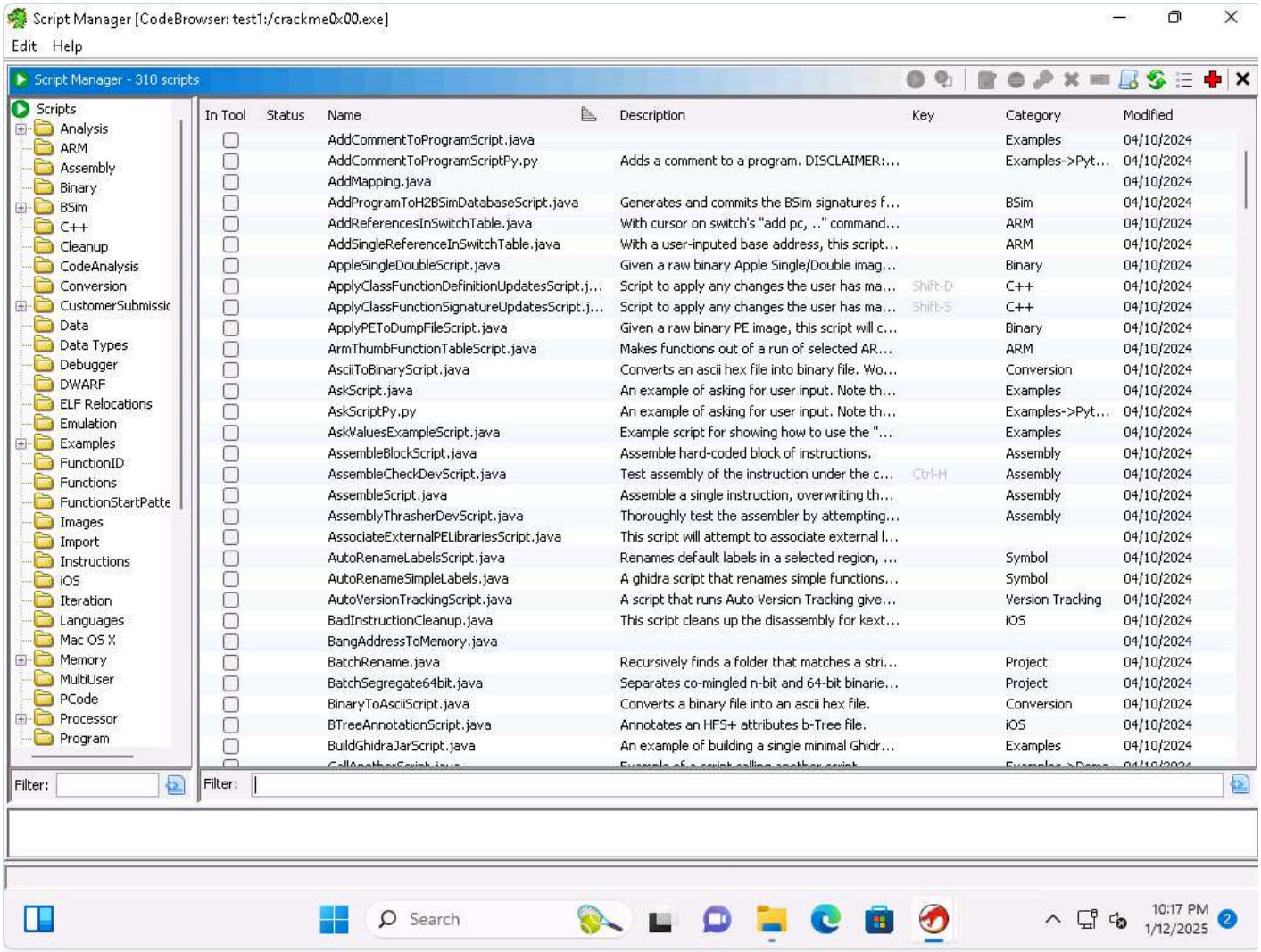

1. Inspect Memory Map window (**Window -> Memory Map**).

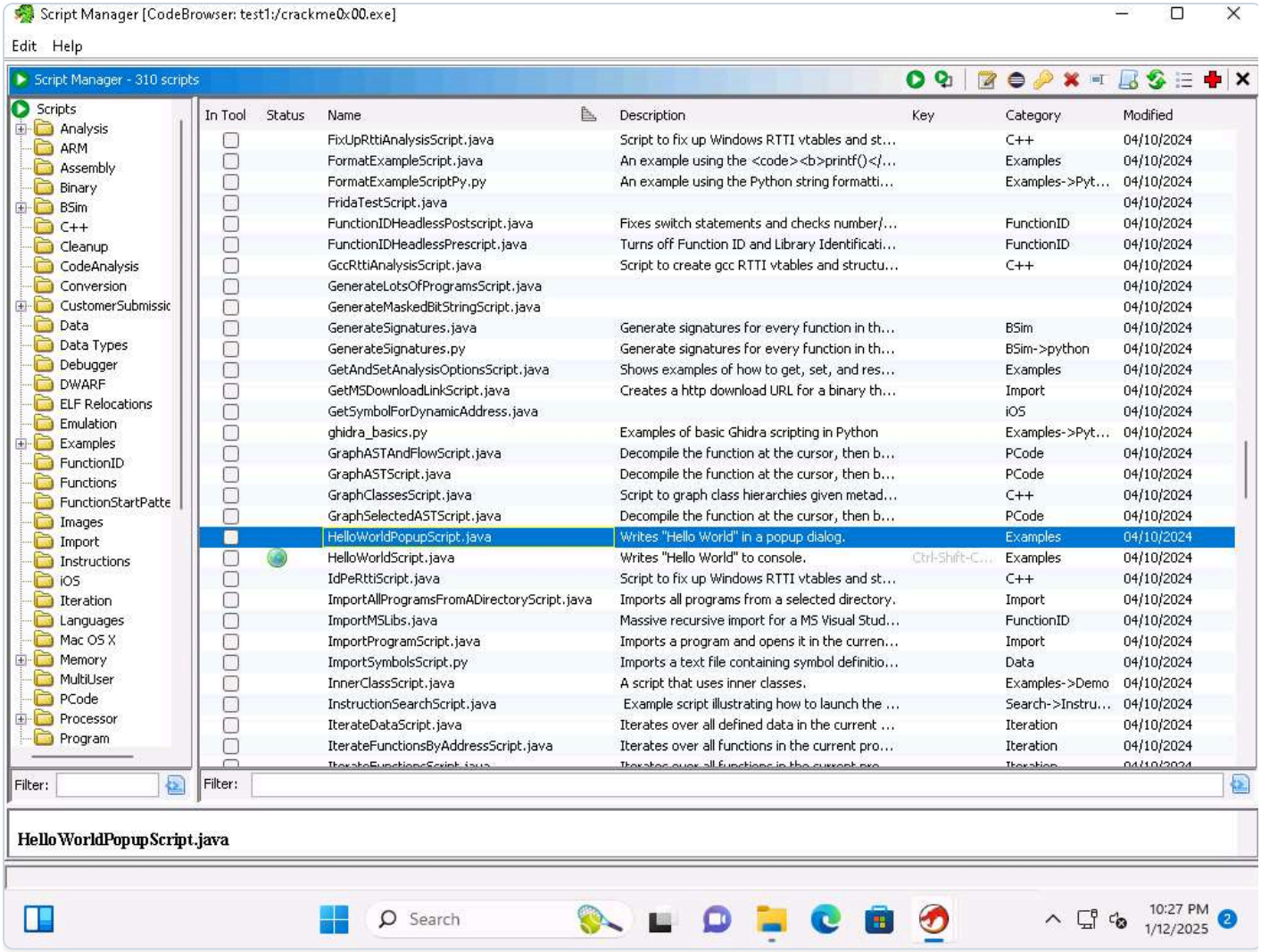



1. Open integrated Python interpreter (**Window -> Python**).

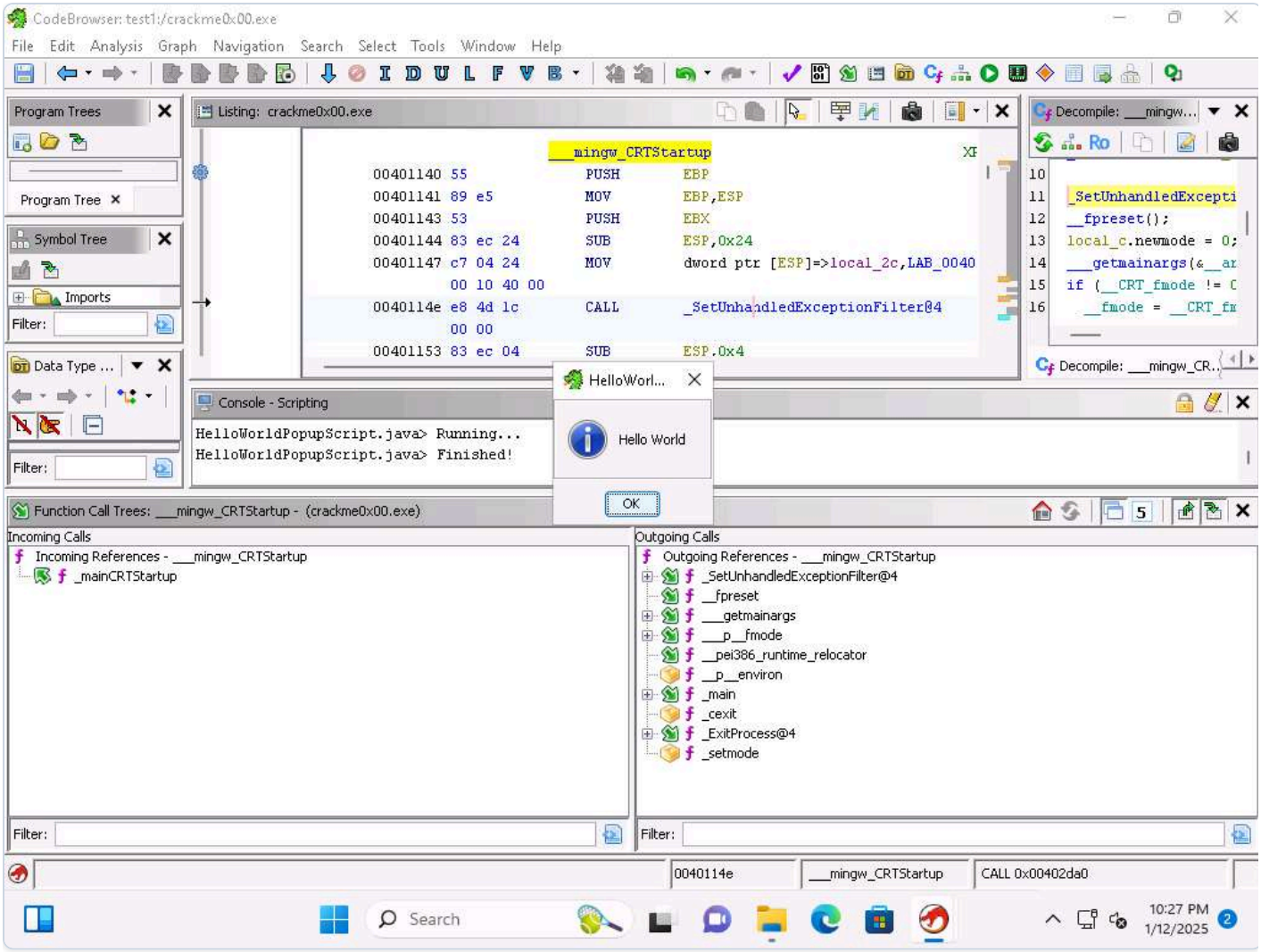



1. Open Script Manager (**Window -> Script Manager**) and execute `Hello World` script.

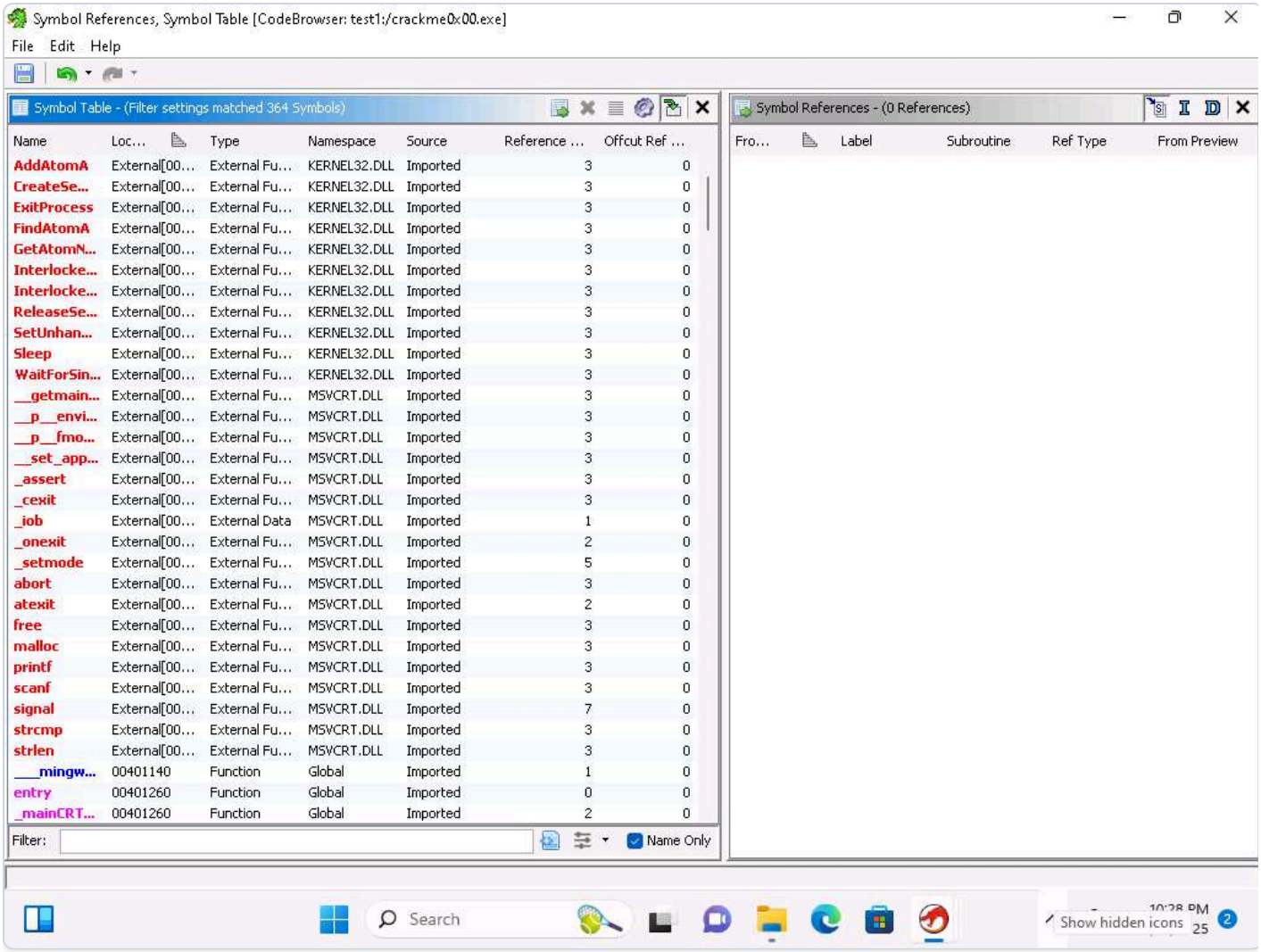

Script Manager [CodeBrowser: test1:/crackme0x00.exe]
Edit Help
Script Manager - 310 scripts
Scripts
Analysis
ARM
Assembly
Binary
BSim
C++
Cleanup
CodeAnalysis
Conversion
CustomerSubmissic
Data
Data Types
Debugger
DWARF
ELF Relocations
Emulation
Examples
FunctionID
Functions
FunctionStartPatte
Images
Import
Instructions
iOS
Iteration
Languages
Mac OS X
Memory
MultiUser
PCode
Processor
Program
In Tool
Status
Name
Description
Key
Category
Modified
AddCommentToProgramScript.java
Examples
04/10/2024
AddCommentToProgramScriptPy.py
Adds a comment to a program. DISCLAIMER:...
Examples->Pyt...
04/10/2024
AddMapping.java
04/10/2024
AddProgramToH2BSimDatabaseScript.java
Generates and commits the BSim signatures f...
BSim
04/10/2024
AddReferencesInSwitchTable.java
With cursor on switch's "add pc, .." command...
ARM
04/10/2024
AddSingleReferenceInSwitchTable.java
With a user-inputed base address, this script...
ARM
04/10/2024
AppleSingleDoubleScript.java
Given a raw binary Apple Single/Double imag...
Binary
04/10/2024
ApplyClassFunctionDefinitionUpdatesScript.j...
Script to apply any changes the user has ma...
Shift-D
C++
04/10/2024
ApplyClassFunctionSignatureUpdatesScript.j...
Script to apply any changes the user has ma...
Shift-S
C++
04/10/2024
ApplyPEToDumpFileScript.java
Given a raw binary PE image, this script will c...
Binary
04/10/2024
ArmThumbFunctionTableScript.java
Makes functions out of a run of selected AR...
ARM
04/10/2024
AsciiToBinaryScript.java
Converts an ascii hex file into binary file. Wo...
Conversion
04/10/2024
AskScript.java
An example of asking for user input. Note th...
Examples
04/10/2024
AskScriptPy.py
An example of asking for user input. Note th...
Examples->Pyt...
04/10/2024
AskValuesExampleScript.java
Example script for showing how to use the "...
Examples
04/10/2024
AssembleBlockScript.java
Assemble hard-coded block of instructions.
Assembly
04/10/2024
AssembleCheckDevScript.java
Test assembly of the instruction under the c...
Ctrl-H
Assembly
04/10/2024
AssembleScript.java
Assemble a single instruction, overwriting th...
Assembly
04/10/2024
AssemblyThrasherDevScript.java
Thoroughly test the assembler by attempting...
Assembly
04/10/2024
AssociateExternalPELibrariesScript.java
This script will attempt to associate external l...
04/10/2024
AutoRenameLabelsScript.java
Renames default labels in a selected region, ...
Symbol
04/10/2024
AutoRenameSimpleLabels.java
A ghidra script that renames simple functions...
Symbol
04/10/2024
AutoVersionTrackingScript.java
A script that runs Auto Version Tracking give...
Version Tracking
04/10/2024
BadInstructionCleanup.java
This script cleans up the disassembly for kext...
iOS
04/10/2024
BangAddressToMemory.java
04/10/2024
BatchRename.java
Recursively finds a folder that matches a stri...
Project
04/10/2024
BatchSegregate64bit.java
Separates co-mingled n-bit and 64-bit binarie...
Project
04/10/2024
BinaryToAsciiScript.java
Converts a binary file into an ascii hex file.
Conversion
04/10/2024
BTreeAnnotationScript.java
Annotates an HFS+ attributes b-Tree file.
iOS
04/10/2024
BuildGhidraJarScript.java
An example of building a single minimal Ghidr...
Examples
04/10/2024
Filter:
Filter:
Search
10:17 PM
1/12/2025
2



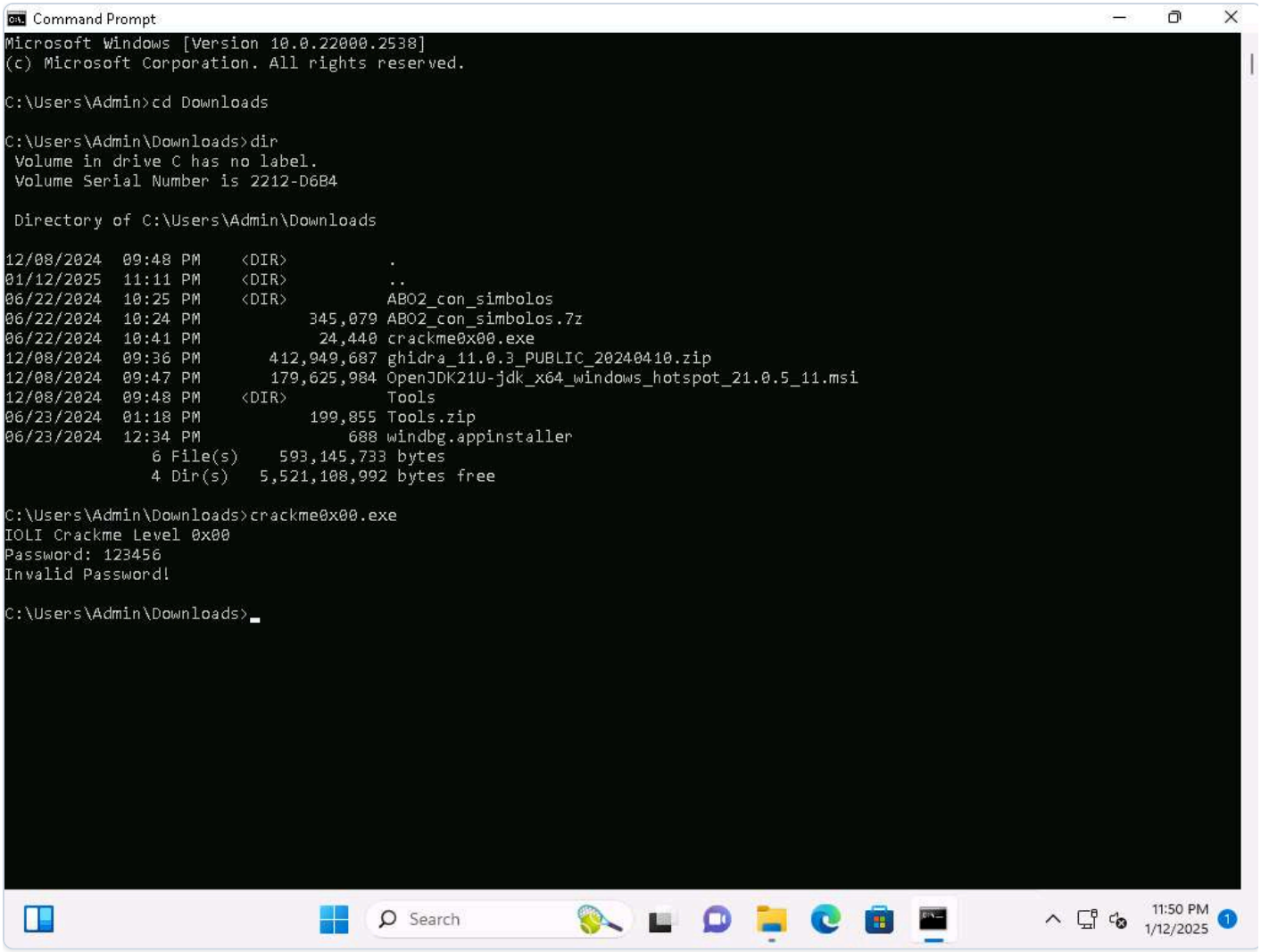

1. Open Symbol References window (**Window -> Symbol References**).

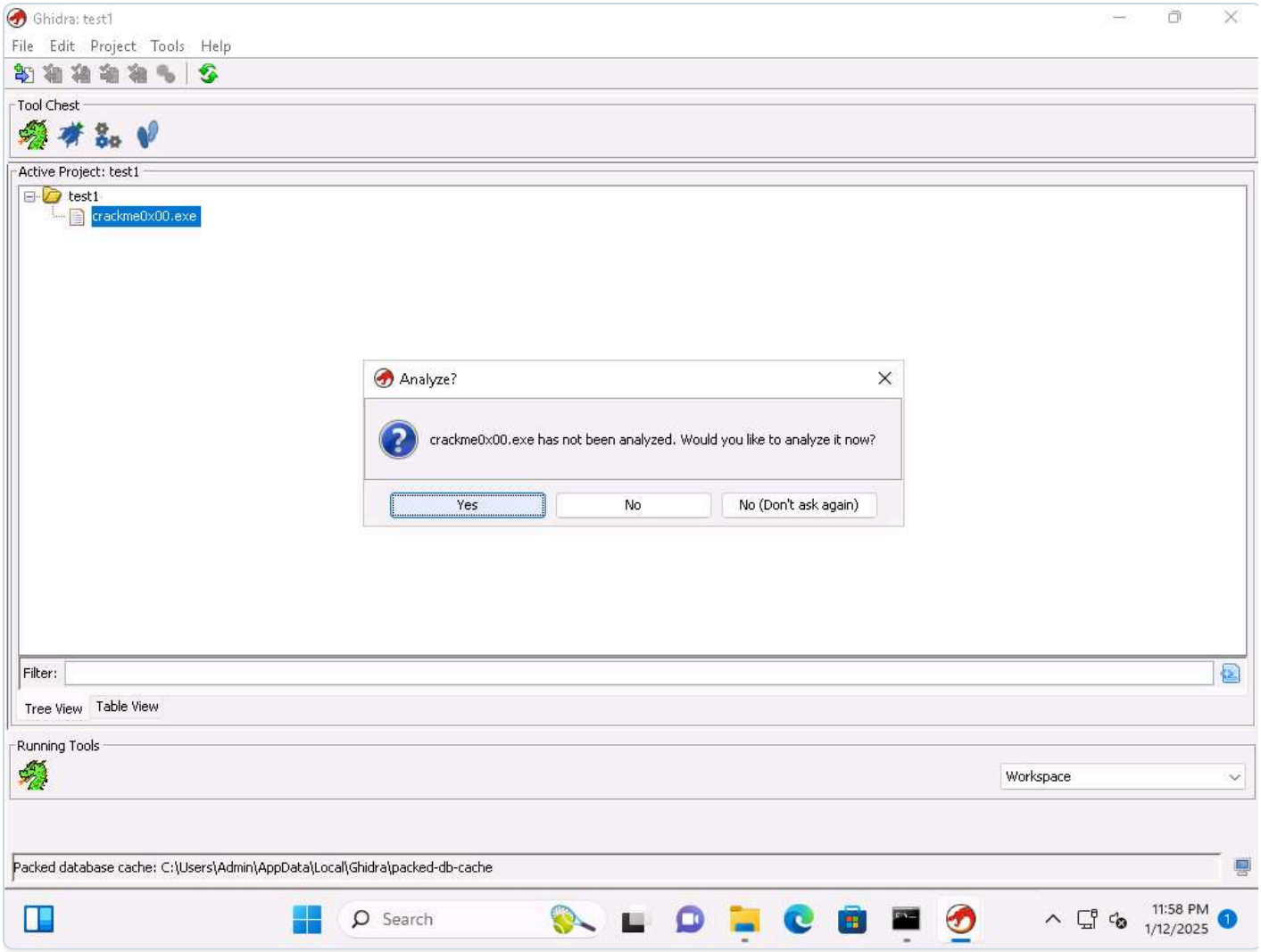



1. Open Command Prompt and test initial password `123456` against `crackme0x00.exe`:
   **`cd C:\Users\Admin\Downloads\Tools`**
   **`crackme0x00.exe`**

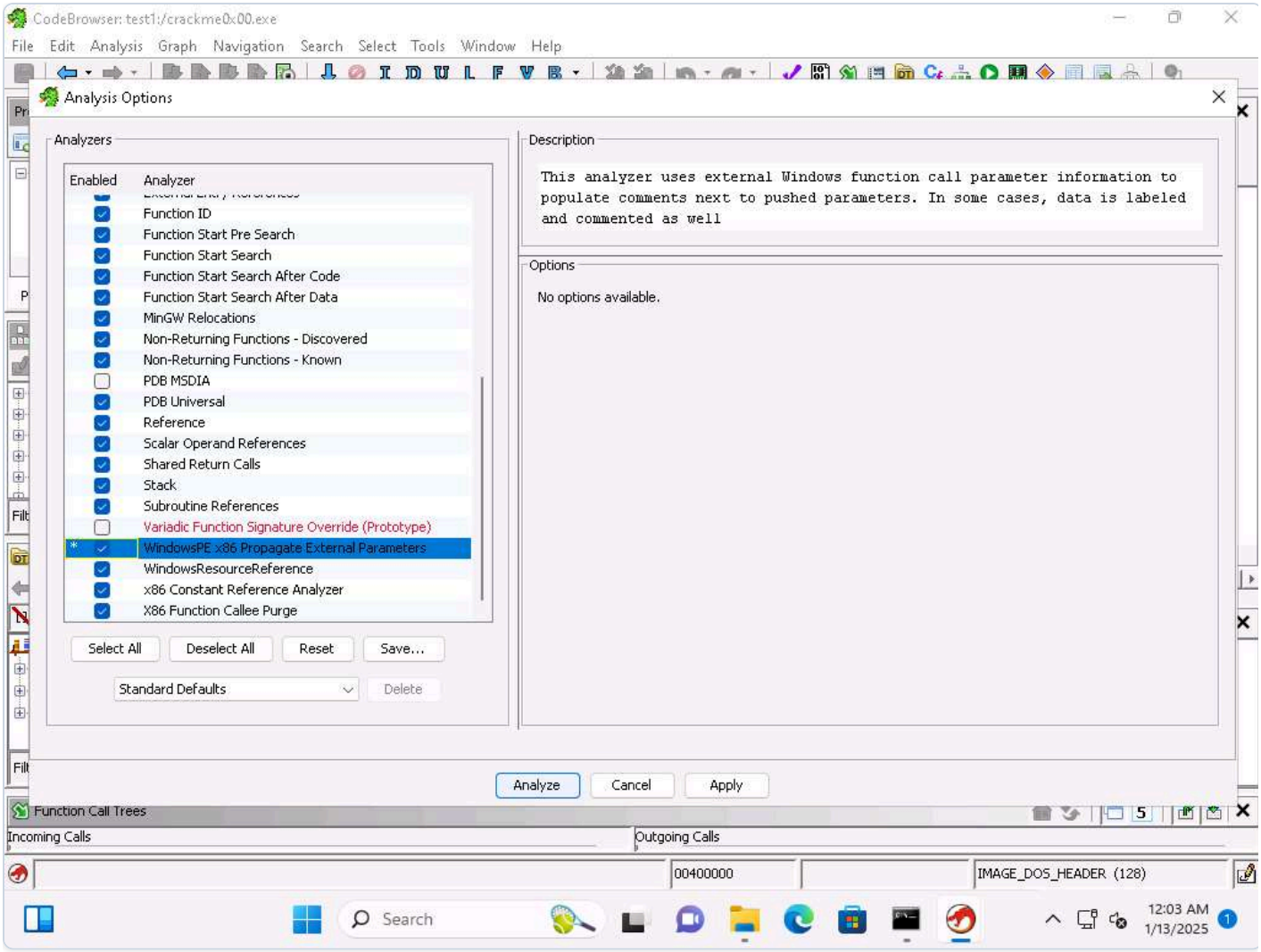



1. Re-import `crackme0x00.exe` into Ghidra and enable `WindowsPE x86 Propagate External Parameters` analyzer.

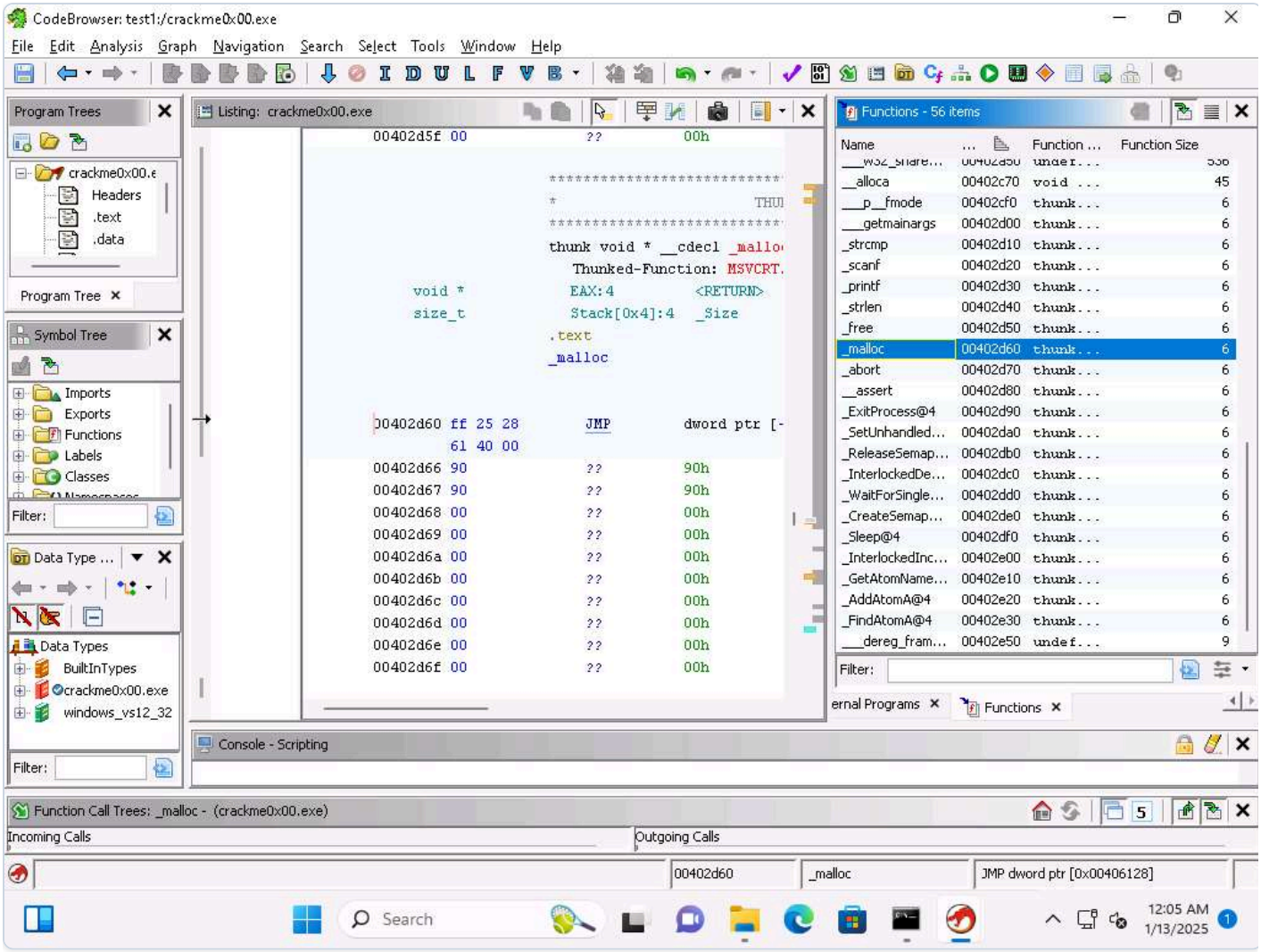

Command Prompt
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Admin>cd Downloads
C:\Users\Admin\Downloads>dir
Volume in drive C has no label.
Volume Serial Number is 2212-D6B4
Directory of C:\Users\Admin\Downloads
12/08/2024 09:48 PM <DIR> .
01/12/2025 11:11 PM <DIR> ..
06/22/2024 10:25 PM <DIR> ABO2_con_simbolos
06/22/2024 10:24 PM 345,079 ABO2_con_simbolos.7z
06/22/2024 10:41 PM 24,440 crackme0x00.exe
12/08/2024 09:36 PM 412,949,687 ghidra_11.0.3_PUBLIC_20240410.zip
12/08/2024 09:47 PM 179,625,984 OpenJDK21U-jdk_x64_windows_hotspot_21.0.5_11.msi
12/08/2024 09:48 PM <DIR> Tools
06/23/2024 01:18 PM 199,855 Tools.zip
06/23/2024 12:34 PM 688 windbg.appinstaller
6 File(s) 593,145,733 bytes
4 Dir(s) 5,521,108,992 bytes free
C:\Users\Admin\Downloads>crackme0x00.exe
IOLI Crackme Level 0x00
Password: 123456
Invalid Password!
C:\Users\Admin\Downloads>
Search
11:50 PM
1/12/2025



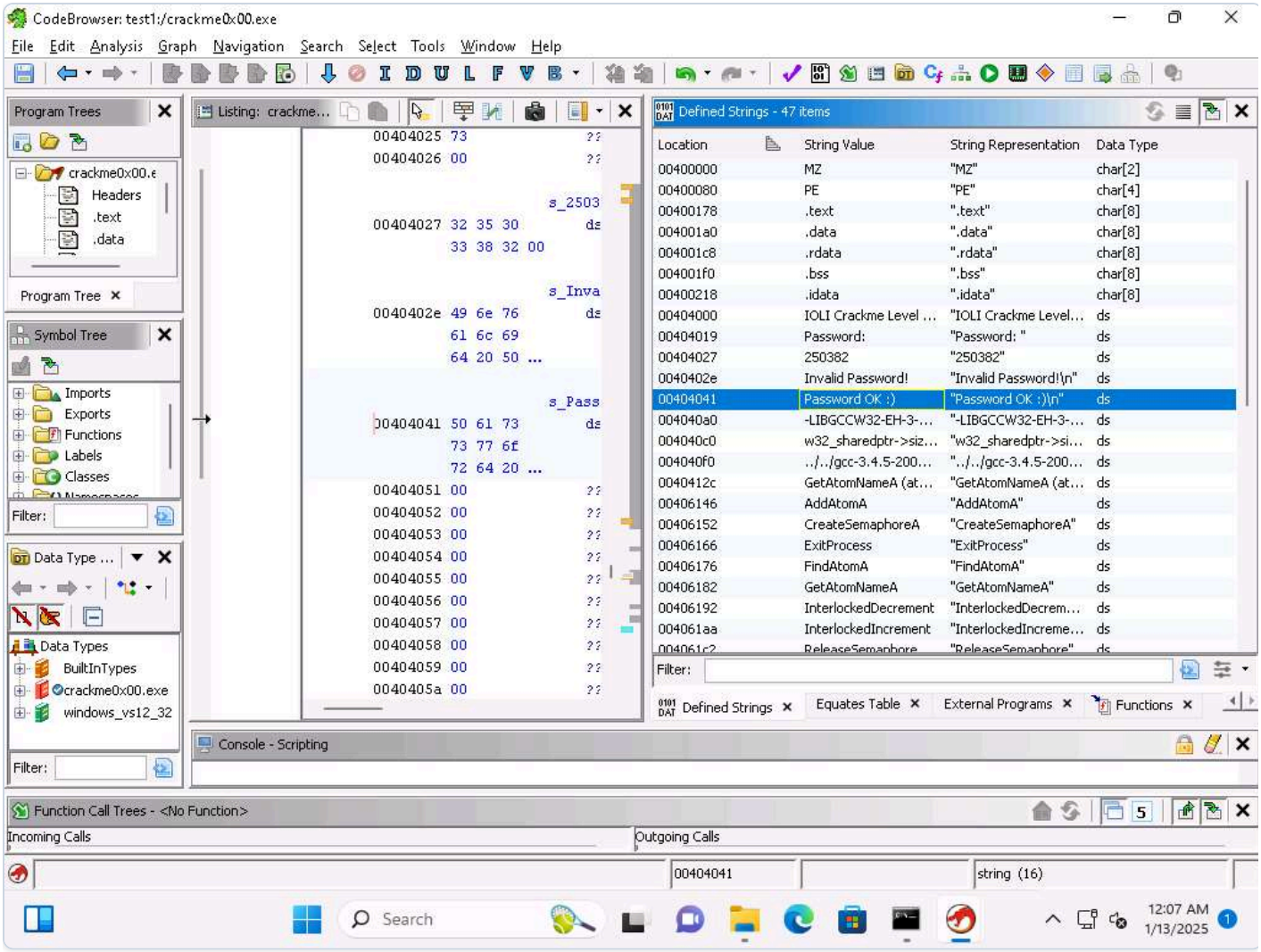

1. Inspect imported function listings including `strcmp` and `malloc`.



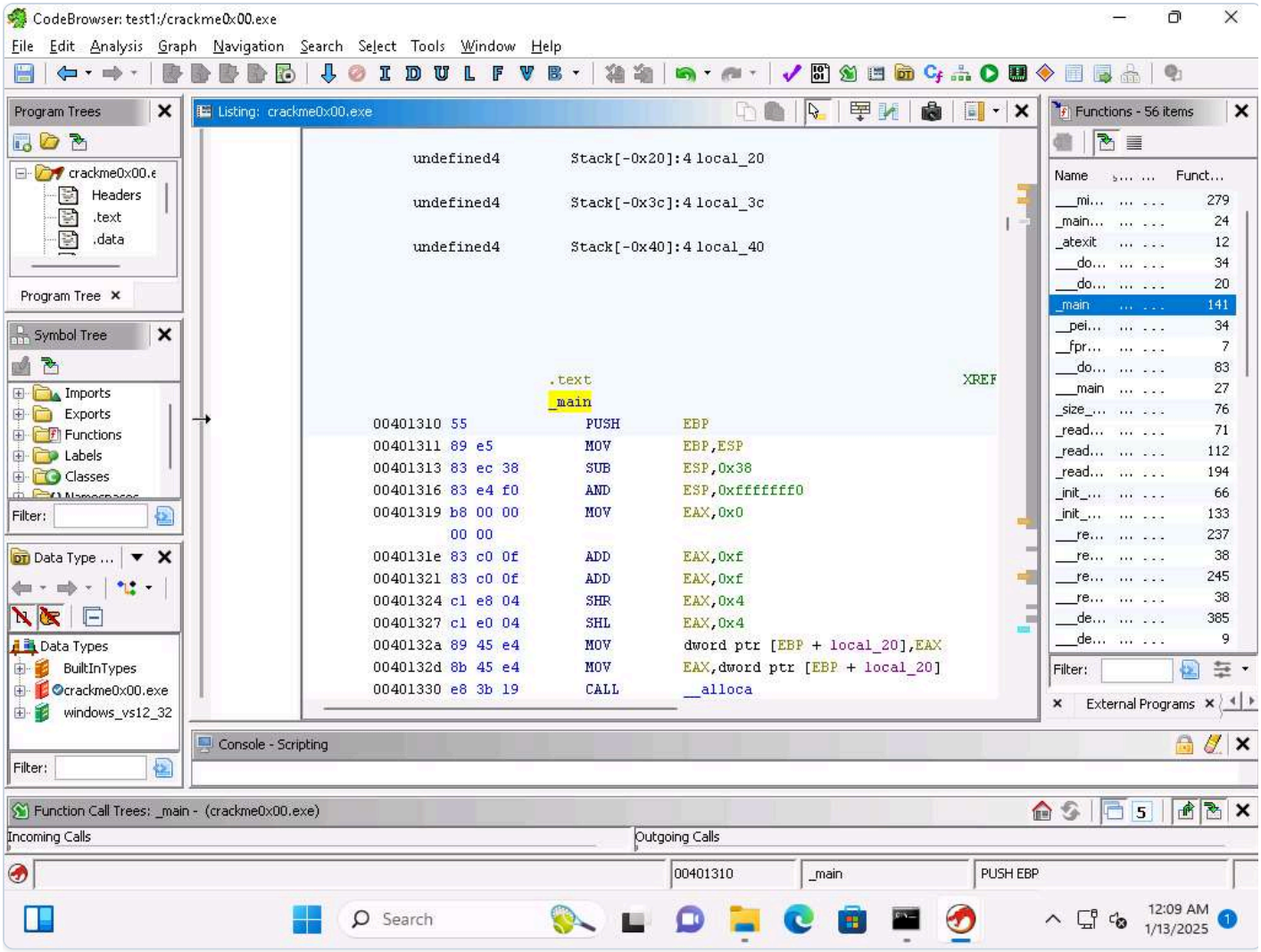

1. Open Defined Strings window and inspect password strings.



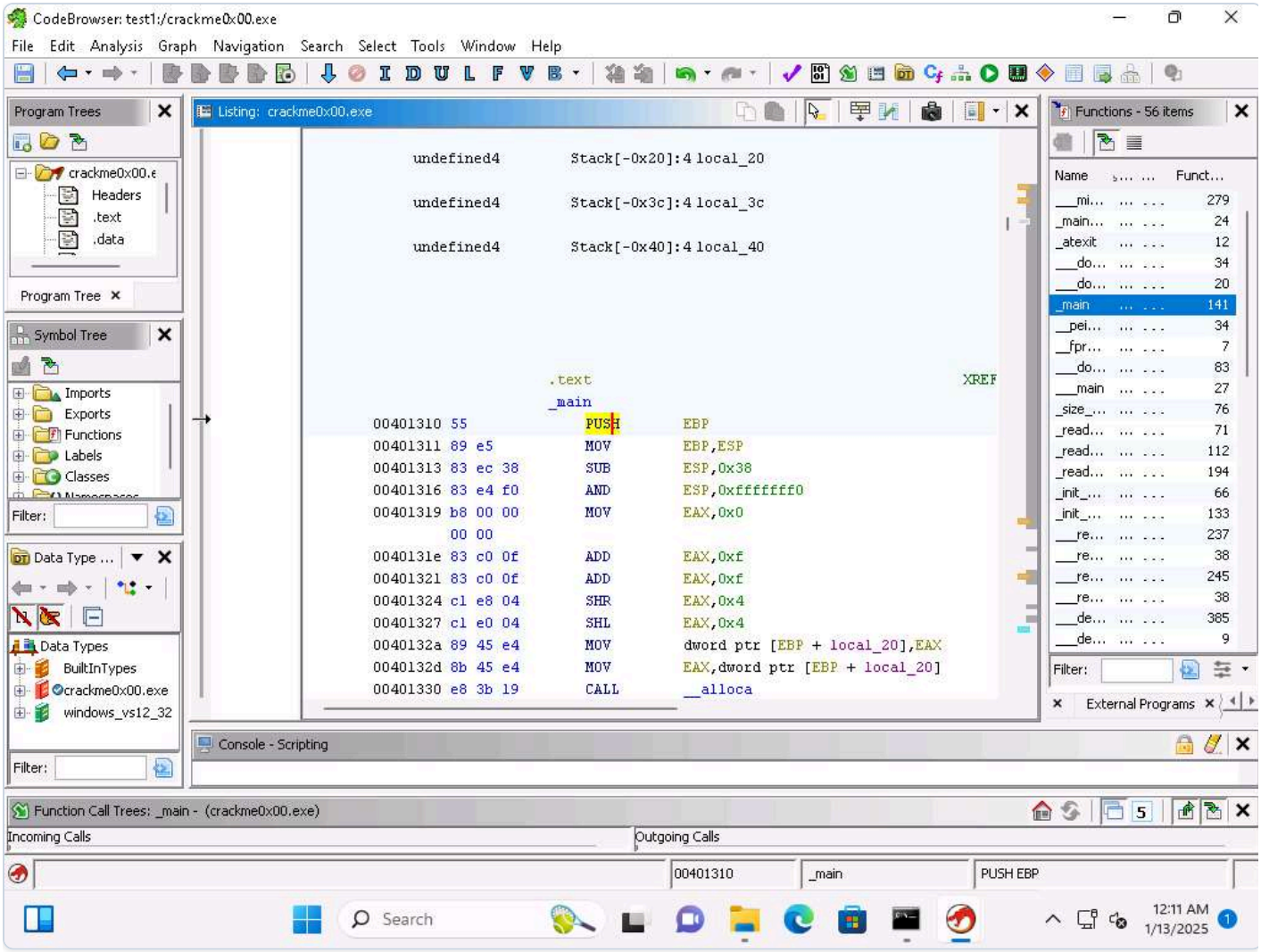

1. Select `main` function in Listing window and analyze stack allocation disassembly:
   - `PUSH EBP`: Save previous base pointer
   - `MOV EBP, ESP`: Set frame pointer
   - `SUB ESP, 0x38`: Allocate stack frame space



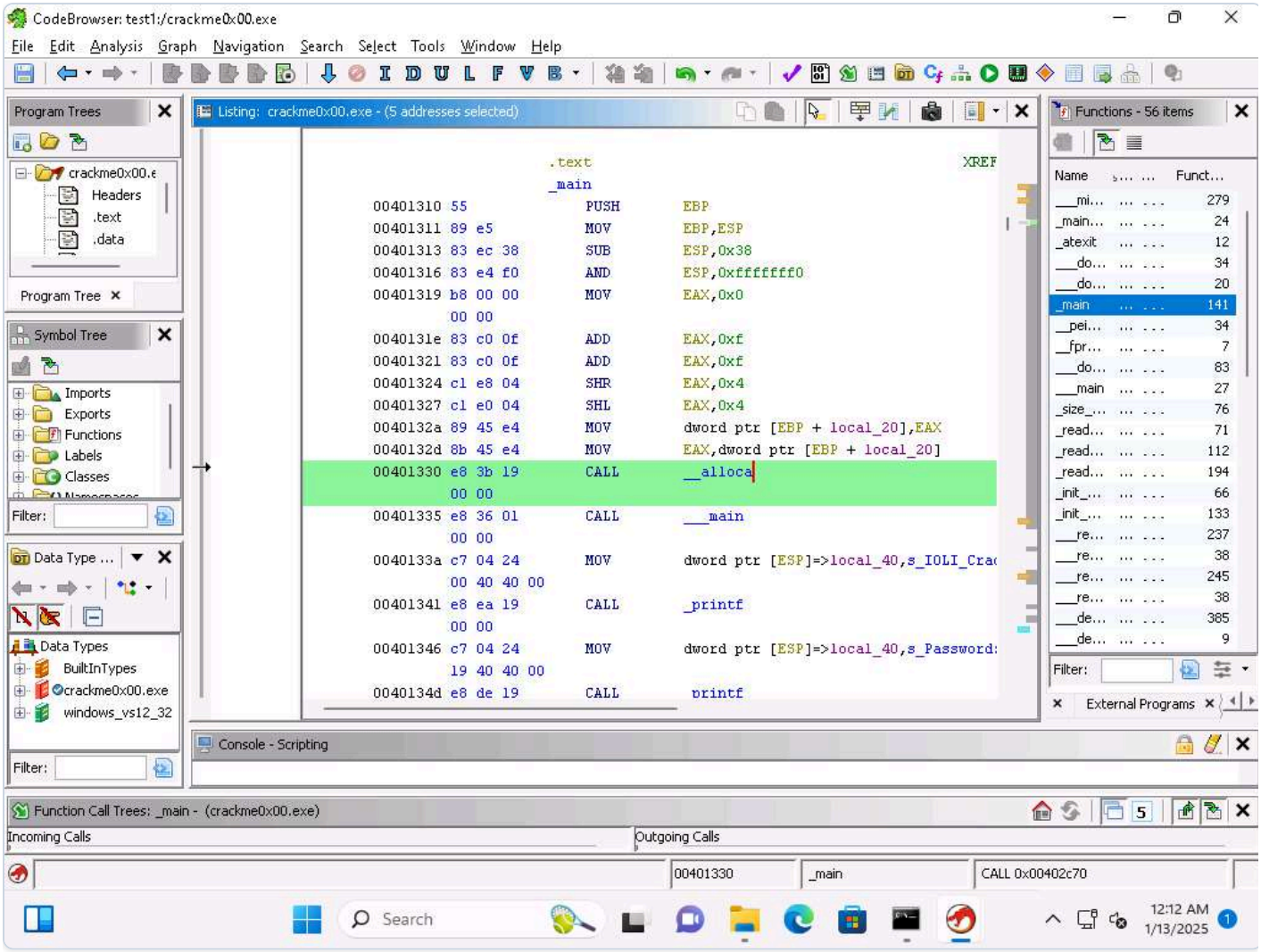

1. Inspect 16-bit stack alignment instructions ( `AND ESP, 0xfffffff0` ).



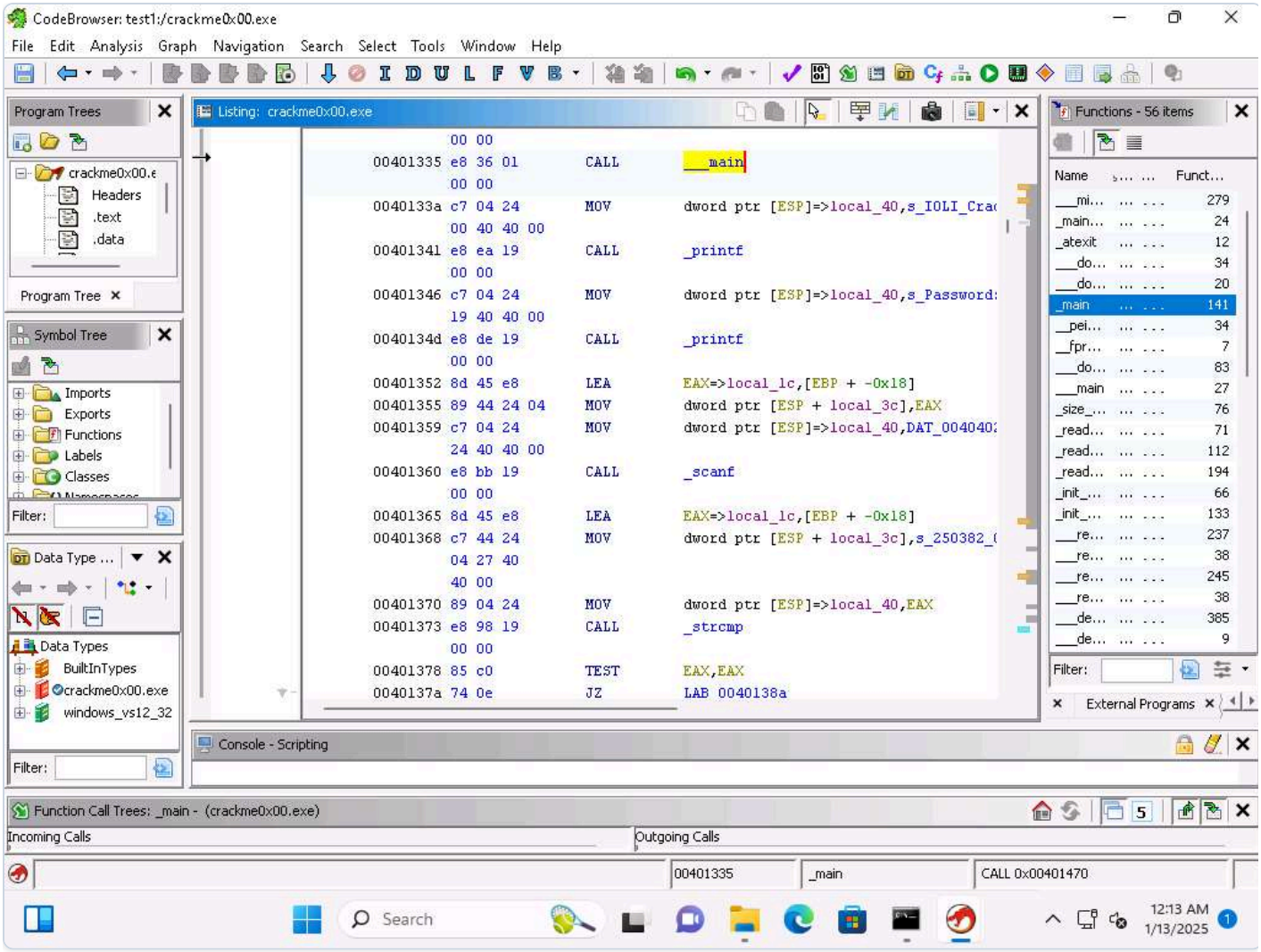

1. Review compiler setup calls preceding user logic.



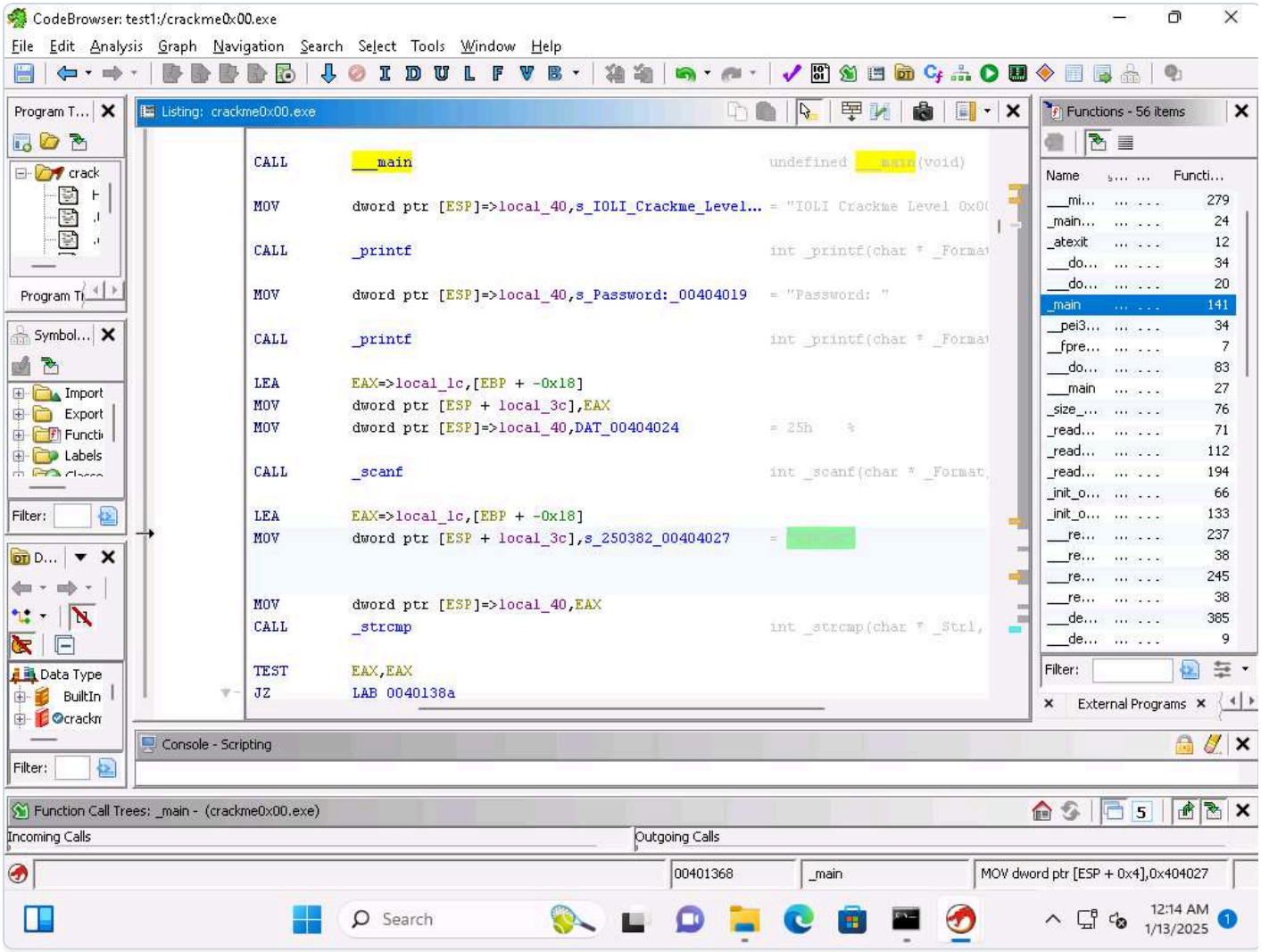

1. Analyze Load Effective Address ( `LEA` ) instructions loading `EBP - 0x18` into `scanf` argument parameters.



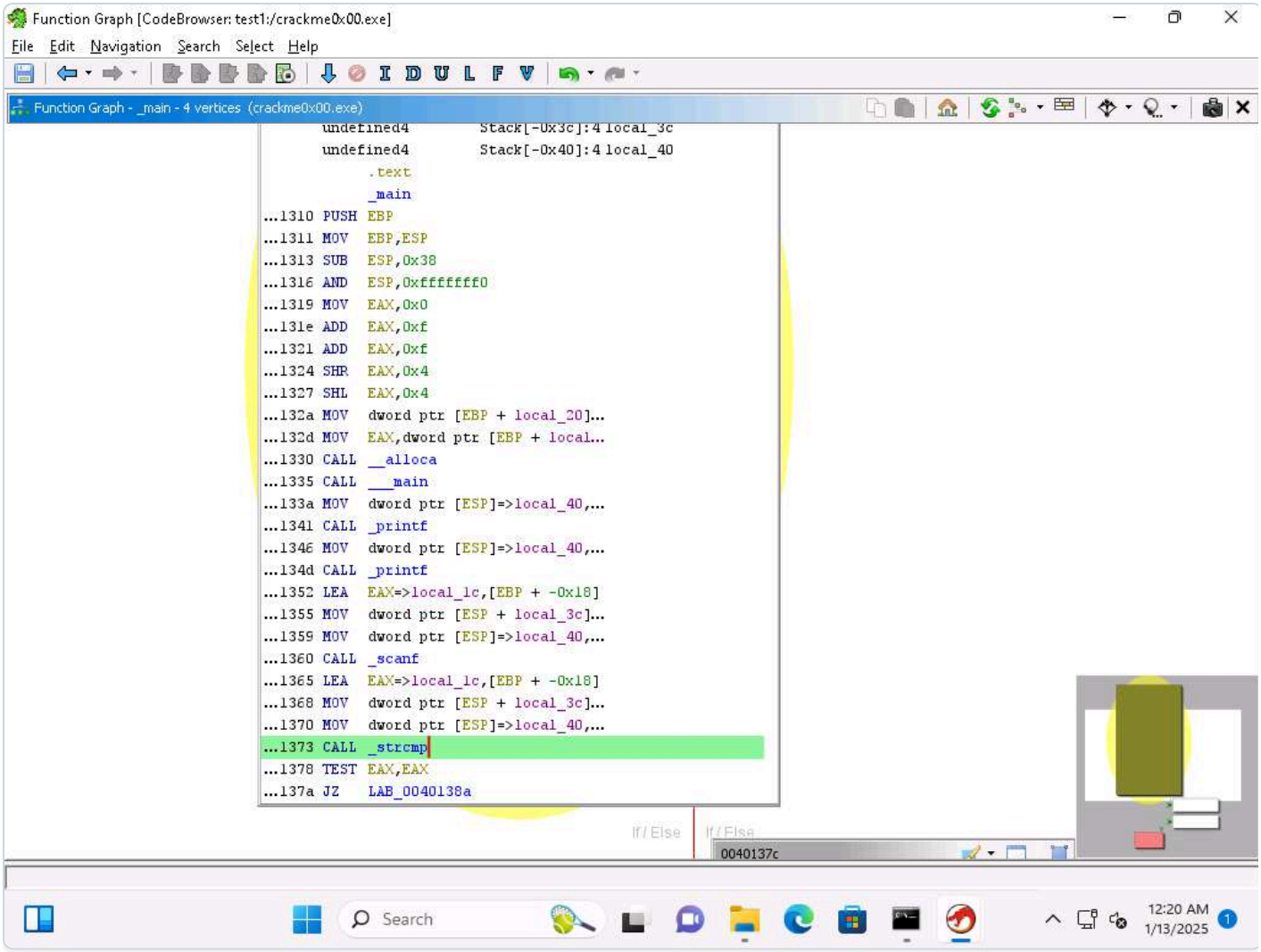

1. Identify hardcoded string constant value: **250382**.
   Observe assignment into `local_40` variable preceding `strcmp` call.



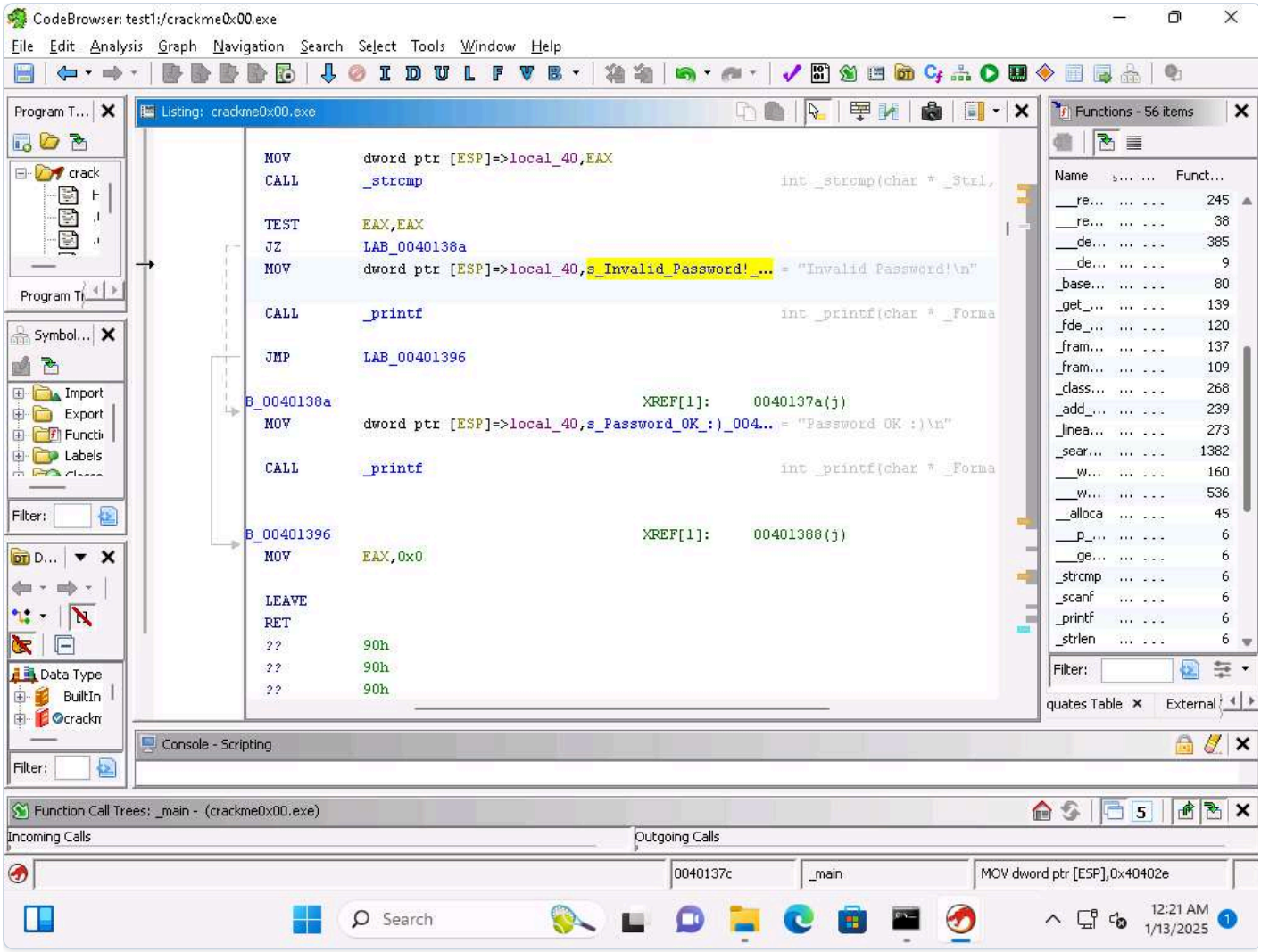

1. Inspect Function Graph basic block for `strcmp` jump branches.



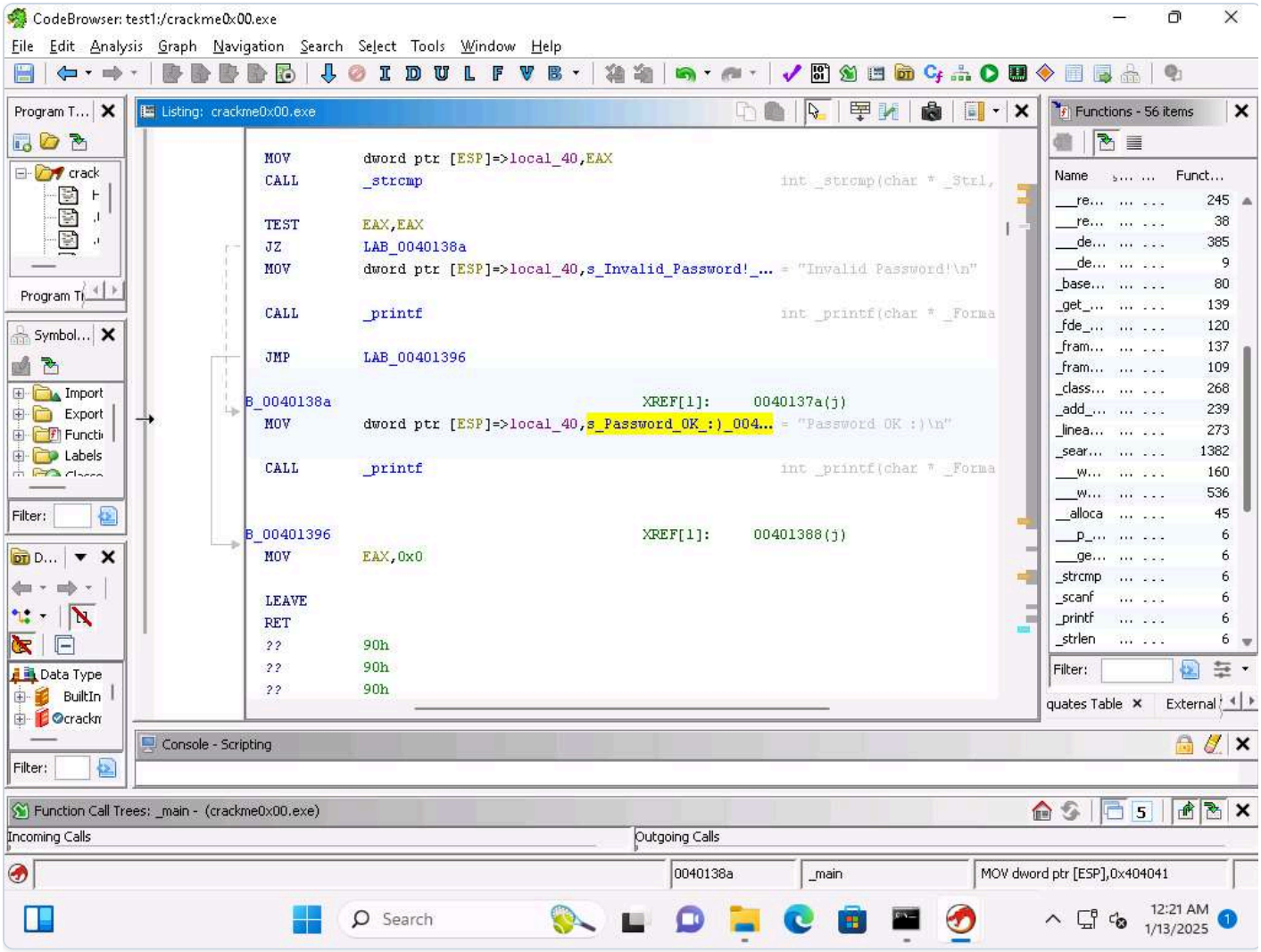

1. Observe Jump Zero ( `JZ 0040138a` ) branch triggered upon matching password string.



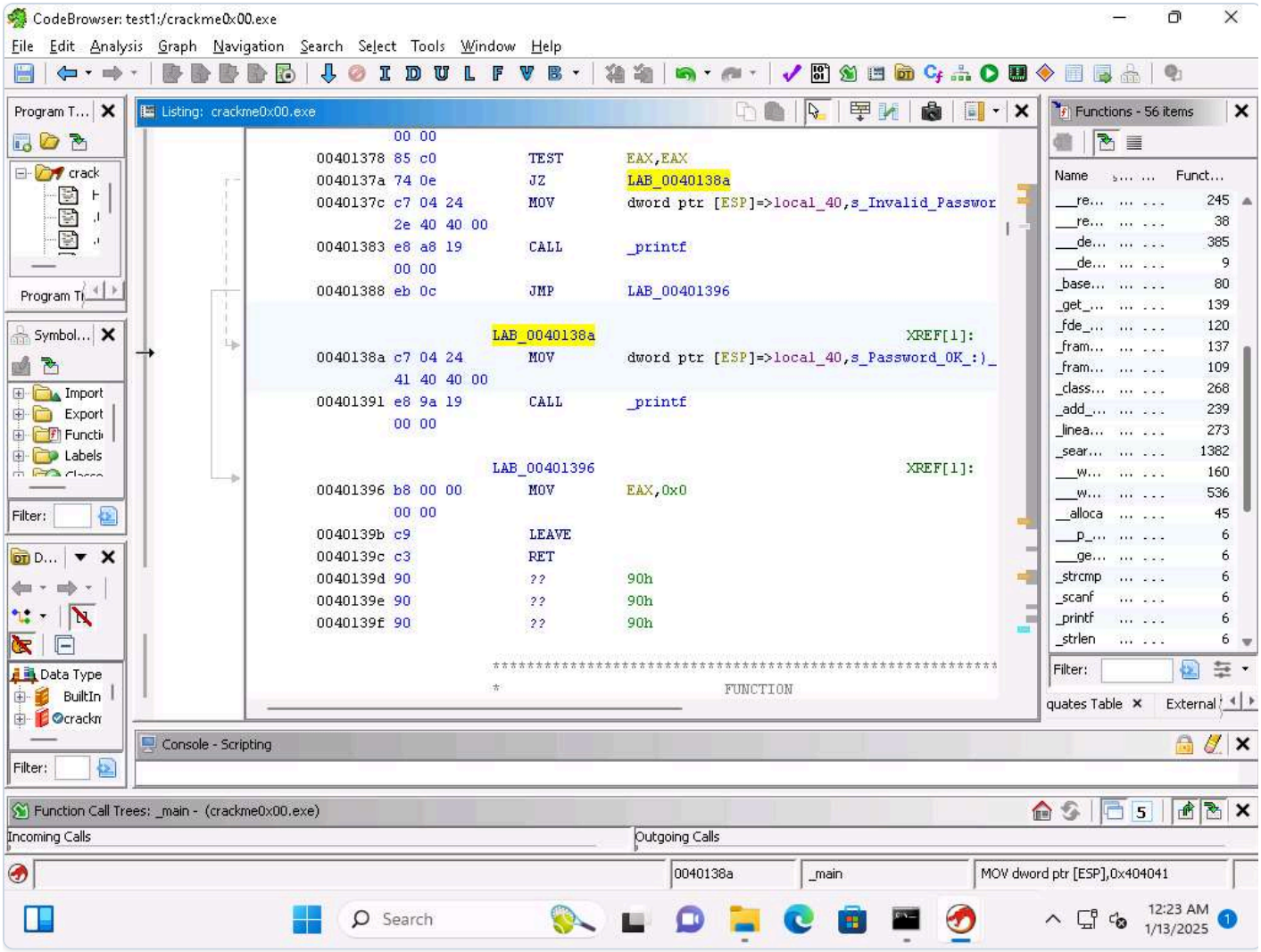

1. Inspect `Invalid Password` failure block branch.




1. Inspect `Password OK` success block branch.



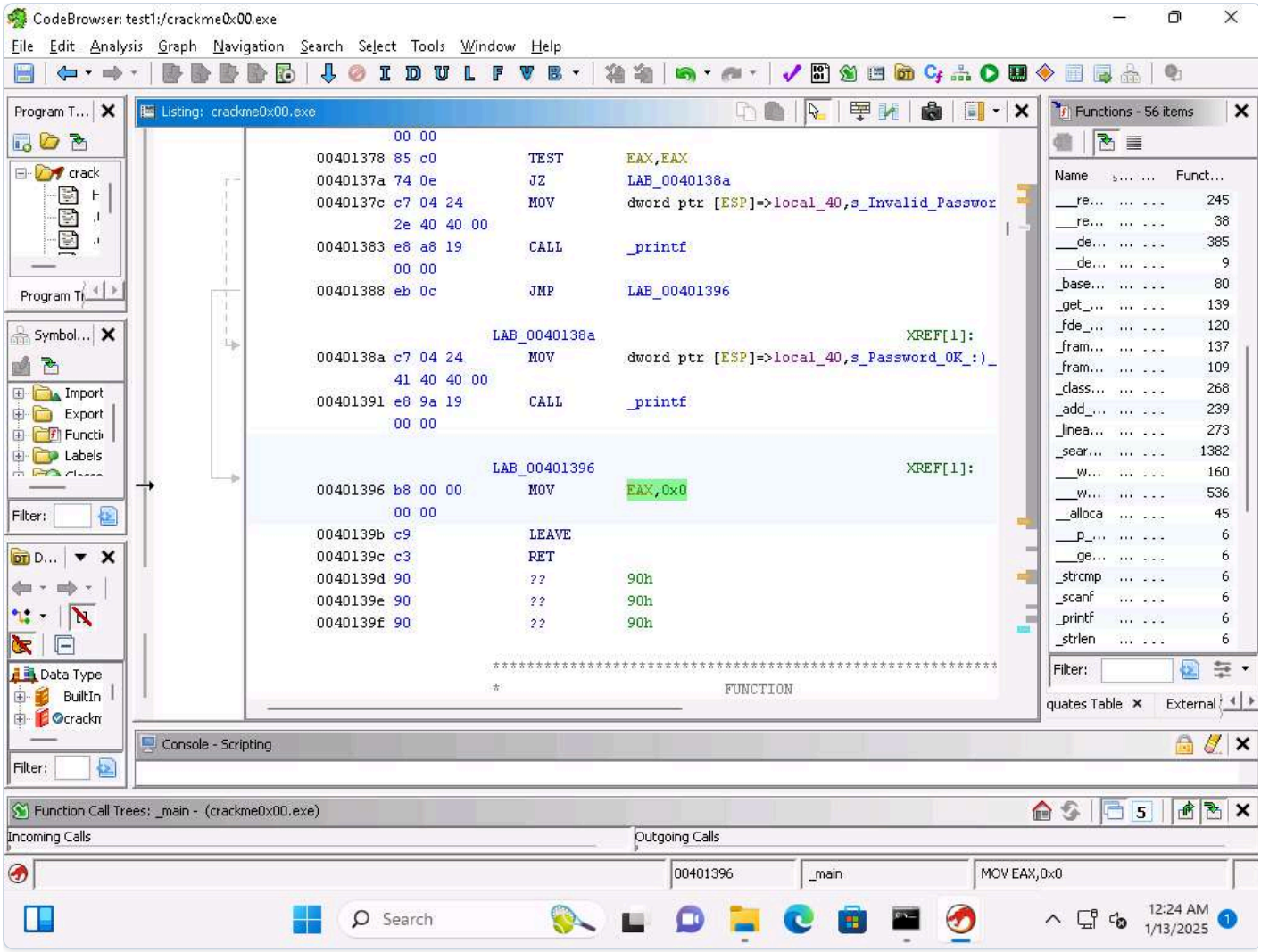

1. Inspect function exit block returning `0x0` in `EAX`.



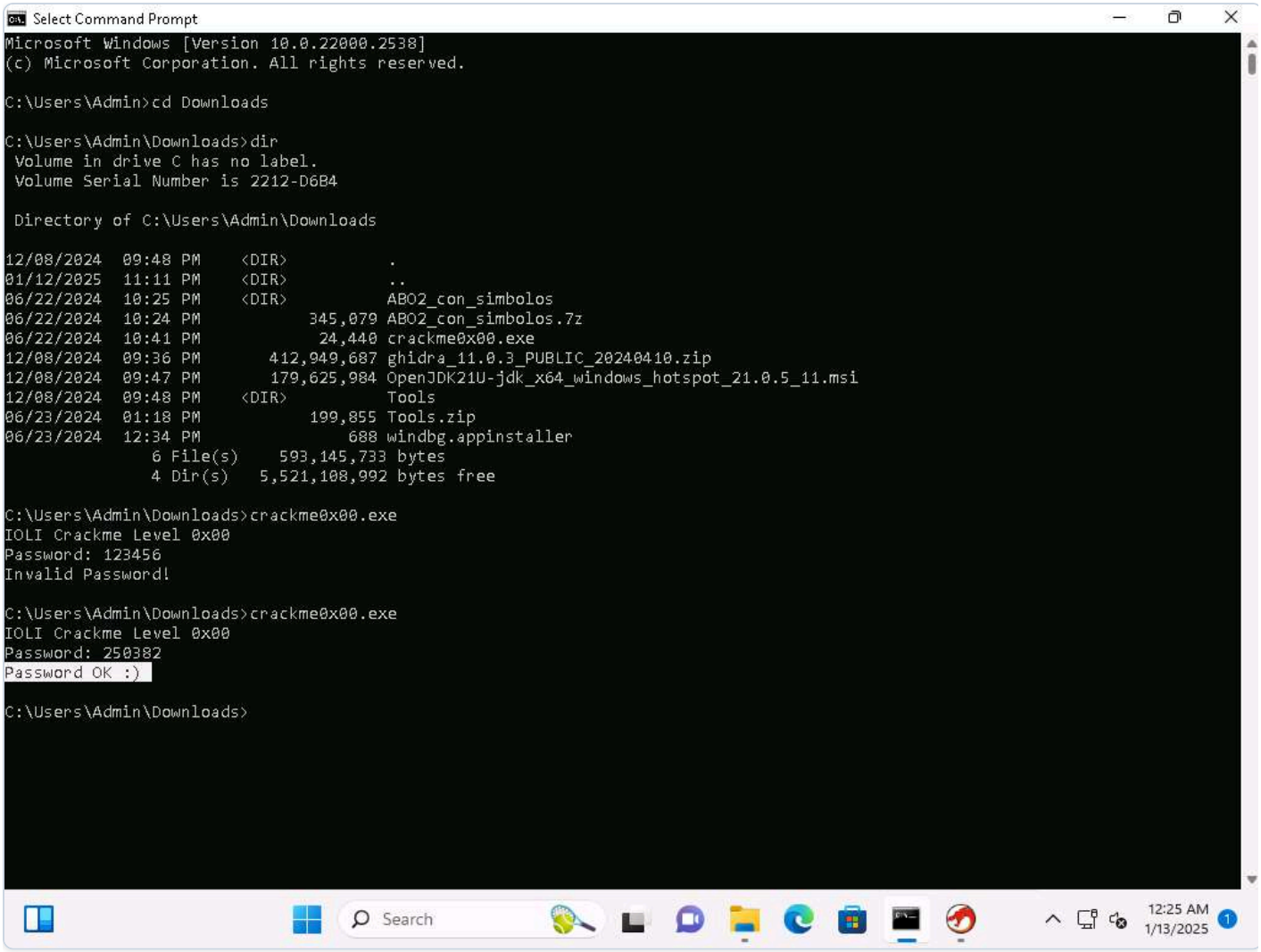

1. Return to CMD prompt and input recovered password:

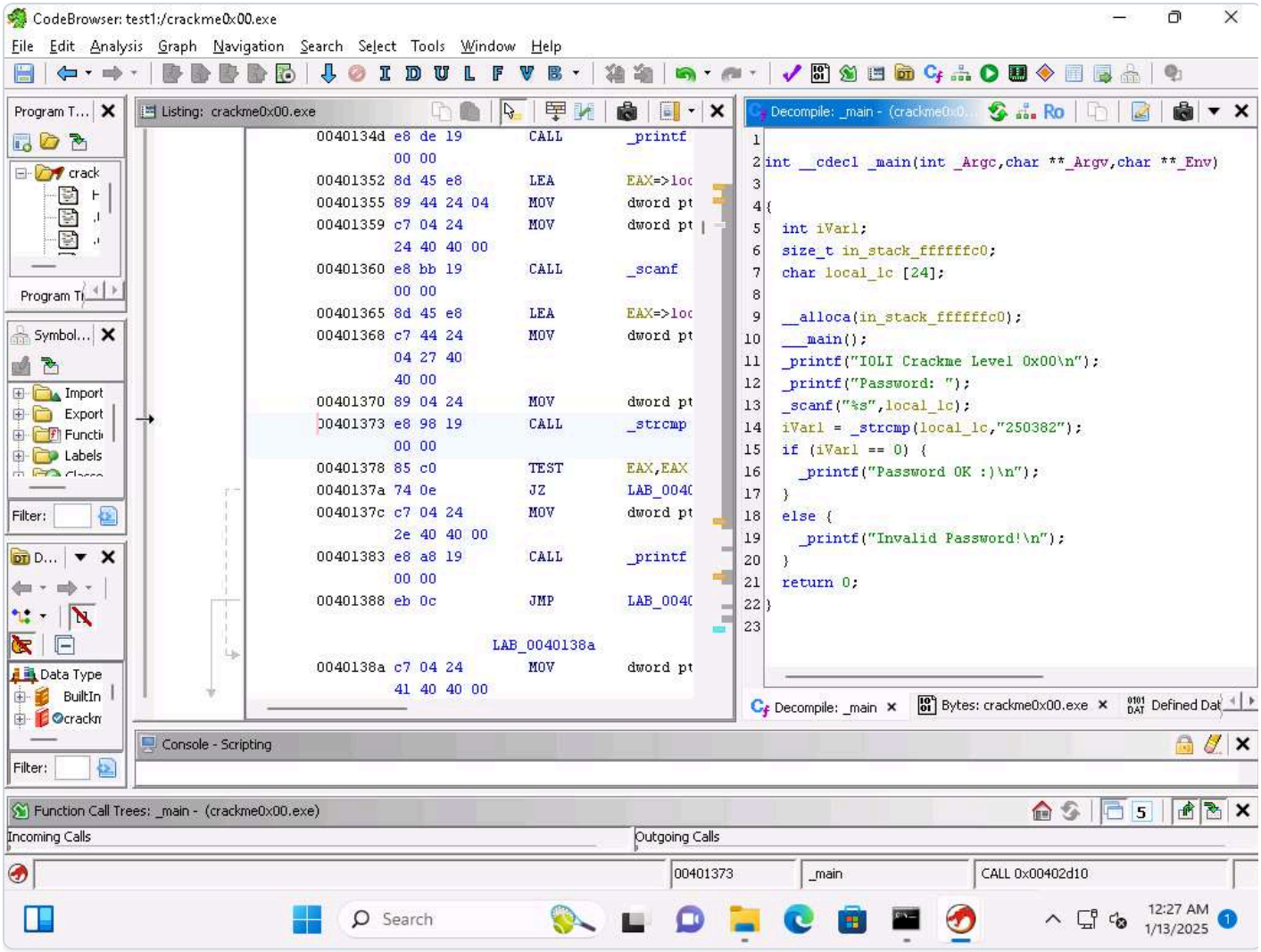
   **250382**

   *Observe successful output:* **Password OK!**




1. Decompile `main` function in Ghidra Decompiler window to validate C pseudo-code comparison logic (`strcmp(local_1c, "250382")`).


Select Command Prompt
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Admin>cd Downloads
C:\Users\Admin\Downloads>dir
Volume in drive C has no label.
Volume Serial Number is 2212-D6B4
Directory of C:\Users\Admin\Downloads
12/08/2024 09:48 PM <DIR> .
01/12/2025 11:11 PM <DIR> ..
06/22/2024 10:25 PM <DIR> ABO2_con_simbolos
06/22/2024 10:24 PM 345,079 ABO2_con_simbolos.7z
06/22/2024 10:41 PM 24,440 crackme0x00.exe
12/08/2024 09:36 PM 412,949,687 ghidra_11.0.3_PUBLIC_20240410.zip
12/08/2024 09:47 PM 179,625,984 OpenJDK21U-jdk_x64_windows_hotspot_21.0.5_11.msi
12/08/2024 09:48 PM <DIR> Tools
06/23/2024 01:18 PM 199,855 Tools.zip
06/23/2024 12:34 PM 688 windbg.appinstaller
6 File(s) 593,145,733 bytes
4 Dir(s) 5,521,108,992 bytes free
C:\Users\Admin\Downloads>crackme0x00.exe
IOLI Crackme Level 0x00
Password: 123456
Invalid Password!
C:\Users\Admin\Downloads>crackme0x00.exe
IOLI Crackme Level 0x00
Password: 250382
Password OK :)
C:\Users\Admin\Downloads>
Search
12:25 AM
1/13/2025




This concludes the lab exercise.

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**IMPORTANT:** ❓ **Question 11.10.1.1** In Windows 11 machine, use Ghidra to analyze ABO2.exe and Crackme0x00.exe and find the password of crackme0x00.exe file. Enter the password of crackme0x00.exe that is found in this lab. [!SUCCESS] ✅ **Answer:** **250382**